Vanguard 8ch Breast Table for 1.5/3T GE HDx Installation and Service Manual

Manufactured By:

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Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric or Sentinelle Medical representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

To file a report with GE:
Call 1–800–548–3366 and use option 6.
Fill out a report on http://edq.health.ge.com/edq/home.jsp. Contact your local service coordinator for more information on this process.

To file a report with Sentinelle:
Call 1-866-735-3744 and use option 1.

ATTENTION!

A GE field engineer or a Sentinelle Medical Field Service Technician MUST perform the installation of this coil. Please contact the field engineer from the company which sold the product to arrange for installation.

Attempts to perform installation of this coil without proper system configuration will result in system errors.

ПРЕДУПРЕЖДЕНИЕ (BG)

Това упътване за работа е налично само на английски език.

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод.
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告 (ZH-CN)

本维修手册仅提供英文版本。

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。

警告 (ZH-HK)

本服務手冊僅提供英文版本。

- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
- 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。
- 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他的危險。

警告 (ZH-TW)

本維修手冊僅有英文版。

- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

UPOZORENJE (HR)

Ovaj servisni priručnik dostupan je na engleskom jeziku.

- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod.
- Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik.
- Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

VÝSTRAHA (CS)

Tento provozní návod existuje pouze v anglickém jazyce.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL (DA)

Denne servicemanual findes kun på engelsk.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING (NL)

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)

This service manual is available in English only.

- If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

HOIATUS (ET)

See teenindusjuhend on saadaval ainult inglise keeles

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS (FI)

Tämä huolto-ohje on saatavilla vain englanniksi.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION (FR)

Ce manuel d’installation et de maintenance est disponible uniquement en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d’installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG (DE)

Diese Serviceanleitung existiert nur in englischer Sprache.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS (HU)

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése.
- Ne próbálja elkezdeni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fáanleg á ensku. · Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu. · Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. · Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. · Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. · Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. · Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
警告 (JA)	このサービスマニュアルには英語版しかありません。 · サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 · このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 · この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. · 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. · 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. · 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. · Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. · Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. · Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. · Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. · Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. · Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. · Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. · Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. · Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE (PL)

Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

AVISO (PT-BR)

Este manual de assistência técnica encontra-se disponível unicamente em inglês.

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

ATENÇÃO (PT-PT)

Este manual de assistência técnica só se encontra disponível em inglês.

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

ATENŢIE (RO)

Acest manual de service este disponibil doar în limba engleză.

- Dacă un furnizor de servicii pentru clienţi necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercaţi să reparaţi echipamentul decât ulterior consultării şi înţelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО! (RU)

Данное руководство по техническому обслуживанию представлено только на английском языке.

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

UPOZORENJE (SR)

Ovo servisno uputstvo je dostupno samo na engleskom jeziku.

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja servisera, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

UPOZORNENIE (SK)

Tento návod na obsluhu je k dispozícii len v angličtine.

- Ak zákazníkov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

ATENCION
(ES)

Este manual de servicio sólo existe en inglés.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING
(SV)

Den här servicehandboken finns bara tillgänglig på engelska. .

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

OPOZORILO
(SL)

Ta servisni priročnik je na voljo samo v angleškem jeziku.

- Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod.
- Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli.
- Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.

DİKKAT
(TR)

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

Table of Contents

Note:

There are sections in this manual that apply only to Sentinelle Medical (SMI) personnel or GE Healthcare (GEHC) personnel. These sections are noted using the following symbols.

SMI ONLY	

OR

GEHC ONLY	

GEHC sold and serviced products will have a "Distributed by: GE Heathcare" label next to the label on the stretcher assembly and the rating plate on the patient support assembly.

1 Purpose

This manual provides detailed instructions to an SMI Service Technician or a GEHC field engineer for installing, troubleshooting and repairing the Vanguard (VG) System including the stretcher and RF Coils (mechanical and electrical parts). The Troubleshooting Table details the required steps to methodically assess and correct the reported problem(s).

Information reported from the site contact explaining the problem is paramount and the Service Technician must record the details. The Service Technician must carefully gather and interpret the information received from the site contact, to ensure all details are enclosed to enable the problem to be quickly diagnosed and repaired.

2 Compatibility

This device is compatible with GE Signa HDi/HDx/HDxt/Vibrant 8ch 1.5T and HDx/HDxt 3T MRI Scanners with a 60cm bore diameter, operating on the software versions indicated below. The table is not compatible with CRM platforms.

- 1.5T (GEHC part 5318678, SMI part 4000001-11): Software version 14.0.
- 3T (GEHC part 5419434, SMI part 4000700-11): Software version HD16.0_V02 with Service Pack 1 is required to use the coil; however, software version HD23.0_V01 with Service Pack 1 or later is recommended for better performance.

CAUTION **The Vanguard uses different RF systems with 1.5T and 3T scanners. Ensure that the correct coils are present before scanning.**

3 Service Tools Required

The tools defined in this section are all that are required to perform all service procedures in this manual. Many of these tools are included in the standard toolkit.

3.1 Primary Tools

Many of these tools are included in the standard toolkit. Equivalent tools may be used. Note that only non-magnetic tools may be used in the magnet room.

Item	***GEHC Part #***	***SMI Part #***	***Qty***	***Description***
1	5112574	1-927-16537	1	Allen wrench set, stubby, w/ball ends
2	5109892-6	03-L-4	1	Non-magnetic Allen Wrench 1/8"
3	5109892-8	03-L-5	1	Non-magnetic Allen Wrench 5/32"
4	5109892-9	03-L-6	1	Non-magnetic Allen Wrench 3/16"
5	5109892-11	03-L-8	1	Non-magnetic Allen Wrench 1/4"
6	See ***Note**	5000274-11	1	Non-magnetic wrench caster bolt 1.5"
7	5109895-13	926-3A1-07B	1	Non-magnetic screwdriver 8mm (cabinet) tip, 6" OAL
8	5109900-5	913-610-10B	1	Non-magnetic combination wrench 1/2” box/open
9	5109900-6	913-610-12B	1	Non-magnetic combination wrench 9/16” box/open
10	5109896-2	912-510-04B	1	Non-magnetic adjustable wrench, 3/4" max open, ~6" OAL
11	5109888	911-810-02B	1	Non-magnetic ratchet, hinged 3/8" drive, 6" OAL
12	5109890-5	910-310-10B	1	Non-magnetic 9/16" socket, 3/8" drive
13	5109890-4	910-310-08B	1	Non-magnetic 1/2" socket, 3/8" drive
14	5109897-2	925-410-02B	1	Non-magnetic pliers, needle nose, 6"
15	5109897-3	See ****Note**	1	Non-magnetic pliers, side cutter, titanium
16	5437045	NA	1	Scissor jack to support stretcher when changing castors

Note: Wrench 1.5” (show in green in Figure 1) can be found inside the rear support assembly of the VG stretcher.

****Note:*** Only applicable to GEHC personnel. SMI Service technician should use SMI equivalent in procedures that call for this tool.

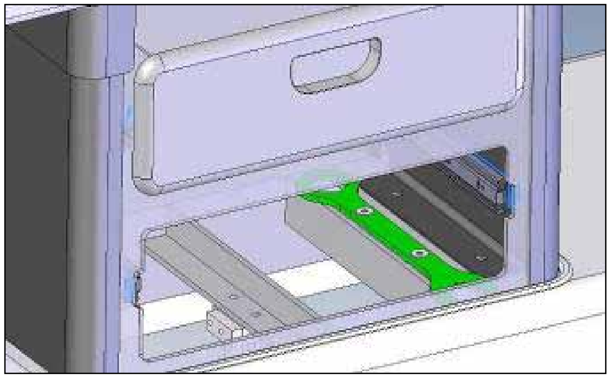


Figure 1: 1.5” Wrench Location

3.2 Consumables

All of these items are included in the standard service toolkit and/or the consumables FRU kit (P/N 4001158-11, GEHC P/N 5408278).

Item	***GEHC Part #***	***SMI Part #***	***Qty***	***Description***
1	N/A	20104C0140002	6	1/4-20 X .5 FHMS
2	N/A	20104C0140000	1	1/4-20 X .215 FHMS
3	N/A	20305FN100006	4	10-32 X 1 RHMS Slot
4	N/A	20305FN100006	4	10-32 X 1 RHMS Hex
5	N/A	21501FN100002	2	LBS cover screws 10-32 X .5"
6	N/A	20201CN080001	1	8-32 X .325" STD
7	N/A	20901N5160000	1	Lock washer 5/16 316 SS
8	N/A	20601C5160000	1	Hex nut 5/16-18 316 SS
9	N/A	20301CN100002	2	10-24 X .5 RHMS
10	N/A	1-444-29021	1	Loctite Green 290 Wicking, 50mL
11	N/A	250689-02	10	Cable End
12	N/A	253662-01	2	1.22 mm bicycle cable
13	N/A	AR10BDR-GL-1/2	2	Patient support wheels, double row
14	N/A	5000261-11	1	Cable Elbow 105 service ass'y
15	N/A	5000260-11	2	Cable Elbow 75 service ass'y
16	N/A	250689-02/ S-11712	1m	Velcro (one length)
17	N/A	E0500-069-3500-S	1	Undock Spring SPRING EXT 0.5OD .069D 3.5L0
18	N/A	E0750-105-3000-S	2	Pedal Spring SPRING EXT .75OD .105D 3.0L0
19	N/A	E0500-063-2000-S	1	No Dock Spring SPRING EXT 0.5OD .063D 2.0LO
20	N/A	C0480-042-1750-S	2	Bridge spring block Spring SPRING 0.48OD .042D 1.75L0 304SS
21	N/A	CS-23	2	Emergency Release Actuator Spring SPRING .345OD .045D 1.125L0 BC

3.3 General Safety Requirements

NOTICE

Follow all required safety and PPE procedures customary for your organization when working on this product.

- Wear safety shoes for foot protection.
- Wear gloves for hand protection.
- Wear safety glasses for eye protection.

3.3.1 *Occupational Exposure to Infectious Agents*

Installing and servicing the Vanguard requires visiting medical facilities which present infection hazards, especially from possible exposure to pathogens found in blood and other potentially infectious bodily materials. Pathogens are microorganisms that can cause disease when transmitted from an infected individual to another individual through blood and certain other body materials. The most common illnesses caused by blood borne pathogens (BBPs) are hepatitis B (HBV), hepatitis C (HCV), and acquired immunodeficiency syndrome (AIDS) from HIV, human immunodeficiency virus. These blood borne pathogens are capable of causing serious illness and death.

WARNING **Failure to adhere to these procedures and precautions may result in infection, serious illness and death.**

Types of Hazardous Exposure

Hazardous exposures which may occur in medical facilities include:

- Percutaneous innoculations or punctures with blood or body fluid by a sharp instrument or needle.
- Contact with blood or body fluid through a fresh cut (less than 24 hours from occurrence) or mucous membrane contact (e.g., a splash to the eye or mouth, or mouth-to-mouth resuscitation).
- Skin exposure involving large amounts of blood or prolonged contact with blood, especially when the exposed skin is chapped, abraded, or afflicted with dermatitis.

Avoiding Exposure

Before beginning work at a medical facility where you may encounter potential infection hazards:

- Ensure that you have had a Hepatitis B vaccination.
- Before beginning to work on equipment, check carefully for signs of blood or body fluids.
- If you notice possible blood or body fluids on the equipment, report it to the responsible facility personnel and have them clean the equipment using their institutional infection control procedures.
- If there is a reasonable expectation of the presence of infectious materials, do not work on the equipment until it is disinfected.
- As a precautionary measure, wear personal protective equipment including gloves, protective eye covering and disposable clothing whenever the potential to be exposed to blood or other potentially infected materials exist.

After Exposure

If you contact a potentially infectious blood or body fluid:

- Seek immediate medical assistance.
- Advise your supervisor of the exposure event.

GEHC ONLY	· If you have previously refused vaccination to Hepatitis B, GE must offer the vaccination to you at this time at no charge. · As part of the investigation process, contact the GE medical team to assist in identifying the status of source individuals.

- Document the incident for your employer and the facility where the incident occurred:
  - If feasible, document the name and address of the individual source of the possible infection.
  - Document the time, place, and circumstances of the contact.

4 Definitions and Abbreviations

Throughout this document the following terms and abbreviations will be used:

316 SS	Stainless Steel Grade 316
Foot End	Patient Foot end of the Vanguard
FRU	Field Replaceable Unit
FSA	Front Support Assembly
Head End	Patient Head end of the Vanguard
HYP	Hypertronics
LBS	Lower Body Support
Left/Right	Directions relative to the patient’s position
LPCA	Low Profile Carriage Assembly
MCQA	Multi Coil Quality Assurance
Patient Support	Top portion of the Vanguard
RSA	Rear Support Assembly
Service Toolkit	SMI-0294 Toolkit utilized by SMI Field Service Technician
SMI	Sentinelle Medical
Stretcher	Bottom portion of the Vanguard
UBS	Upper Body Support
User Manual	SMI-1238 for 3T systems or 5000398-11 for 1.5T systems.

5 Performing Service

5.1 Precautions

Before entering the MRI Room with the Vanguard system the following activities must be performed:

- Thoroughly inspect the system to ensure loose ferrous/magnetic items are not present anywhere on the system. Loose objects can become airborne and/or travel at high velocity in the presence of the scanner, potentially causing harm to the Patient, Operator or equipment.
SMI ONLY	Do not attempt to make any changes or adjustments to the GE scanner or accessories. If any damage to the GE scanner is observed, report it to customer.
- Do not touch metal pins of any connectors. Leave the protective packaging on all connectors until install is complete to prevent electrostatic discharge damage.
- Patient data is confidential – be sure not to examine, copy or photograph any printed or electronic matter containing patient names, personal details, medical images or other confidential data to ensure compliance with the Health Insurance Portability and Accountability Act (HIPAA).

CAUTION

- ***When replacing any springs on the Vanguard, be sure to do so outside of the scanner room. Springs in the Vanguard are Stainless Steel may be mildly magnetic***

5.2 SMI ONLY Preparing for the Service Call

Prior to performing the service call the following are required:

- Installation must be booked before application training is scheduled to start. Follow up 3 days before the installation date to remind the customer and confirm that the scanner time is booked off and the Vanguard has been delivered to the immediate area of the scanner.
- A copy of the P.O. or work order for the unit to be installed.
- Contact information for the site (usually the charge tech or administrator) and back-up contact person.
- Determine if any nearby sites can be visited during the same trip to perform preventative maintenance.
- Bring a complete service kit, camera (if possible), notebook, service report cards and personal business cards.
- When international travel is required, a signed letter from the service manager or other manager explaining the purpose of the trip, the destination, dates and the employment status of the service technician.
- ID badge, to be kept on person at all times, and passport/visa (as needed).
- Hotel info, customer site address, directions to/from site, airport and hotel.
- If possible, contact the site specific Service Personnel to inform them of the pending service call and request information on any recent problems with or service to the scanner (SMI service calls can be generated from recent service to the scanner involving adjustment of mechanical parts or software upgrades and patches). Request they be present to provide the service key (code).
- Ensure that only Sentinelle-authorized service personnel make adjustments or repairs to the system.
- Review Sentinelle service logs for the site for previous service reports and service pending.

5.3 Recording the Service Call

Record the results of all activities performed including:

- Any non-conformances that were discovered during unpackaging and installation or service and any remedial action taken.
- Leave a completed service card and inform relevant staff members who are present that service is complete and indicate state of functionality. If the system fails any of the functionality tests in the installation procedure, it must be taken out of service.
- Unless prior arrangements have been made with the Applications Support Specialists, the system will normally be “Out of Service – Pending Applications training” immediately following installation.
- A SMI or affiliate service log must be filled out to officially record the installation.
- If any non-conformances or problems are discovered during installation, it is imperative that these are recorded, reported and entered into the Problem Management / Complaint Database as soon as possible.

Record the rating plate number and serial number when recording the service call.

5.4 Departure

- Ensure work area is clean and no tools or garbage are left behind.
- Replace any coils and padding previously on the scanner bed.
- Place all Vanguard components and accessories in a common storage area.
- If possible, inform staff member on site of completion before leaving.
- Ensure doors (especially scanner room door) are locked and lights are off in accordance with the customer site’s requirements.
-

SMI ONLY	Complete a Sentinelle calling card and leave it in a prominent place, preferably at the console.

6 System Contents

This section outlines the shipping contents of this system and locates the rating plate labels for product identification.

6.1 Shipping Contents

The Vanguard components are shipped in major three groups; the Patient Support with Stretcher, RF system and Padding Kit. All should be verified with included packing list. Refer to ***Damage Transportation*** information section on page 3 of this manual if damage is found.

6.1.1 Patient Support and Stretcher

Item	GEHC Part #	SMI Part #	Description
1	5326168	4000008-11	Patient Support Assembly*****
2	-	4000064-11	Stretcher Assembly
3	5408280	SVC-00301	Head Rest Assembly
4	5399108	4000049-51	Compression Frame Assembly L
5	5399109	4000050-51	Compression Frame Assembly R
6	5408239	SVC-00307	Compression Plate (2)
7	5326153	4000095-11	Compression Slider (2) (Enlarged)
8	5408284	SVC-00302	Shoulder Bridge (2)
9	-	SVC-00303	Contralateral Support
10	5408279	SVC-00306	Sternum Support
11	5419434-9	SVC-00401	GE Spherical Phantom Positioner******
12	-	5000268-11	Sentinelle Vanguard QA Phantoms (2) (not shown)†
13	-	MME-01439	Step Stool Folding (not shown)

Shown in Figure 2 with VG 1.5T 8ch Cable System installed.
***** This part is only available with the GE 3T 8Ch HDx Table System.***
† This part is only available with the GE 1.5T 8Ch HDx Table System.

GEHC ONLY	*** Note: Step Stool Folding (Item 11 in table above) is not a service tool.*

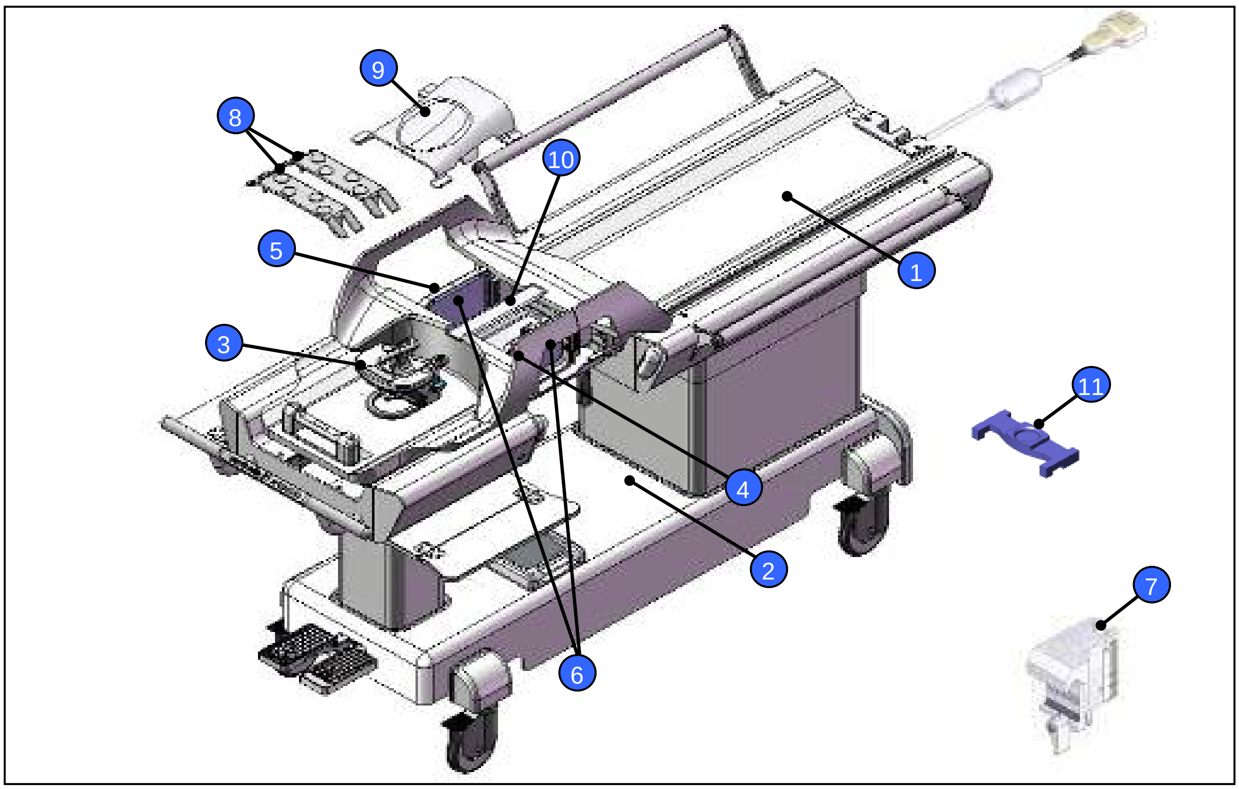


Figure 2: Patient Support and Stretcher

6.1.2 RF System - 1.5T

Item	GEHC Part #	SMI Part #	Description
1	5326161	4000012-12	Lateral Single Loop Coil Left VG
2	5326163	4000013-12	Medial Single Loop Coil Left VG
3	5326165	4000019-12	Lateral Array Coil Left VG
4	5326166	4000006-11	Medial Array Coil VG
5	5326164	4000019-11	Lateral Array Coil Right VG
6	5326162	4000013-11	Medial Single Loop Coil Right VG
7	5326160	4000012-11	Lateral Single Loop Coil Right VG
8	5326169	4000015-11	Vanguard HYP Cable System
9	5326167	4000058-11	Medial Plug

VG 1.5T 8ch Cable System is shipped installed in Patient Support.

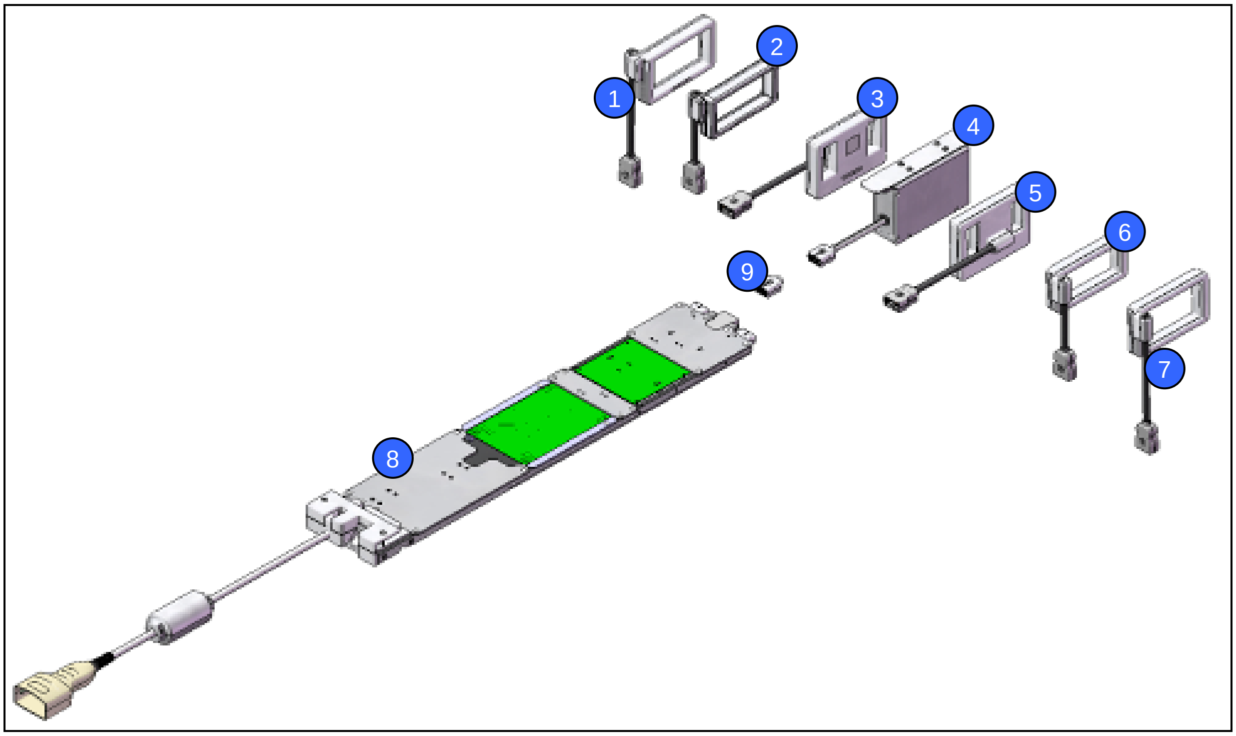


Figure 3: 1.5T RF Components

6.1.3 RF System – 3.0T

Item	GEHC Part #	SMI Part #	Description
1	5419434-6	4000172-11	Coil Assy Medial Array VG 3T
2	5419434-3	4000173-11	Coil Assy Single Loop Right VG 3T
3	5419434-2	4000180-11	Coil Assy Single Loop Left VG 3T
4	5419434-8	4000181-11	Vanguard HYP Cable System 3T
5	5419434-4	4000182-11	Coil Assy Lateral Array Right VG 3T
6	5419434-5	4000183-11	Coil Assy Lateral Array Left VG 3T
7	5419434-7	4000184-11	Assy Medial Plug 3T

***VG 3T 8ch Cable System is shipped installed in Patient Support**.

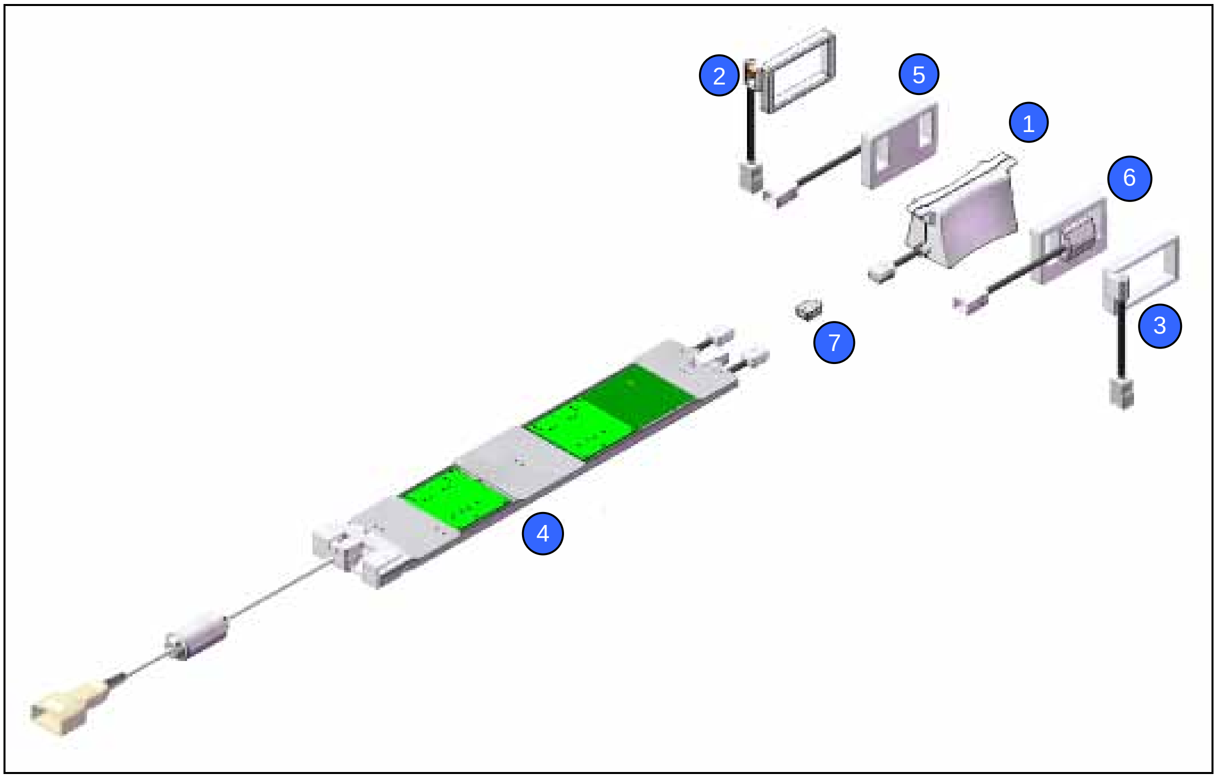


Figure 4: 3T RF Components

6.1.4 Padding Kit

Item	Box #	GEHC Part #	SMI Part #	Description
1	1 of 2	In 5408292 Kit	SVC-00311	Pad Shoulder Wedge Right
2	1 of 2	In 5408292 Kit	SVC-00312	Pad Shoulder Wedge Left
3	1 of 2	5408293	SVC-00304	Pad Arm Support
4	1 of 2	5408294	SVC-00305	Pad Sub Body
5	1 of 2	In 5408292 Kit	SVC-00313	Pad Foot Rest
6	1 of 2	In 5408292 Kit	SVC-00314	Pad Body Wedge
7	1 of 2	5408287	SVC-00308	Pad Headrest Thin
8	1 of 2	In 5408292 Kit	SVC-00315	Pad PS Guard Right
9	1 of 2	In 5408292 Kit	SVC-00316	Pad PS Guard Left
10	1 of 2	In 5408292 Kit	SVC-00317	Pad Medial Array VG
11	1 of 2	5408288	SVC-00309	Pad Hip Wedge

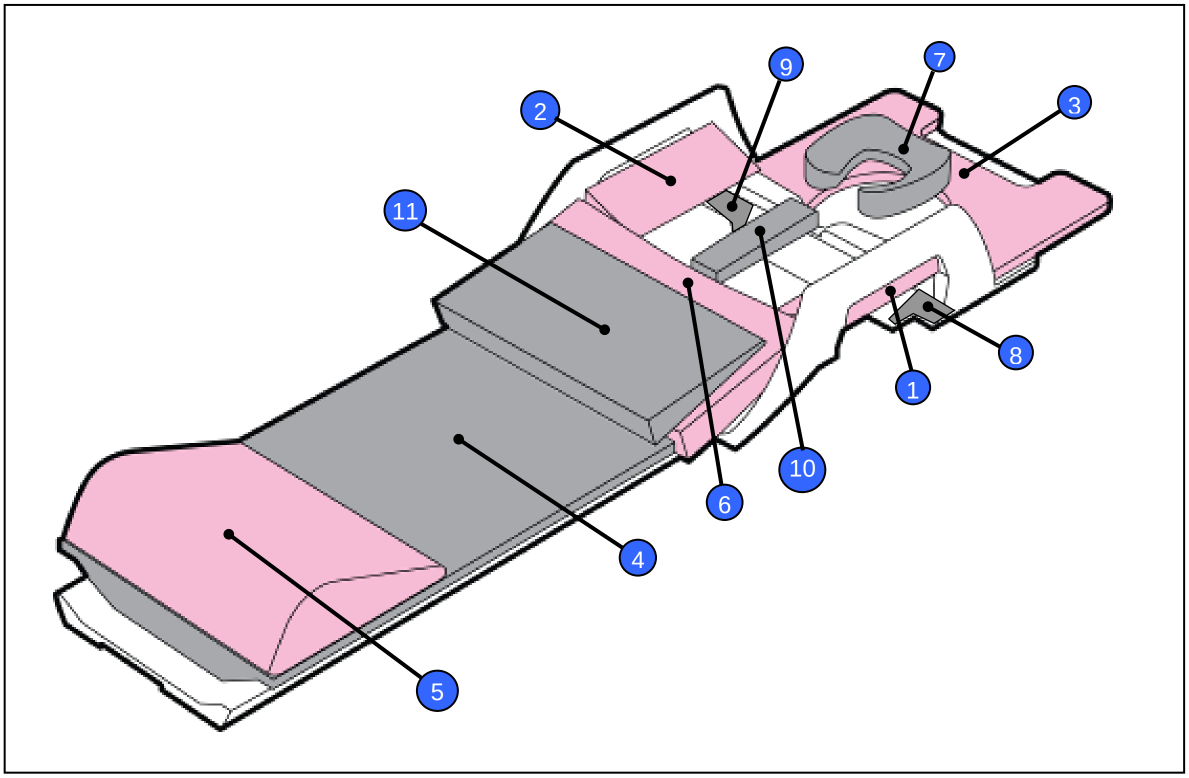


Figure 5: Padding Components

6.1.5 Miscellaneous Items

Item	Box #	GEHC Part #	SMI Part #	Description
1	2 of 2	-	SMI-0247	Service Manual (Distributed by GE only)
2	2 of 2	-	5000398-11 OR SMI-1238	Operator's Manual (for 1.5T Table System) Operator's Manual (for 3T Table System)

6.2 Product Identification and Label Locations

The system is uniquely identified using the labels below. For clarity, the labels are surrounded by a red box in the figures below. Products sold and serviced by GEHC will have a *"Distributed by: GE Heathcare"* label next to the label on the stretcher assembly (Figure 6) and the rating plate on the patient support assembly (Figure 7).

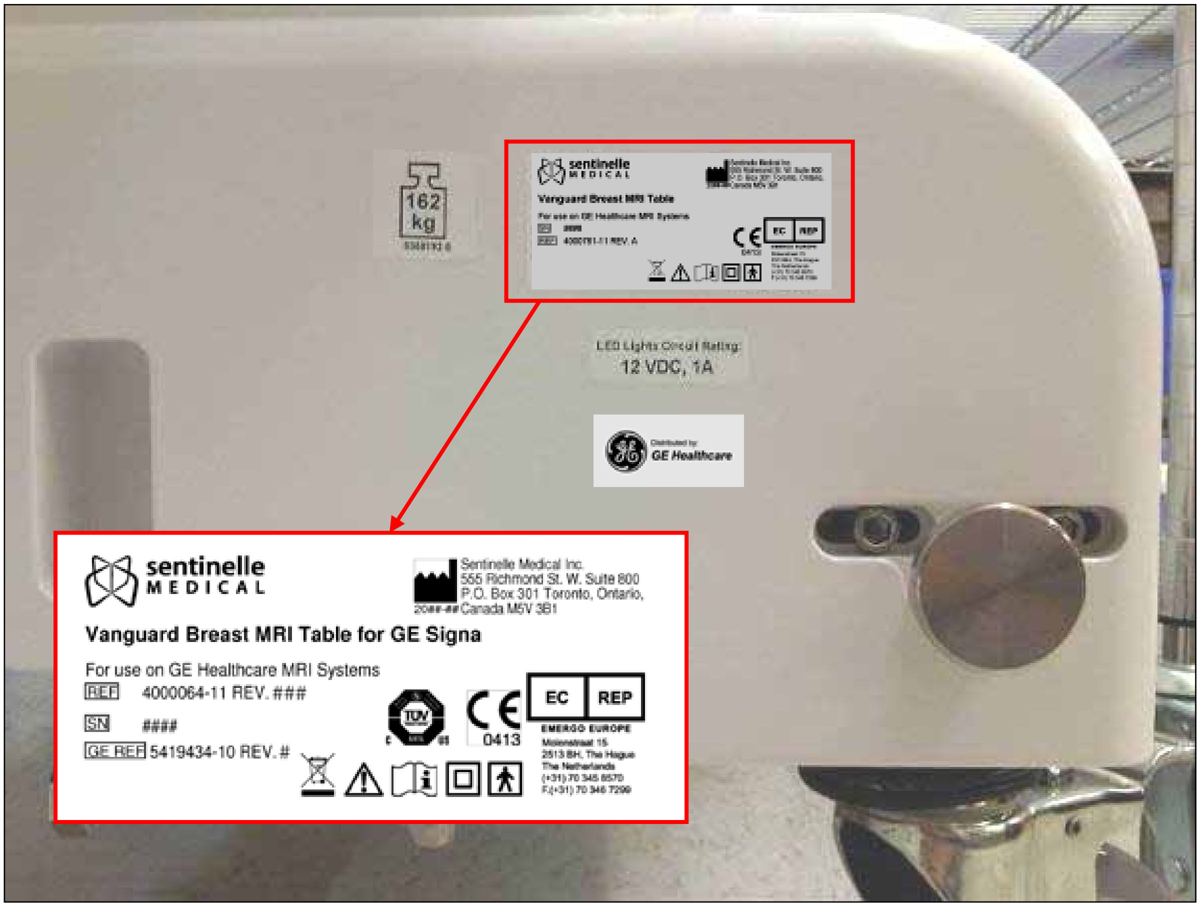


Figure 6: Stretcher Bumper Label

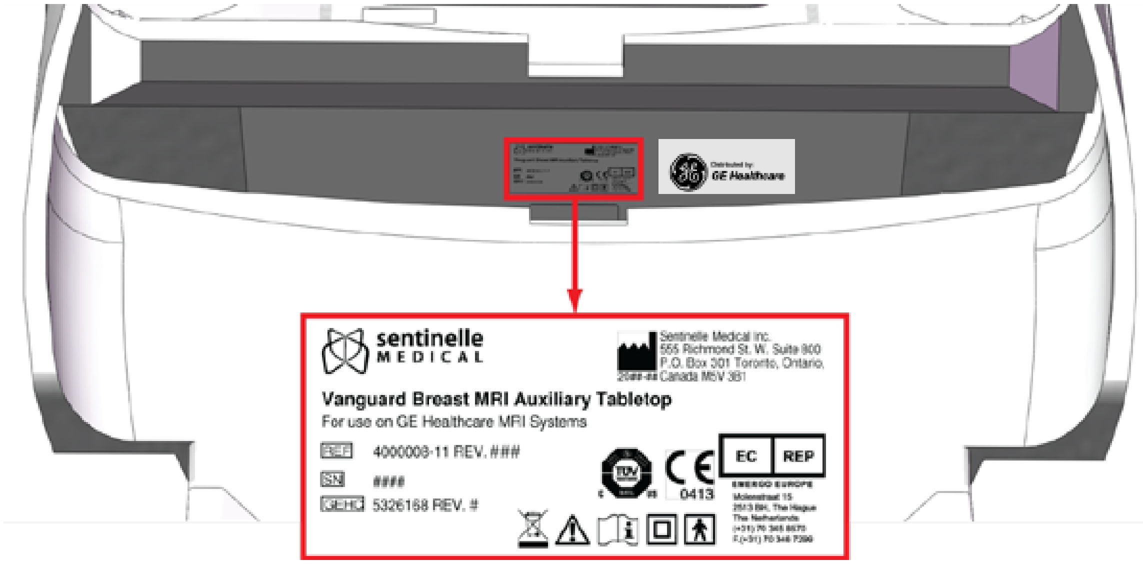


Figure 7: Stretcher Patient Support Label

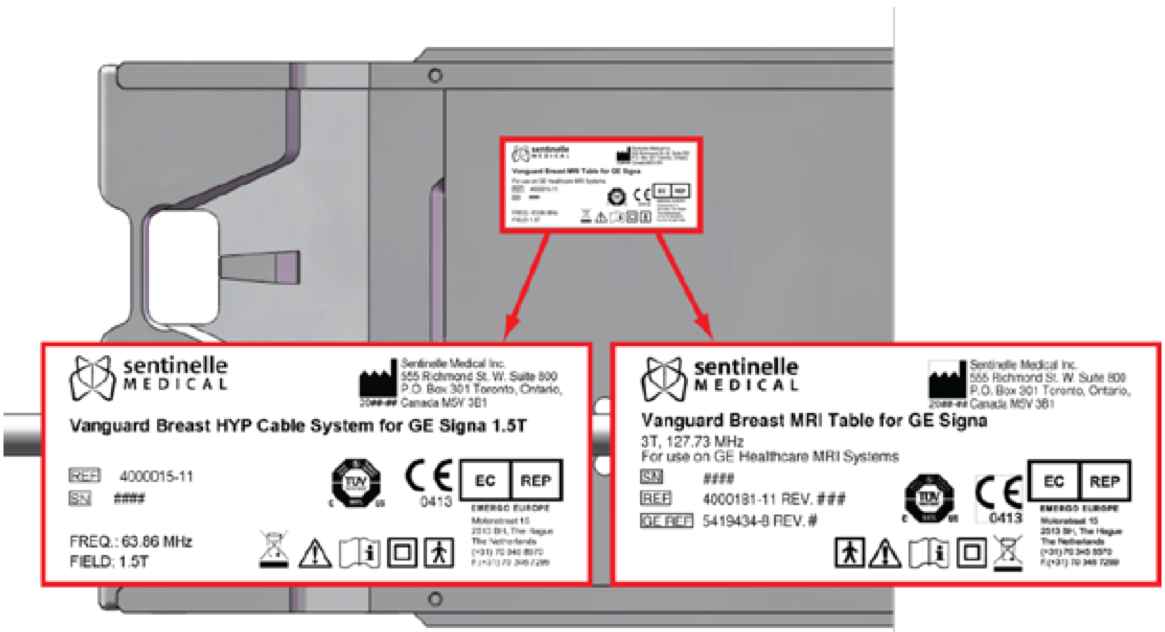


Figure 8: Cable Tray Labels – 1.5T (Left) and 3T (Right)

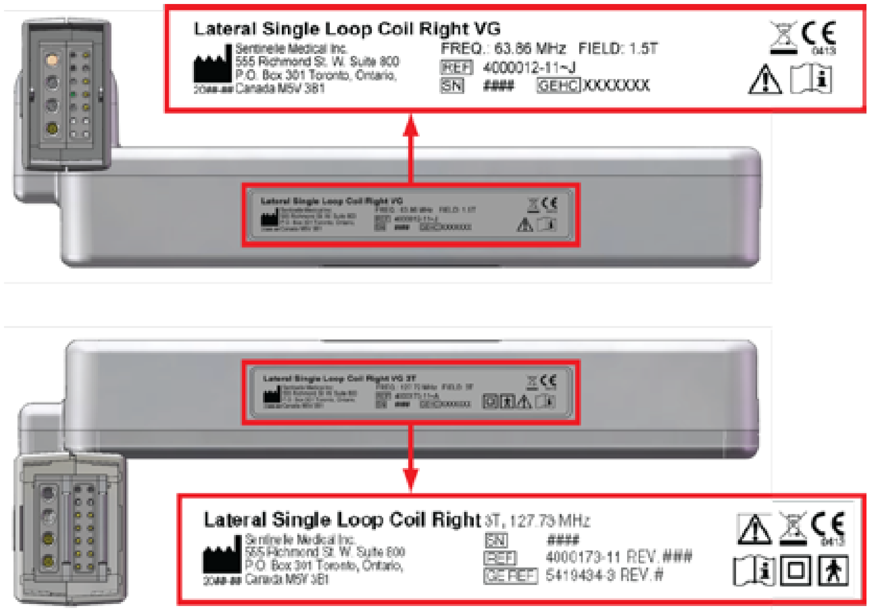


Figure 9: Lateral Single Loop Coil Right Labels - 1.5T (Top) and 3T (Bottom)

Figure 10: Lateral Single Loop Coil Left Labels - 1.5T (Top) and 3T (Bottom)

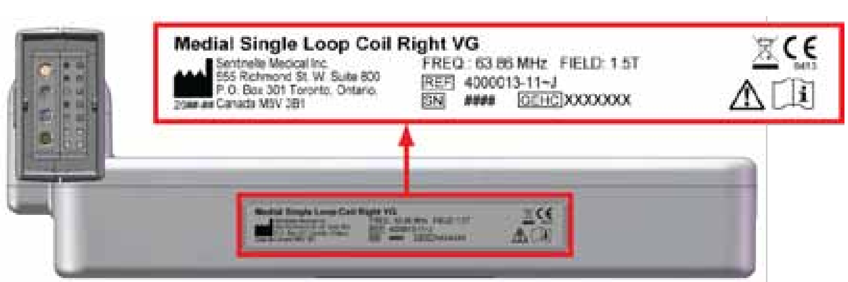


Figure 11: Medial Single Loop Coil Right Label - 1.5T Only

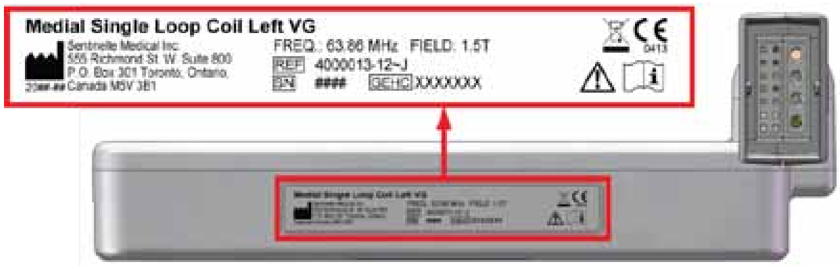


Figure 12: Medial Single Loop Coil Left Label - 1.5T Only

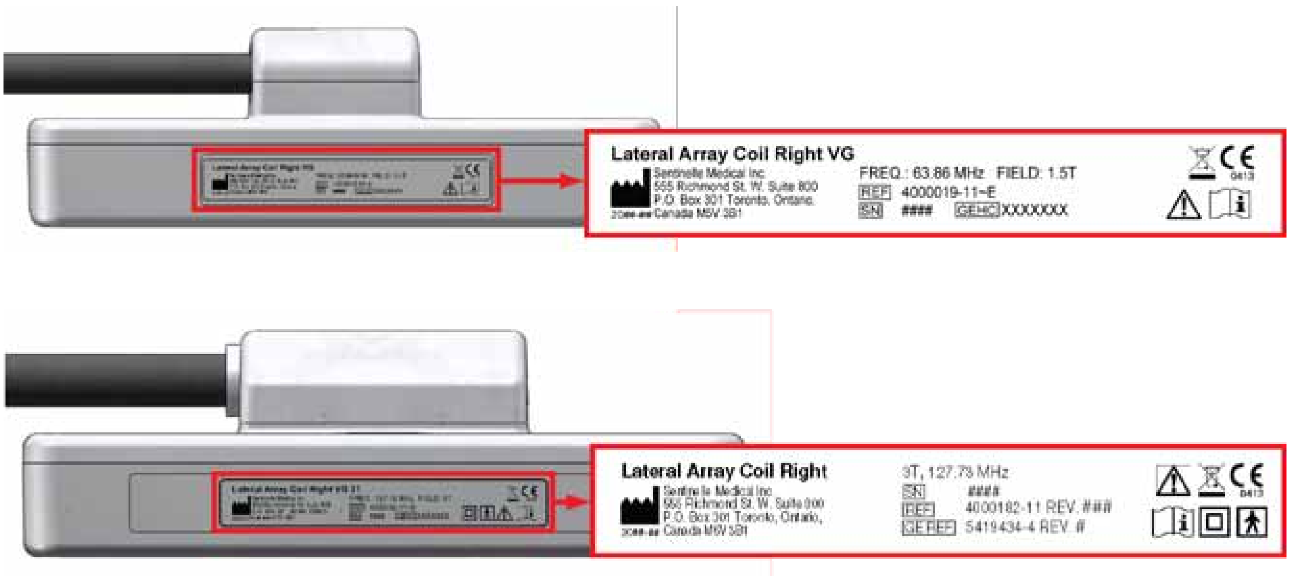


Figure 13: Lateral Array Coil Right Labels - 1.5T (Top) and 3T (Bottom)

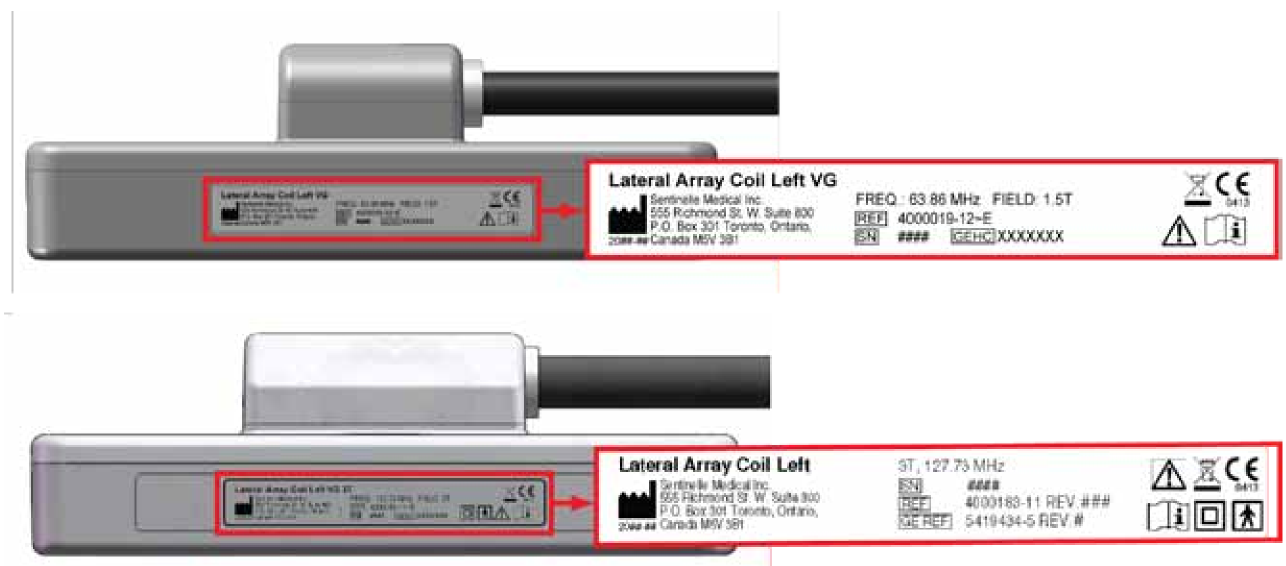


Figure 14: Lateral Array Coil Left Labels - 1.5T (Top) and 3T (Bottom)

Figure 15: Medial Terminator Plug Label - 1.5T (Left) and 3T (Right)

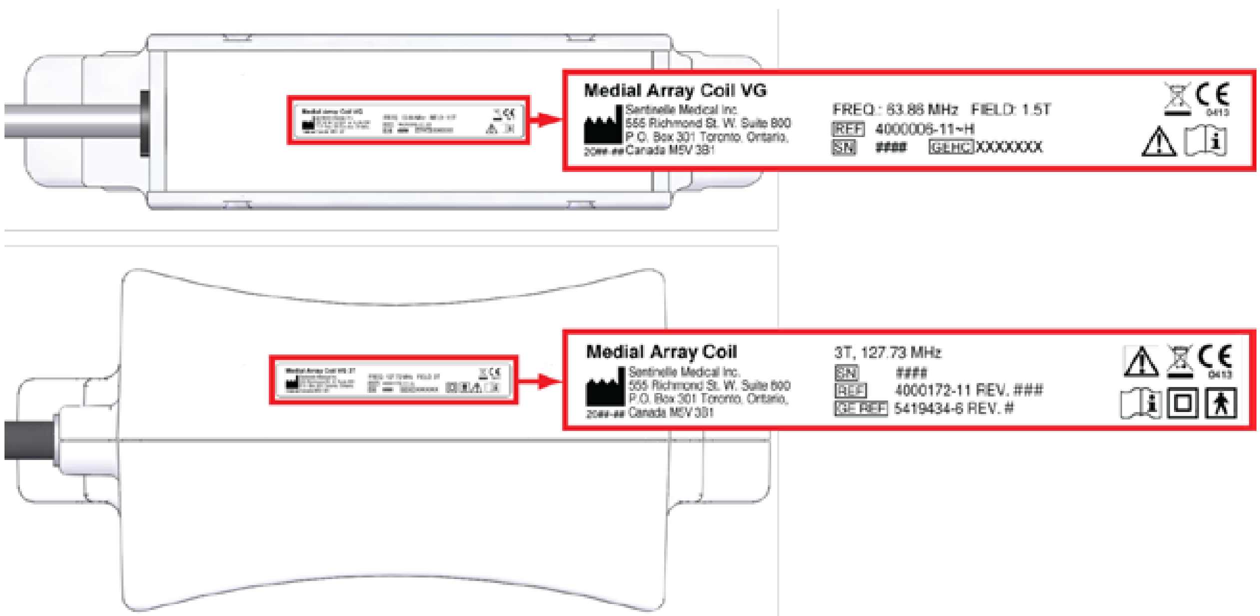


Figure 16: Medial Array Coil Label - 1.5T (Top) and 3T (Bottom)

7 Installation Procedure

Time required to complete the installation procedure: 3 - 4 hours.
Personnel required to complete the installation procedure: 2 people.

Installation of the Vanguard requires performing a series of mechanical adjustments and tests of the RF system. The installation procedure will have to be repeated if the Vanguard is to be used with a different scanner.

Note: The Vanguard is shipped with the screws on the docking mechanism loose. Be sure to tighten these screws after checking the alignment and performing any adjustments.

7.1 Initial Alignment

The initial alignment sets up the Vanguard for install and assesses the Vanguard for height adjustment. The Vanguard table has been pre-set at time of manufacture to the correct approximate height for the Signa scanner. Coarse height adjustment will always be required when installing to a Signa scanner. No attempt to dock the Vanguard should be made at this time. The patient support is heavy and fragile. Exercise caution when moving the patient support and have another person assist in lifting if possible.

1. Undock the GE Signa table and move it out of work area.
2. Remove the patient support from the stretcher and place in a secure location by following the method in steps 3 to 9.

HEAVY ITEM

⚠WARNING ***Use proper heavy lifting techniques and be careful when lifting the patient support. Ensure that two people are lifting, because the patient support weighs approximately 55lbs.***

3. Begin by ensuring that there is enough space to remove the patient support from the foot end of the stretcher. There must be about 4ft or 1.2m free at the foot end of the stretcher to allow for the patient support to be removed. If there is not enough space, unlock the caster wheels (if locked) and move the stretcher to a position that meets this requirement.
4. One person should be at the head of the stretcher and the second person should be at the foot end.
5. The person at the head end should come around the side of the stretcher, close the bridge, and return to the head end. This same person should press and hold the dock pedal.
6. The person at the foot end should press and hold the LPCA interlock plunger and begin pulling the patient support towards the foot end. Meanwhile, the person at the head end should assist in pushing the patient support towards the foot end.
7. Once the end of the patient support has cleared the foot end of the stretcher, the person at the head end should release the dock pedal and move to the side of the stretcher, while keeping a firm grip on the patient support. As the patient support is moving away from the end of the stretcher, the person at the foot end can release the LPCA interlock plunger.
8. Once the patient support has been pushed past the foot end of the stretcher, both people should lift the patient support clear of the stretcher and place it on any horizontal surface, such as a counter top or the Signa table, taking extra care not to damage any of the patient support hardware or the table. Use packing material or linens to protect the patient support, Signa table or other items it may be in contact with. Be careful not to pinch fingers when removing or replacing the patient support.
9. Remove the drawers by sliding them completely open pressing the two plastic tabs on either side, upward on the left and downward on the right, and pulling the drawers out of the rear support assembly (Figure 17). Place them away from the work area. The drawer sliders have a light oil coating which may get on hands, and if so, should be washed off to avoid fingerprints on the system.

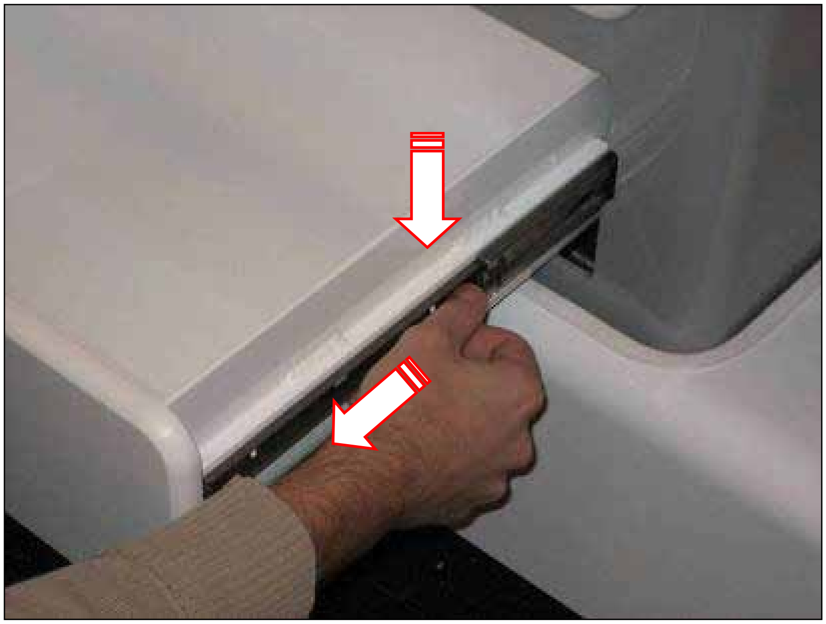


Figure 17

10. Check to ensure the LPCA sensor actuator in the middle of the bumper is screwed in as far as possible turning it in a clockwise direction by hand (Figure 18).

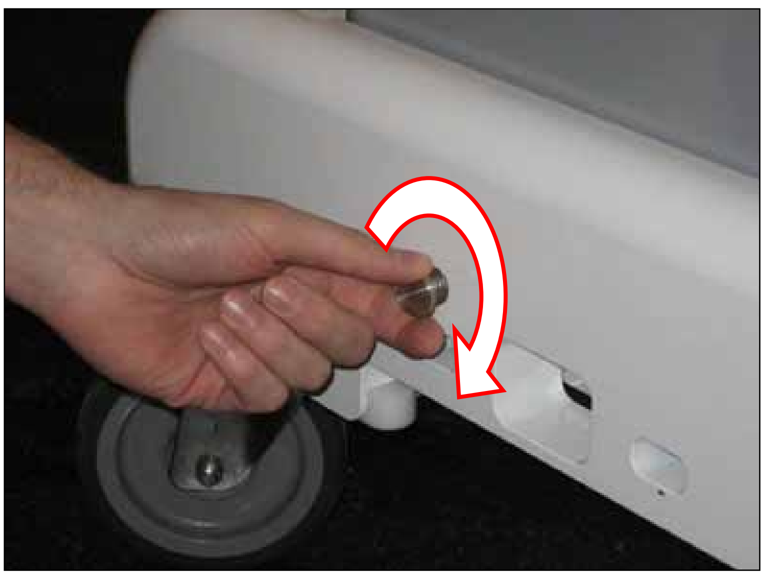


Figure 18

11. Use your finger to depress the MRI's docking pins (circled in Figure 19) until a slight click is heard and hold it while pressing the scanner control button to advance the LPCA into the bore of the scanner until it stops.

PINCH HAZARD

CAUTION

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

12. Align the Vanguard to the scanner carefully by rolling it up to the centre of the scanner, ensuring that the bumper does not contact the static pin protruding from the dock (Figure 19). The static pin should be aligned with the corresponding clearance pocket in the bumper. Measure the height difference of the

stretcher LBS and the scanner bridge (Figure 20) with an MRI compatible ruler and straightedge or other available means. Do not use surfaces other than the ones specified.

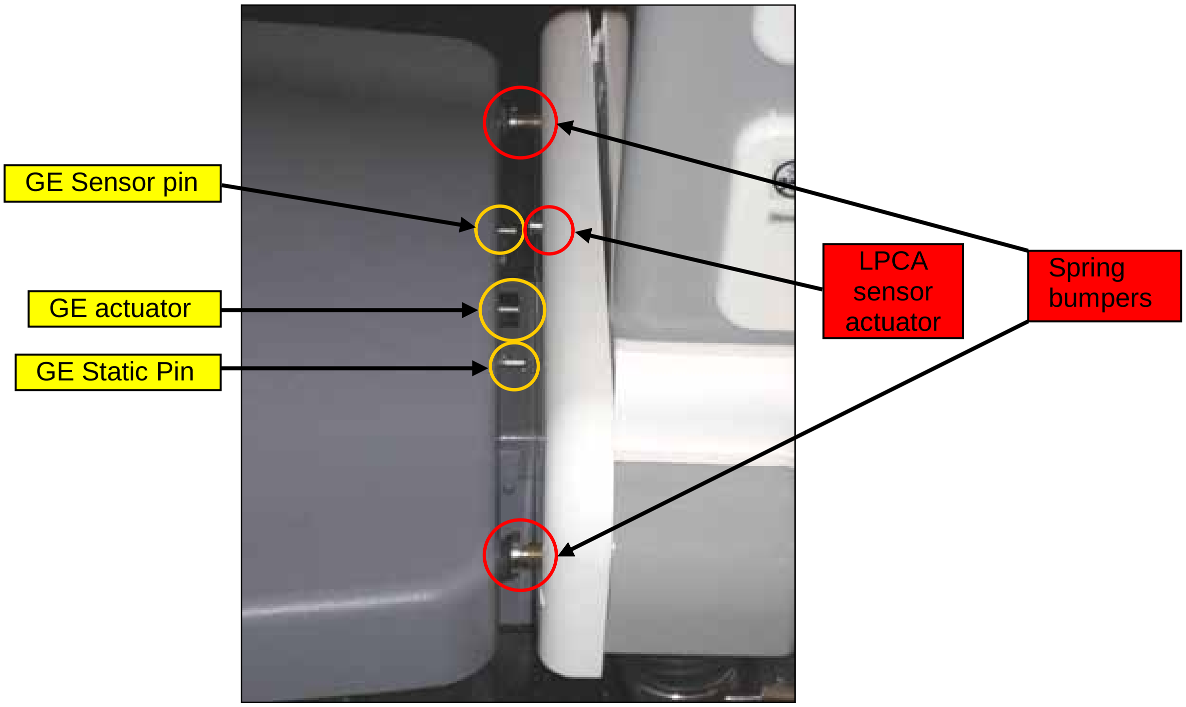


Figure 19

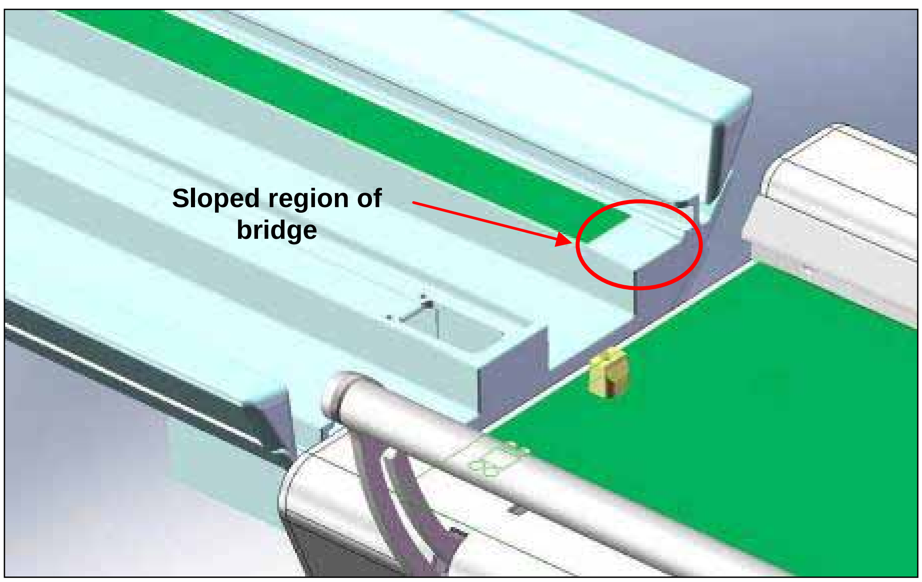


Figure 20

13. If the height difference between the LBS and the scanner bridge (highlighted green for identification) is greater than ½” (12.7mm), perform coarse height adjustment (Section 7.2), followed by fine height adjustment (Section 7.3). If the height difference is less than ½” (12.7mm), skip the coarse height adjustment and proceed to fine height adjustment.

7.2 Coarse Height Adjustment

Coarse height adjustment is performed when the difference between the scanner bridge and surface of the LBS on which the patient support rolls is greater than ½" (12.7mm).

1. Loosen the cable which actuates the no-undock hook by loosening the round head screw which clamps the cable to the hook with a 5/32" Allen key. Leave this screw loose until Section 7.12.

PINCH HAZARD

CAUTION

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

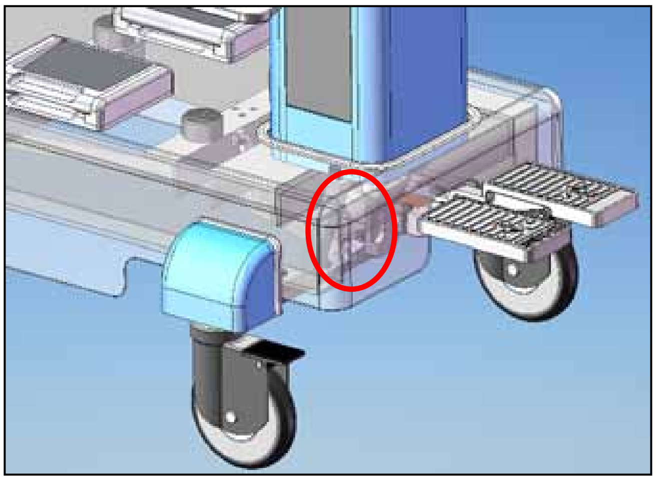


Figure 21

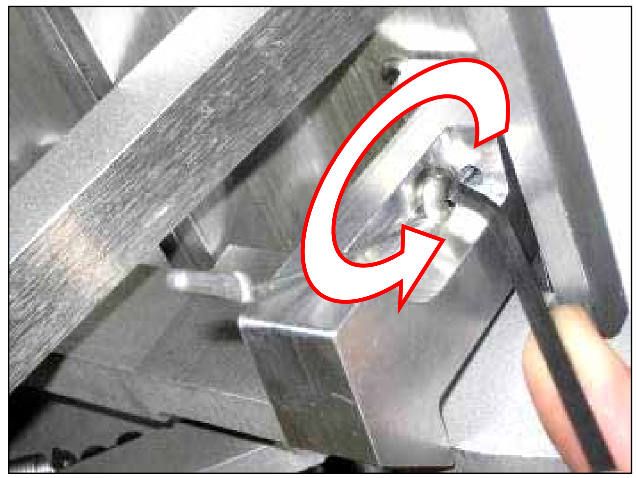


Figure 22

2. Loosen the cable which actuates the latch tab by loosening with a 3/16" Allen key the socket head cap screw which fastens the cable clamp to the swing assembly (Figure 24).

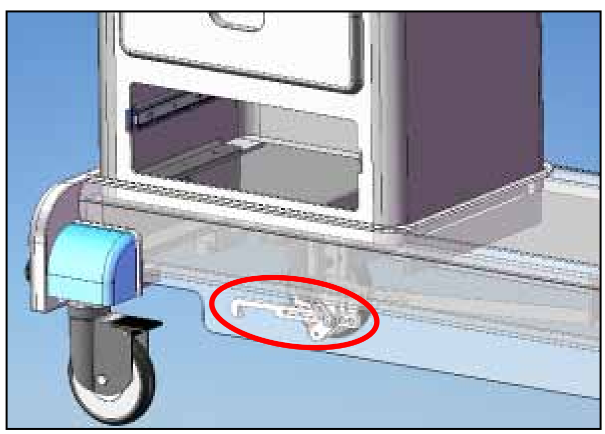


Figure 23

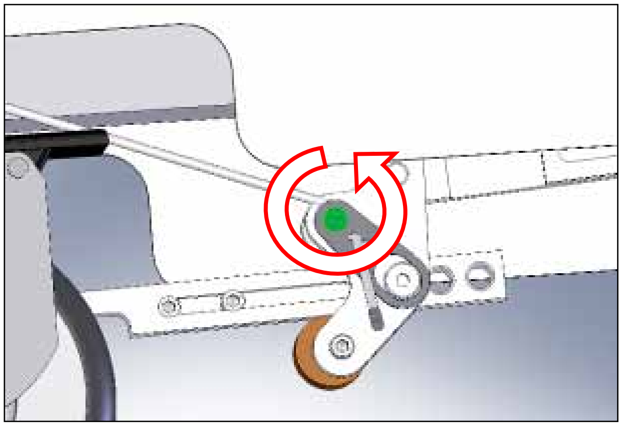


Figure 24

3. Using a 1/8” Allen Key remove the LBS skirt by unscrewing the two external mounting round head screws and loosening three more screws (Figure 25). Slide the skirt down to the bottom of the RSA and remove it by bending it slightly open.

Figure 25

4. Remove the two half skirts from the FSA of the stretcher by unscrewing the four round head screws on each side using a 5/32” Allen key (Figure 26). To gain access to these screws, the tray tables can be extended to their outermost position. These tray tables may also be removed by unscrewing and removing the screws at the end of their sliding arms using a 3/16” Allen key (Figure 26). Since each tray table has two sliding arms, a screw must be removed from each sliding arm before the table can be pulled out.

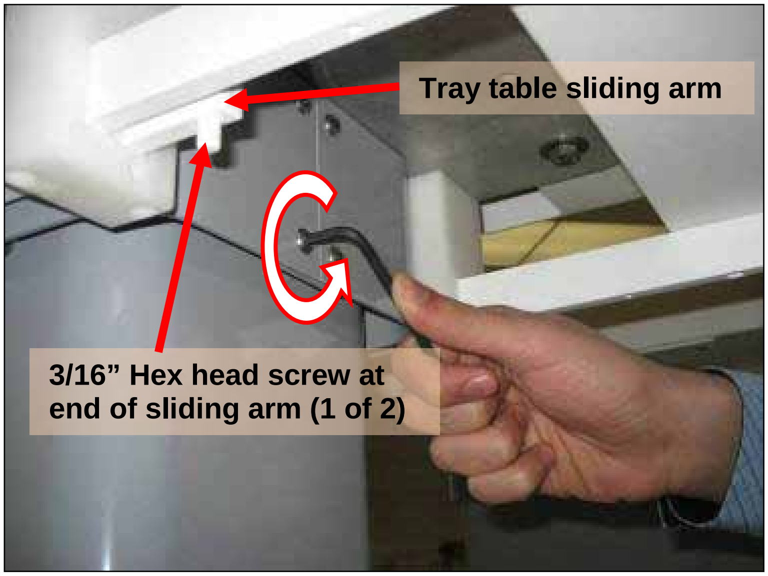


Figure 26

5. Open the battery compartment access cover on the FSA by unscrewing the thumb fastener (Figure 27) and then pivoting the assembly into its open position (Figure 28).

Figure 27

Figure 28

6. Take note of the position of the 4 bolts in the series of vertical holes inside the FSA and RSA. A marker may be used if necessary (Figure 29).

Figure 29

7. Inside the RSA, completely loosen four nuts and lock washers using a 1/2" wrench. From the outside of the RSA remove two of the four carriage bolts which are nearest to the bumper while supporting the weight of the LBS (Figure 30). The vertical distance between the holes is 1/2" (12.7mm). For safety reasons, always heed the warning below and always move only one hole level at a time.

CAUTION

Do not remove all four of the carriage bolts at the same time as the LBS will be unsupported and could cause injury or damage.

The following steps (9, 10) detail, and will only permit, height adjustment by one increment at a time. In the rare case when more than one increment is required, repeat each step as required.

Figure 30

PINCH HAZARD

CAUTION

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

8. Adjust the LBS height by tilting the LBS up or down while positioning one of the carriage bolts in its mounting hole so it will slide in as the adjustment holes lines up. Insert the other bolt ensuring the square base of the shank is correctly oriented to the square hole in the RSA. Repeat the process on the other side of the LBS and fasten with the four sets of nuts and lock washers.

9. Adjust the UBS while facing the open battery access door in the side of the FSA. Remove the 2 closest and one of the farther bolts while supporting the UBS. Once the bolts are removed this will cause the UBS to tilt down from the point where the carriage bolt remains. Tilt the UBS up or down one adjustment increment in both axes (diagonally) while positioning the opposite carriage bolt in its mounting hole so it will slide in as the adjustment holes line up. Remove the bolt which remained at the initial height and level the UBS in the same manner. Finally, push the remaining carriage bolts into place when the correct height is set, re-install the 4 sets of nuts and lock washers.

10. Upon completion of coarse height adjustment, verify the height adjustment with the scanner and then replace the skirts on the UBS and LBS.

7.3 Fine Height Adjustment

Fine height adjustment is performed when height adjustment of less than 1/2" (12.7mm) is required or after coarse height adjustment has been performed. This adjustment is essentially the “ride height” determined from adjusting the casters, which are mounted to the head of a large bolt which is then screwed into a plate in the stretcher frame.

1. Align the stretcher to the scanner carefully, ensuring that the bumper does not contact the fixed static pin or the LPCA switch on the dock, and take note of the height difference.

2. Remove the four wheel covers from the stretcher by loosening the two bottom-mounted round head screws several turns using a 5/32” Allen key and sliding the covers off (Figure 31).

Figure 31

3. Loosen each of the four large nuts so they are flush with the end of the bolt, using the 1.5” MRI compatible wrench (Figure 32).

Figure 32

4. Compress the black rubber shroud, place the 1.5” MRI compatible wrench on the exposed height adjustment bolt head and turn clockwise (as seen from above) to raise the stretcher or counter clockwise to lower the stretcher. Ensure that all four hex bolts are initially adjusted by the same number of turns. **Note that a 360 degree turn = 1/8” (3.17mm) of height adjustment**. This can also be accomplished more easily by locking the caster, raising the edge of the stretcher and rotating the caster with one foot.

5. Sighting along the length of the flat surface of the stretcher and the scanner’s bridge (Figure 20), adjust the casters at the head or foot end to correct for any pitch or roll angle of the stretcher.

6. Next, use a flat straight edge to measure the co-planarity of the stretcher LBS surface and scanner bridge at the surfaces on which the wheels roll (see green surfaces in Figure 33). First, lay the straight edge on the scanner bridge, and attempt to slide the tool out over the stretcher. If it clears the stretcher, measure the gap. Repeat this process on both sides (left and right) of the bridge. If the tool will not slide over the stretcher, then lie the straightedge on the stretcher’s Lower Body Support (LBS) and attempt to slide it out over the scanner’s bridge. Ensure that the tool slides freely past the ramped section of the bridge, and continues to slide freely at least 6” (152.4mm) into the scanner. Measure the gap under the tool in both left and right wheel tracks.

7. Adjust the caster height by the measured amount from the higher surface. The lower body support must be level or no more than 3/64” (1mm) lower than the scanner’s bridge, disregarding the sloped regions of the bridge (Figure 33).

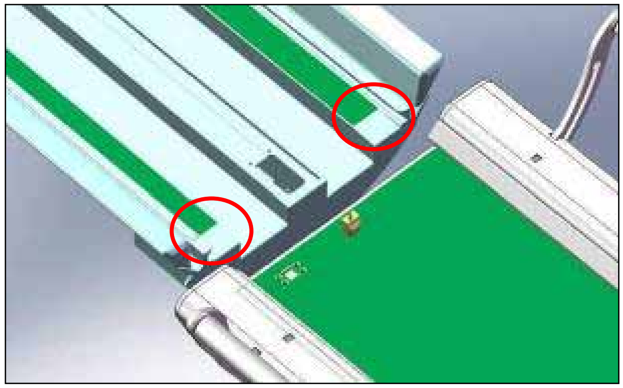


Figure 33

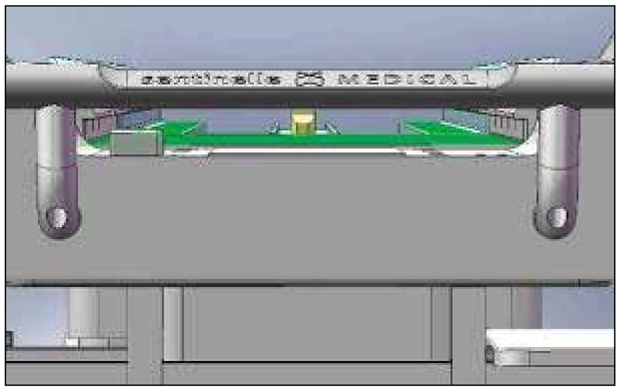


Figure 34

8. Once fine height adjustment has been performed, re-check the pitch of the stretcher relative to the scanner bridge by sighting along the stretcher into the bore from the head end (Figure 34). Perform fine adjustment as required while continually checking for co-planarity. A straight edge can be additionally used.

CRITICAL ITEM

Note: Adjusting the pitch with the head end casters will affect the height of the edge of the LBS foot end relative to the scanner, as the pitch angle will rotate about the casters at the foot end of the stretcher. The fine height adjustment must be verified again within 1mm, and re-adjusted if necessary. Accurate fine height adjustment is critical to smooth and reliable operation.

9. Scanner room floors are not always flat. Verify that all the stretcher wheels are in contact with the floor when the stretcher is in the docked position. This can be checked by trying to swivel the casters with the stretcher's weight on them to determine if it is evenly distributed. Adjust any non-contacting casters to ensure all four wheels make contact. Sometimes this will result in "wobble" of the stretcher when not in the dock position to the scanner, which is unavoidable.

10. Tighten each of the large nuts very tightly using the 1.5" non-magnetic wrench. Ensure the fine height adjustment bolts do not spin when the nuts are tightened. It is necessary to brace against the stretcher and use both hands for very high torque approximately 1/8 turn past tight.

11. Replace the wheel covers on the stretcher and tighten the round head screws using a 5/32" Allen key, ensuring the covers fit tightly to the shroud.

7.4 Docking Hook Height Adjustment

Docking hook height is set from the factory but is affected when fine height adjustment of the stretcher is made and may not correspond to the opening of the scanner dock.

1. Place the stretcher close to the dock so the stretcher docking hook enters the dock.
2. Loosening the socket head cap screw with a 3/16” Allen key and adjust the position of the round eccentrically mounted hook stop so the docking hook is at least ¼” (6.35mm) above the scanner’s bronze dock hook and is not contacting the inside top surface of the dock. (Figure 35).

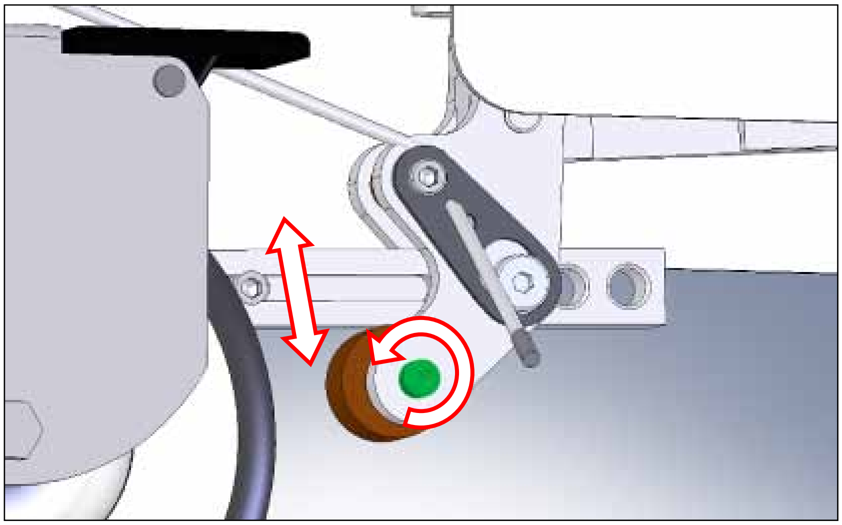


Figure 35

PINCH HAZARD

CAUTION

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

7.5 Bumper Adjustment

The guide rails on the top of the stretcher and on the scanner bridge must be collinear in order for the patient support to advance into the scanner. Simultaneously, the stretcher bumper must be centered vertically and horizontally with respect to the Dock for the Vanguard to function properly.

1. Loosen the two ¼"-20NC bolts on the back face of the bumper which lock the LR adjustment bumpers in place with a ¼" Allen key (Figure 36).

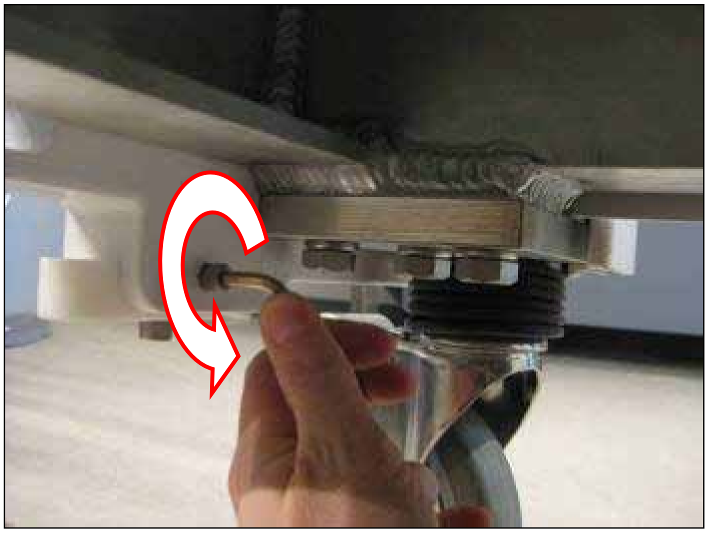


Figure 36

2. Loosen the two left-right fine adjustment bumpers which are located in slots on the bottom of the bumper with a ¼" Allen key and slide them into the bumper as far as possible (Figure 37).

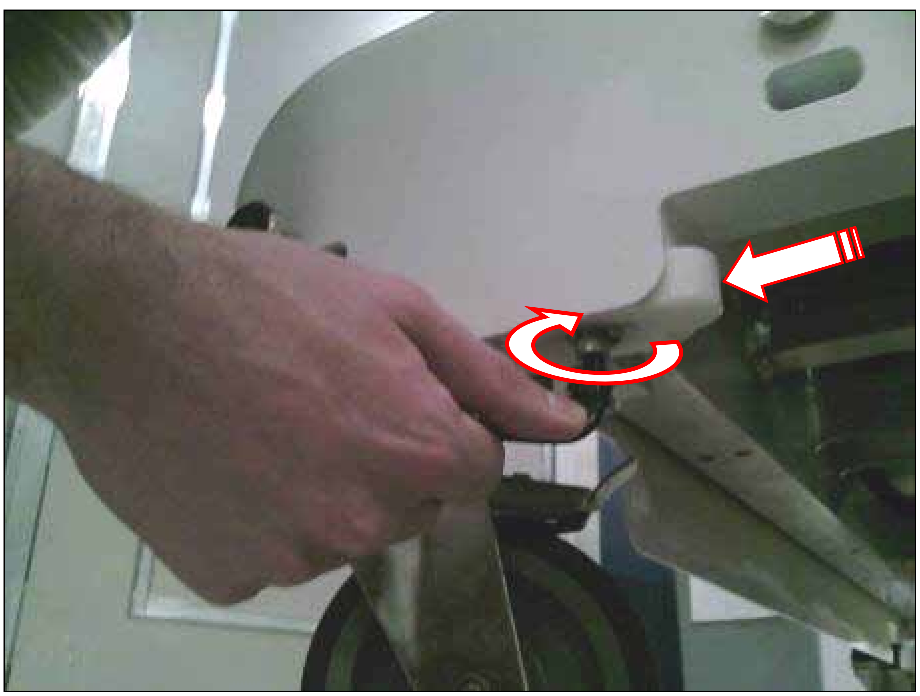


Figure 37

3. Move stretcher forward to the dock position against the scanner and then backward so the bumper is roughly 2” (50.8mm) from contacting the scanner dock to permit an Allen wrench to be used to adjust the bumper. Kneel at the head end of the stretcher and sight along the guide rails and position the stretcher so the guide rails are collinear (Figure 38). Lock all of the casters to ensure stretcher position is not affected after alignment has been achieved.

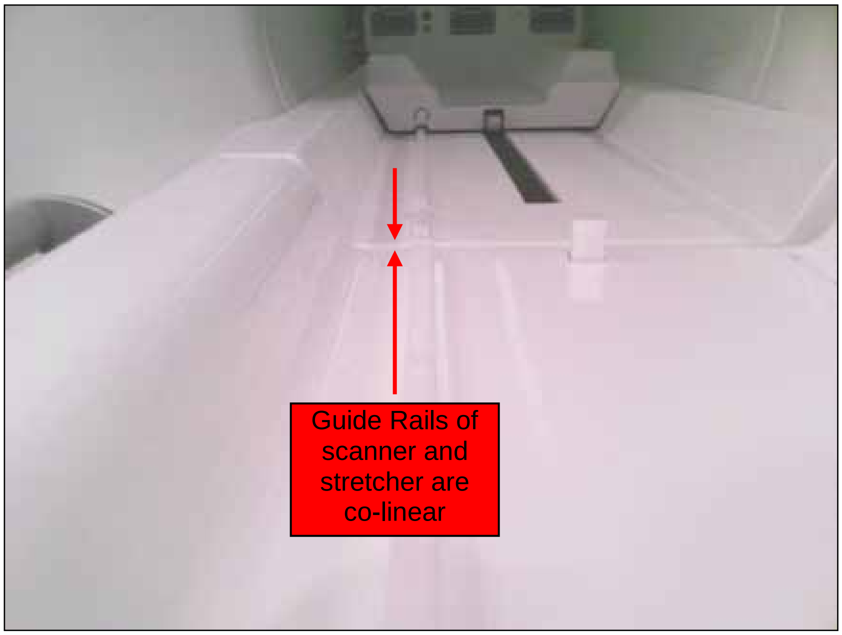


Figure 38

4. The bumper spring plungers of the stretcher bumper must now be centered to the corresponding rubber features of the dock. Take note of the amount that the bumper must be moved in order to centre it. Loosen the four socket head cap screws with a ¼” Allen key (Figure 39).

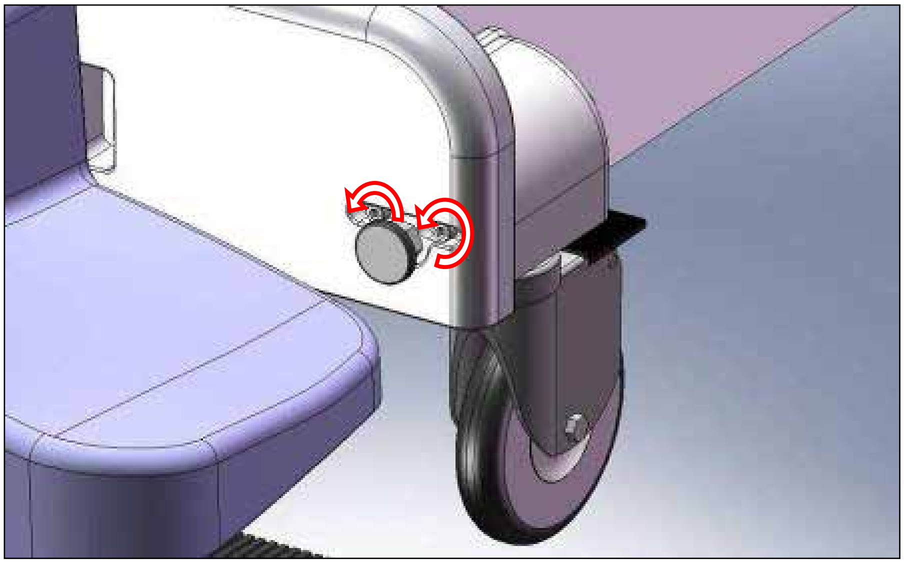


Figure 39

5. Centre the bumper horizontally while looking from above (Figure 40), then centre one side vertically while looking from the side (Figure 41). Tighten the outermost screw so the bumper will rotate about this point and then centre the other side vertically. After the bumper is centered, firmly tighten the four socket head cap screws to secure the bumper plate to the stretcher using a ¼” Allen key, supporting it to ensure it

doesn’t shift while being tightened.

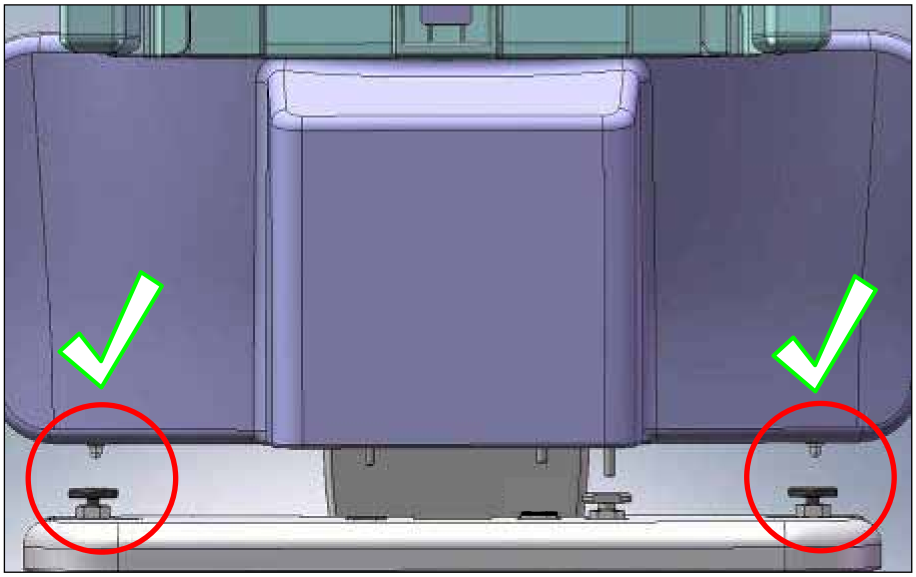


Figure 40

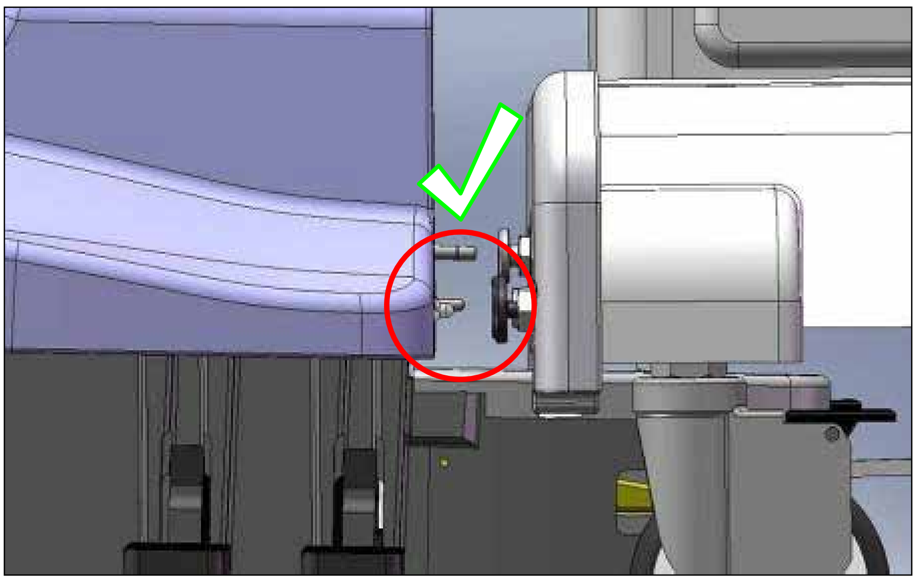


Figure 41

7.6 Docking Force Adjustment

Docking force is critical to function of the Vanguard. Setting the docking force requires incremental adjustments of the docking hook until adequate docking force is achieved. If the docking force is too high it will result in premature wear or damage to the Vanguard docking mechanism. Too little docking force will not be sufficient to keep the stretcher immobilized during use and can cause it to undock unexpectedly.

1. Loosen the two socket head cap screws on the docking hook assembly (Figure 42) using a 3/16” Allen key and adjust the hook by sliding it to the shortest position. Tighten one of the screws by hand.

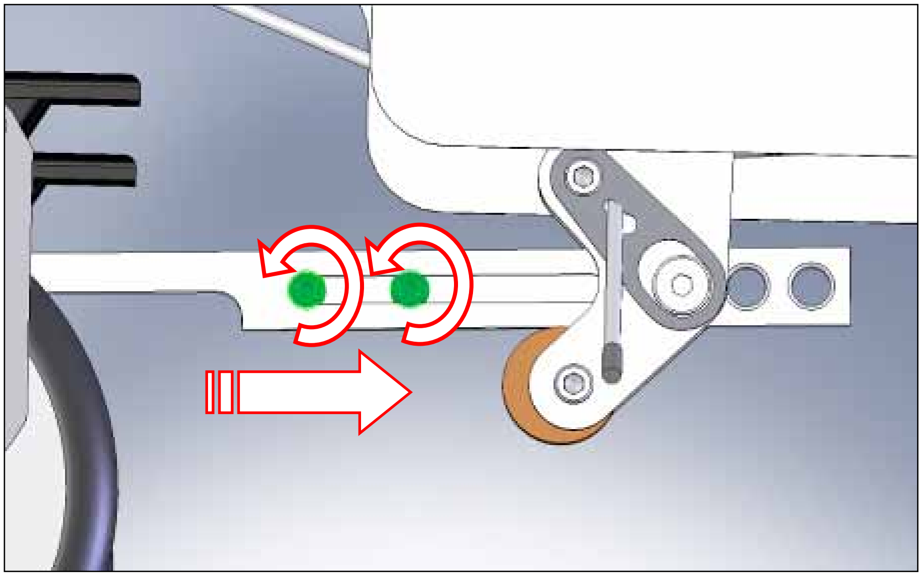


Figure 42

PINCH HAZARD

CAUTION

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

2. While applying constant force toward the scanner from the UBS handle of the stretcher to compress the bumper spring plungers, depress the docking pedal and then release it slowly to prevent any jarring motion to the docking hook, then release force from the UBS handle. This sets the hook to the point where it just contacts the bronze hook in the dock while the dock pedal is completely depressed.

3. Adjust the hook by sliding it approximately 1/16" (1.59mm) away from the dock, and tighten one of the two socket head cap screws using a 3/16" Allen key.

4. Take note of the docking pedal travel with the stretcher away from the scanner. Place the stretcher against the dock and begin depressing the docking pedal slowly and take note of the peak force required to dock the Vanguard, which will occur in the last quarter of pedal travel. If the force required to dock will require more than the weight the foot and leg being placed on the pedal, stop at once and do not press any further—loosen and extend the docking hook approximately 1/32" (0.79mm) and retighten.

CRITICAL ITEM

***Note: The docking pedal force must be between 18 and 22 lbs, *which is provided by the weight of an average foot and leg on the pedal and only very light muscular force.* If the docking force is set too high, requiring force to be applied to pedal, this will increase wear and can damage docking mechanism.**

5. Test the docking force adjustment while the stretcher is docked by attempting to push the stretcher sharply towards the scanner with considerable force. If it comes undocked increase the force slightly by loosening the bolts, retracting the hook a further 1/32" (0.79mm) and repeat step #4.

6. When the stretcher is docked, verify that the bumper spring plungers remain fully compressed even when pushing on the sides of the stretcher. If the stretcher comes undocked or the bumper spring plungers are free to extend slightly, adjust the docking hook tighter in 1/32" (0.79mm) increments while repeating steps 4 – 6.

7. When adequate docking force is achieved, firmly tighten the two socket head cap screws on the docking hook assembly using a 3/16" Allen key.

7.7 Bumper Spring Plungers and Straightness (Yaw) Adjustment

The bumper springs can be adjusted to correct for any slight misalignment between the guide surfaces of the stretcher and scanner, or to finely adjust docking force.

1. Align the guide rails (as previously completed in *Section 7.5*) and gently dock the stretcher to the scanner. While doing so, gauge by eye or use a long straight edge to assess the straightness or "yaw angle" of the stretcher and scanner guide rails.

2. Undock the stretcher and ensure the guide rails are collinear (Figure 43) by turning the threaded brass body of one bumper spring plunger. For every turn or fractional turn, apply the *opposite* number of turns to the other bumper spring plunger (Fig 7.7.2). Applying the opposite adjustment on each side ensures no increase or decrease of docking force. Dock and undock the stretcher to check the effectiveness of the alignment adjustment.

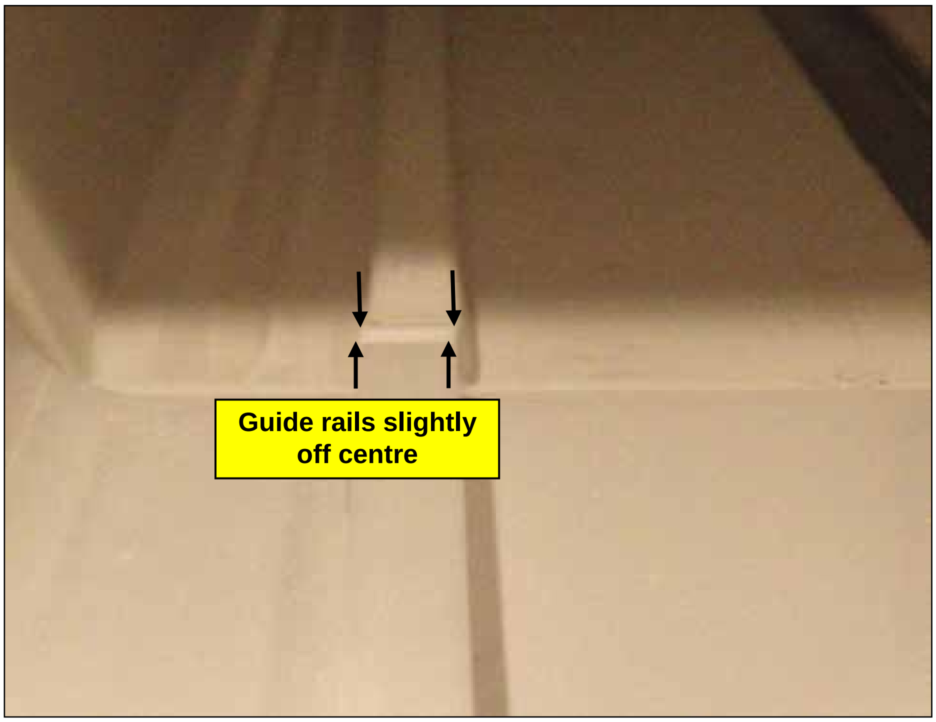


Figure 43

Figure 44

3. Once the stretcher is as straight as possible, double check that the docking force is not too high, and that the bumper spring plungers remain fully compressed even when the stretcher is pushed towards the scanner or side-loaded (see section 7.6)

4. Loosen the white plastic nut on the centre plunger. With the stretcher docked, adjust the height of the center plunger. It is possible to overextend the bumper's center plunger and thereby damage the scanner's dock. It is also possible to depress the switch too little, which causes erratic or no operation. The ideal adjustment is to depress the switch halfway between where it activates (clicks) and where it bottoms out. Extend the center plunger by unscrewing it by hand until it just depresses the sensor pin on the dock - you will hear a faint click -then adjust it out a further 1/8" to 3/16" (3.17mm - 4.76mm). One complete turn advances the plunger 1/16" (1.59mm) outward.

7.8 Final Centering Adjustment

Centering the Vanguard is an adjustment which requires attention to detail and must be executed with precision. This adjustment will ensure that the patient support will transit to the scanner's bridge smoothly.

1. Dock the stretcher by slowly depressing the docking pedal while carefully aligning the guide rails of the stretcher and the Scanner's bridge using a straightedge. It may take more than one attempt to get the guide rails perfectly aligned.

2. Slide the two left-right fine adjustment bumpers (located on the bottom edge of the bumper plate) so they firmly contact the scanner's dock (Figure 45). Fasten them firmly in place by tightening the socket head cap screws with a ¼" Allen key while pressing them against the dock with your other hand.

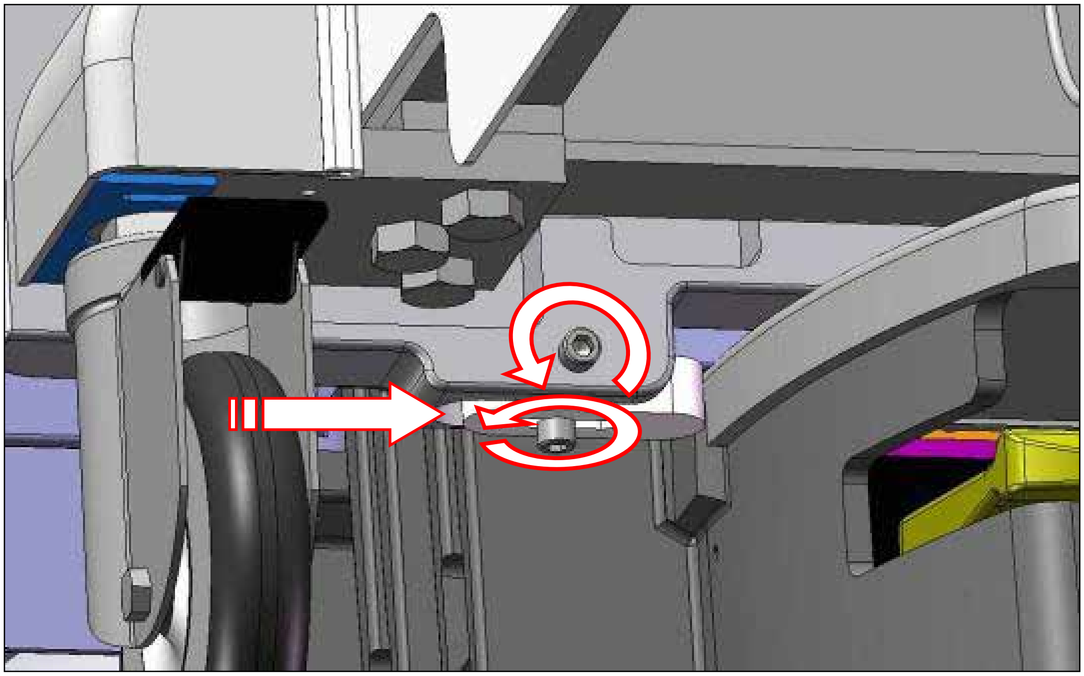


Figure 45

3. Undock, reposition the stretcher and dock the stretcher again to ensure that it is centered. Re-adjust the left-right bumpers if required and check again. When the stretcher is pressed up against the scanner but undocked, there should be no more than 1/16” (1.59mm) of L-R play possible by applying lateral force to the stretcher near the foot end. Once the stretcher is centered with bumpers firmly fastened, undock the stretcher and firmly tighten the two socket head cap screws on the back side of the bumper plate with a 3/16” Allen key.

Note: Subtle left-right misalignment is sufficient to cause serious problems and will prevent long-term smooth and reliable operation.

CRITICAL ITEM

7.9 Stretcher to Scanner Gap Adjustment

The gap adjustment is required to eliminate a possible pinch point between the Vanguard LBS and the MRI’s bridge. This adjustment involves moving the LBS in the head foot direction, and later (Section 7.11) moving the UBS using the patient support as a gauge.

1. Dock the stretcher to the scanner and measure the gap between the stretcher LBS and the scanner with a MRI compatible ruler. This gap must be 1” (+ 0.5/ -0) or 25.4mm (+12.7/ -0) when the system is docked (Figure 46).

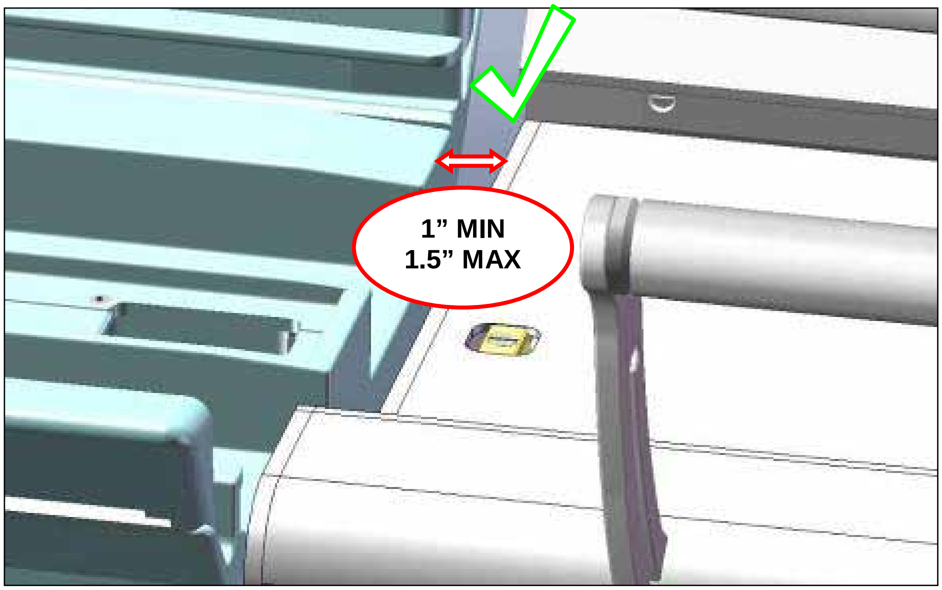


Figure 46

2. If adjustment is required and coarse adjustment was not previously performed, loosen the cable which actuates the latch tab by loosening with a 1/8" Allen key the socket head cap screw which fastens the cable clamp to the swing assembly (Figure 23). Loosen the cable which actuates the no-undock hook by loosening the round head screw which clamps the cable to the hook with a 5/32" Allen key. Leave this screw loose until Section 7.12.

3. Move the LBS by loosening with a 9/16" wrench or ratchet the four hex head bolts which are located inside the LBS. Access these bolts by reaching through the drawer openings in the RSA and upward to locate the bolt heads on the top of the RSA plate on which the LBS is mounted (Figure 47). Set the gap distance of 1" to 1.5" (25.4mm to 38.1mm) (Figure 46). When complete, tighten the four bolts firmly to fasten the LBS in place.

Figure 47

7.10 Latch Tab Height Adjustment

The latch tab keeps the patient support in place when the Vanguard is not docked and like the GE Signa table, unlatches the LPCA from the patient support when the Vanguard is undocked.

1. Dock the Vanguard to the scanner.

2. Using pliers, pull the cable far enough to ensure the latch tab is flush or below the LBS surface (Figure 49). While continuing to pull to maintain maximum tension in the cable, clamp the cable in place by tightening the cable clamp screw with a 5/32" Allen key (Figure 48).

3. Undock and dock the table and take note of the latch tab height. If latch tab still protrudes, undock the stretcher loosen the cable clamp and take up just enough cable to bring the latch tab flush when docked. Ensure that the travel of the latch tab matches that of the dock pedal in that it comes flush with the LBS at the same time the pedal contacts the hard stop. If the tab is flush with LBS and pedal travel remains, do not press the pedal further and adjust the cable to have more slack.

PINCH HAZARD

CAUTION

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

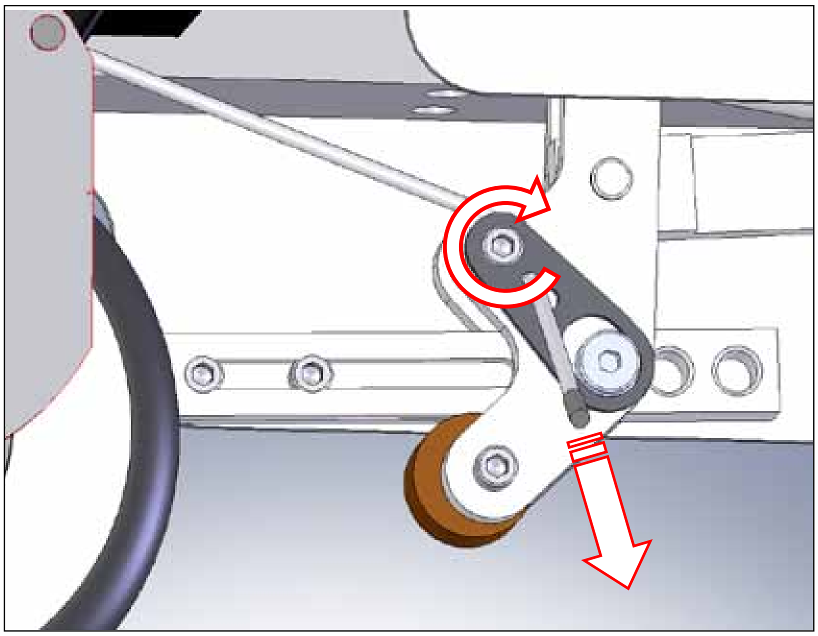


Figure 48

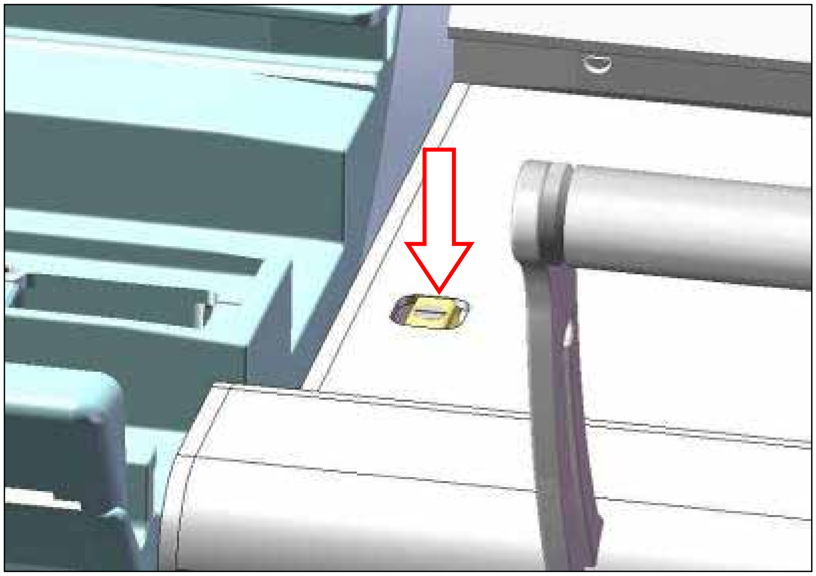


Figure 49

7.11 Head End Position Adjustment

The Upper Body Support (UBS) can be adjusted in the Head-Foot direction, as well as the Left-Right direction. The Left-Right direction helps ensure the bridge latches smoothly, and the Head-Foot direction determines the travel stop position of the patient support.

1. Open the bridge. Prepare to reposition the UBS by loosening the four hex head bolts on the underside of the UBS using a 9/16” socket or wrench (Figure 50). Pull the UBS handle to slide the UBS all the way to the head end.

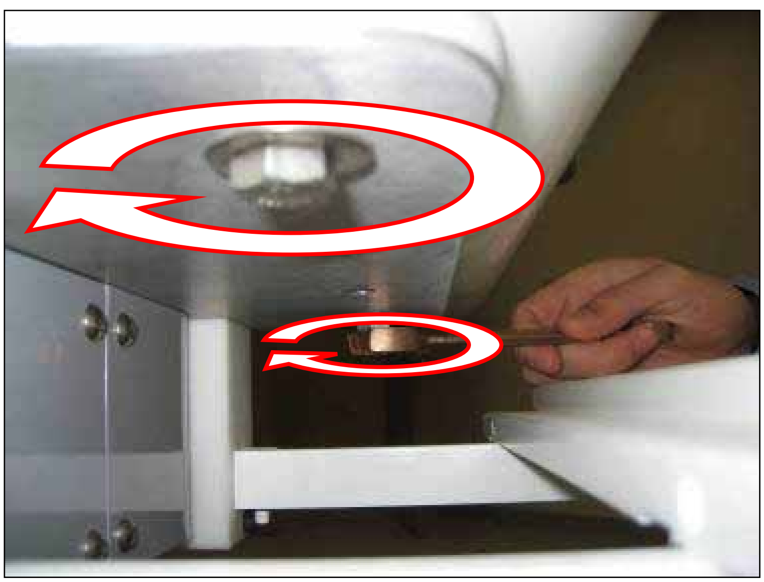


Figure 50

2. Replace the patient support on the stretcher by performing the inverse of the procedure in Section 7.1. The patient support should end up in the home position.

3. Push the UBS gently toward the foot end until the travel stop (Figure 53) on the patient left side of the UBS bottoms out against the patient support.

4. Close the stretcher bridge most of the way so that its spring plungers are just contacting the LBS (Figure 51). Adjust the UBS so that the bridge's nose doesn't rub either side of the square hole in the LBS.

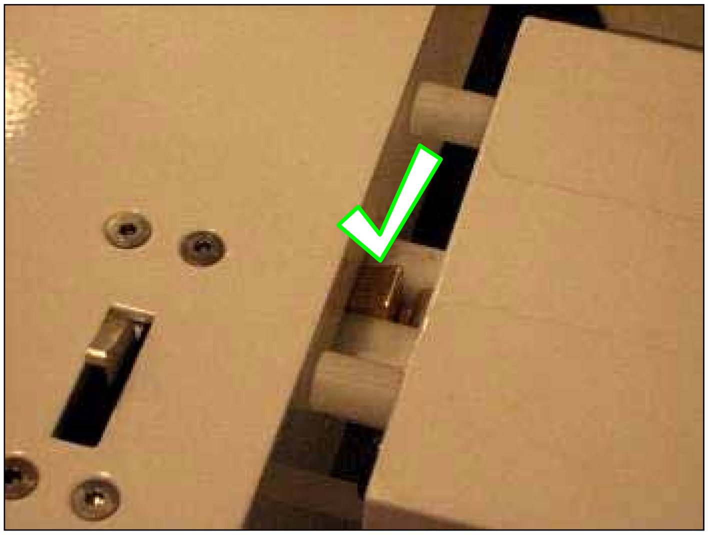


Figure 51

5. Ensure there is a gap between each side of the latch tab and the latch fixture by firmly pushing the patient support towards the head end of the stretcher which will slide the UBS forward and observing the position of the patient support with respect to the latch fixture (Figure 52). The latch tab must not rub any part of the latch fixture when the patient support is at its home position on the stretcher.

Figure 52

6. Ensure the UBS is centered left/right relative to the tabletop by looking at the travel stops position relative to the cervical wheel fascia (Figure 53).

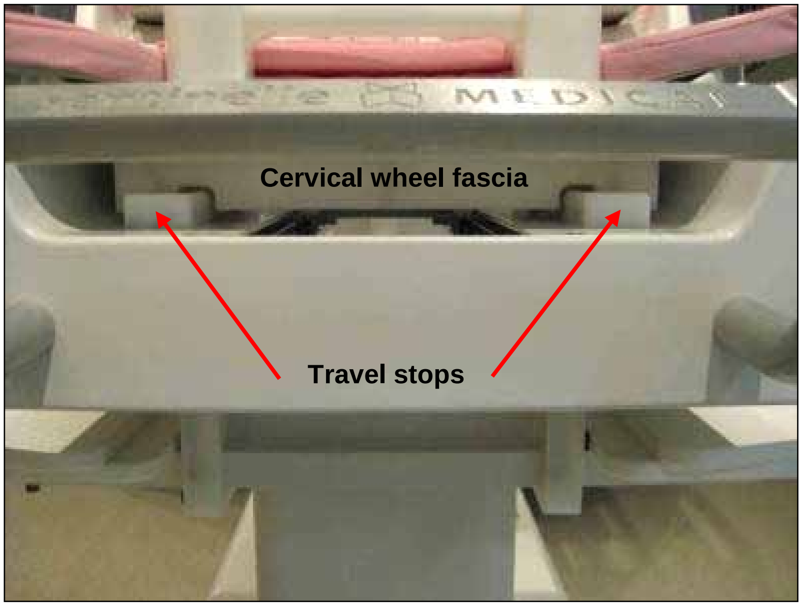


Figure 53

7. Check the UBS to ensure the bridge latch mechanism still slides freely in and out of the square hole in the UBS and adjust UBS angle if required.

8. Fasten the UBS in place by tightening the four hex head bolts using a 9/16" wrench.

9. Verify that the latch tab moves freely by depressing and releasing it by hand and docking and undocking the table. If the latch tab does not fully protract or is rubbing the latch fixture of the patient support, the UBS adjustment must be repeated.

7.12 No-undock Hook Adjustment

The no-undock hook is an interlock which prevents the stretcher from being undocked when the patient support is not in the home position. By hand, move the hook up and down to familiarize yourself with the range of travel. This adjustment must be made *only* if coarse height adjustment or scanner to stretcher gap adjustment was previously performed.

1. Dock the table, close the bridge and advance the patient support roughly 6"- 8" (152.4mm to 203.2mm) into the scanner. While the patient support is advanced and the no-undock hook is in the down position against its travel stop, pull the cable firmly downward and tighten the round head screw with a 5/32" Allen key so the cable can slip behind the screw head if the cable is pulled with moderate force. (Figure 54).

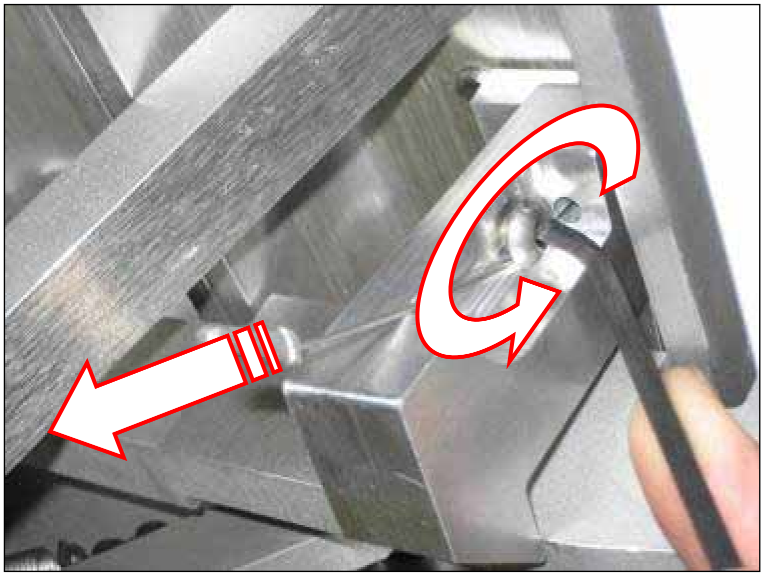


Figure 54

2. Return the patient support to the home position, while viewing the no-undock hook to verify it raises its travel stop when it is actuated. Note that the cable may slip behind the screw if the hook contacts its hard stop when raised before the patient support is completely in the home position. The hook should be at least 1/4" (6.35mm) clear of the undock pedal arm when the patient support is in the scanner, and completely clear of the undock pedal arm when the patient support is at the home position.

3. Check to ensure the hook is not overloaded when the stretcher is in the home position by pulling the patient support in the head-foot direction. There should be a small amount of free play, and the latch fixture should not be pressed against the latch tab. If the no-undock hook is overloaded, the cable should be loosened slightly and these checks repeated.

4. After correct function has been confirmed as per steps 2 and 3, completely tighten the round head screw with a 5/32" Allen key.

7.13 LPCA Interlock Adjustment

The LPCA Interlock is a safety mechanism that prevents the patient support from being rolled off the stretcher if the dock pedal is held down when the stretcher is not docked to the scanner. Adjustment of the LPCA mechanism plunger is always required. Now that the LPCA will need to be moved with the stretcher docked to the scanner, adjustment of the sensor actuator plunger on the stretcher bumper is required in this section.

1. Before docking the stretcher, completely depress the LPCA plunger (brass screw) on the patient support and take note of the amount of protrusion by viewing it from directly above and using the edge of the Patient Support as a reference.

2. Press the sensor pin in the middle of the Dock (Figure 19) and take note of the total amount of travel. Dock the stretcher to the scanner. The sensor actuator plunger in the centre of the bumper should not be contacting the corresponding sensor pin in the dock. Screw the plunger outwards from the bumper until it contacts the pin in the dock and a faintly audible click is heard from the switch inside the dock. Screw the plunger several more revolutions so it is pressing the pin roughly half way. Be careful not to overload the pin in the dock. Press the **Home** button on the scanner to move the LPCA to the patient support.

3. As the LPCA advances to latch with the patient support, watch carefully to see if the white LPCA plunger depresses the LPCA interlock plunger fully and if the LPCA latches freely to the patient support latch fixture. There are two failure conditions that can occur:

- The LPCA plunger can be prematurely depressed. This can prevent the LPCA from latching to the patient support.

OR

- The LPCA will latch to the patient support but the scanner LPCA plunger will not sufficiently depress the patient support LPCA interlock plunger.
  Take note of the distance required for latching to occur (Figure 55).

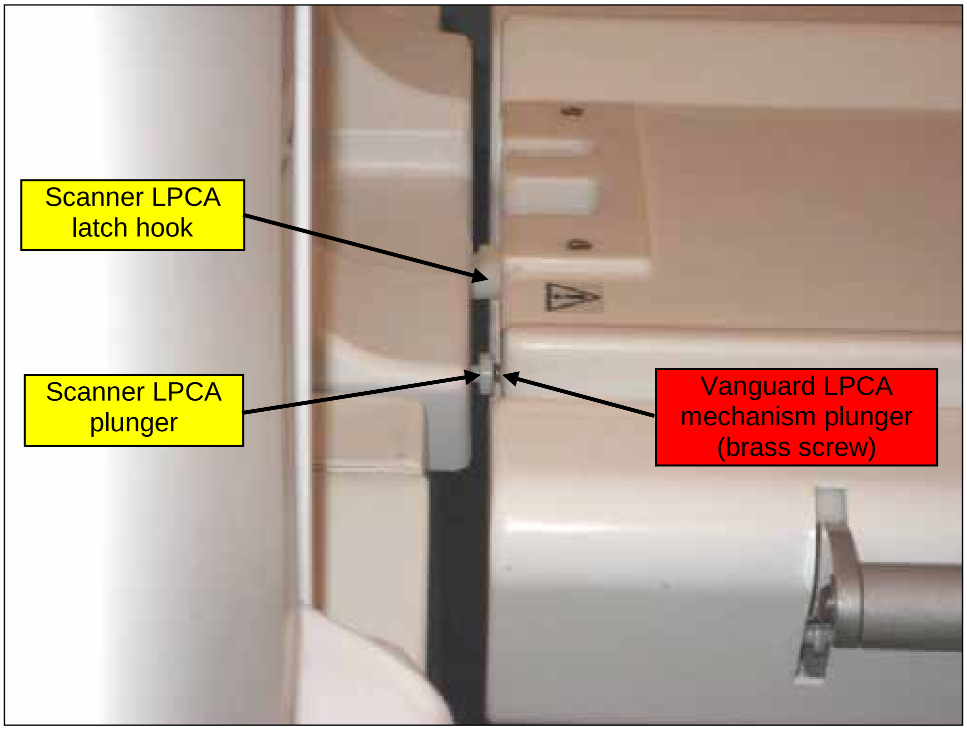


Figure 55

4. Unlatch the LPCA by hand by holding up the LPCA latch hook while advancing the LPCA into the scanner. Loosen the nut on the LPCA Interlock Plunger so that it can be adjusted. Screw the LPCA Interlock Plunger in or out the amount required to permit latching.

5. Return the LPCA to the home position and ensure that the LPCA interlock plunger is fully depressed but not overloaded as it latches. When the LPCA advances to latch to the tabletop, the patient support latch fixture should be latching fully with a click.

PINCH HAZARD

CAUTION

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

6. Advance the patient support ~4” (101.6mm) into the bore. Check to ensure that the LPCA Latch Hook has free play (<1/16” or 1.59mm) in the head-foot direction. If there is no movement, screw in one complete turn the LPCA Interlock Plunger and re-assess free play.

7. Advance the patient support in and out of the scanner. Ensure that the patient support can move in and out of the bore freely and the LPCA mechanism is not rubbing against the scanner’s bridge. Apply some downward pressure with one hand on the foot end of the patient support and ensure the LPCA can freely advance the patient support into the scanner without interference from the LPCA interlock.

8. Once the adjustment is correct, re-tighten the locking nut on the LPCA Interlock Plunger. Use a non-magnetic pliers to ensure the nut is tight.

7.14 Final Verification of Install

Final verification of install is an important series of functional and visual checks which is undertaken to ensure that the installation procedure was executed correctly. If a malfunction is detected during any of the verifications, review and repeat the relevant section of the installation procedure and test again and determine if any parts of the Vanguard or the scanner are damaged. As all of these checks are performed complete the **Installation Verification Checklist** in tandem/addition.

1. Replace the drawers. Test them to ensure they open and close smoothly.

2. Verify the function of the bridge latch and locking mechanism by opening and closing the bridge with the patient support in the home position. The tabletop must be prevented from advancing to scan using the MRI's controls when the bridge is open.

3. Verify the proper function of the travel stop plunger mechanism by advancing the patient support into the scanner and depressing the travel stop plunger by hand. It should not show any signs of friction. Verify the travel stop plunger permits the stretcher to undock only when the patient support is at the home position.

4. Verify the no-undock hook is lifted at least 1/4" (6.35mm) clear of the undock pedal arm when the patient support is in the home position. The hook must also touch its travel stops at either end of its range of motion.

5. Verify the correct function of the latch tab mechanism by stepping on the dock pedal. The latch tab must retract flush or up to 1/32" (0.79mm) above or 1/16" (1.59mm) below the top surface of the LBS when the docking pedal is pressed fully and should not rub against any part of the patient support when it is being actuated.

6. Verify the correct function of the LPCA interlock. It must move freely and fully actuate the LPCA mechanism when the LPCA is latched to the patient support without overloading the white plunger on the LPCA.

7. Ensure the catchment opens fully, does not interfere with any part of the Vanguard and that it closes and stays latched when patient support moves in and out of the scanner.

8. Verify the correct function of the side handles. They must actuate fully between locked positions and the white plastic covers must not leave any gaps larger than 5/16" (8mm)

9. Verify the correct function of the Vanguard docking and undocking. When docked the Vanguard should not come undocked when it is subject to force in any direction. Apply 50lbs of abrupt force to the stretcher handle toward the scanner to ensure the Vanguard stays docked.

   ⚠ CAUTION

   If the Vanguard does not stay docked, applying force may cause imbalance. Exercise caution when applying force to the Vanguard and be prepared for possible movement.

10. Verify the correct function of the patient support advancing to scan and returning to home. While advancing to scan apply enough resistance to the patient support to cause the LPCA to stop and continue to advance to scan and then return to home position. The patient support should not come unlatched or produce any noise or signs of friction with the scanner bridge or bore.

11. Verify the function of the emergency release by advancing the tabletop to the scan position and then actuating the release to free the table. It should release when the handle is twisted 90 to 110 degrees and the patient support is pulled with minimal force. The patient support should latch securely to the LPCA again after the emergency release handle is released gently.

12. Check that all caster roll and swivel lock brakes work properly.

13. Using the included reference card, install the foam padding to the patient support and ensure that a complete set of pads is present.

14. Insert the fuse which is taped to the side of the battery enclosure to the battery PCB. Verify correct function of all four lights on each side of the stretcher. Ensure all LEDs are functioning.

15. Review all technical notes, service bulletins and additional procedures related to the Vanguard for GE Signa HDx MRIs and ensure all actions are complete.

7.15 RF Coil Testing on GE Vanguard 1.5T 8Ch Table System

Sentinelle Medical recommends that the coils be checked for imaging integrity on a regular basis, by performing the Quality Assurance (QA) procedure. The QA procedure for 1.5T systems involves a phantom scan, a visual inspection of the QA images and a calculation of the signal-to-noise ratio.

For 3T systems, see *7.16 RF Coil Testing on GE Vanguard 3T 8Ch Table System* on page 58.

7.15.1 Setting up Coils and Phantoms

1. Install the compression frames, medial array and lateral arrays, as shown in Figure 56.
2. Test the RF system as instructed in this section. Repeat this test for all coil configurations and enter results in the Installation Verification section in Appendix A.

The phantoms used for the QA scan will be provided by Sentinelle Medical. Only use these phantoms when performing QA scans on the Vanguard.

To perform a scan of imaging phantoms for the purposes of analyzing **SNR** and verifying the function of individual **channels** of RF coil systems for the Vanguard for GE. Detailed instructions for setting up the Vanguard to perform a scan can be found in the Vanguard User Manual SMI-0246 which is issued to every customer and is available from Sentinelle.

- Dock the Vanguard to the scanner and plug the main connector into port A of the scanner.
- Set up the coil configuration which is to be tested (8ch will be used in this instruction) and position the phantoms such that they are touching the medial array and that the top surfaces of the medial array and phantoms are co-planar.
- Advance the table into the scanner, landmark over the centre of the phantoms and advance to scan.

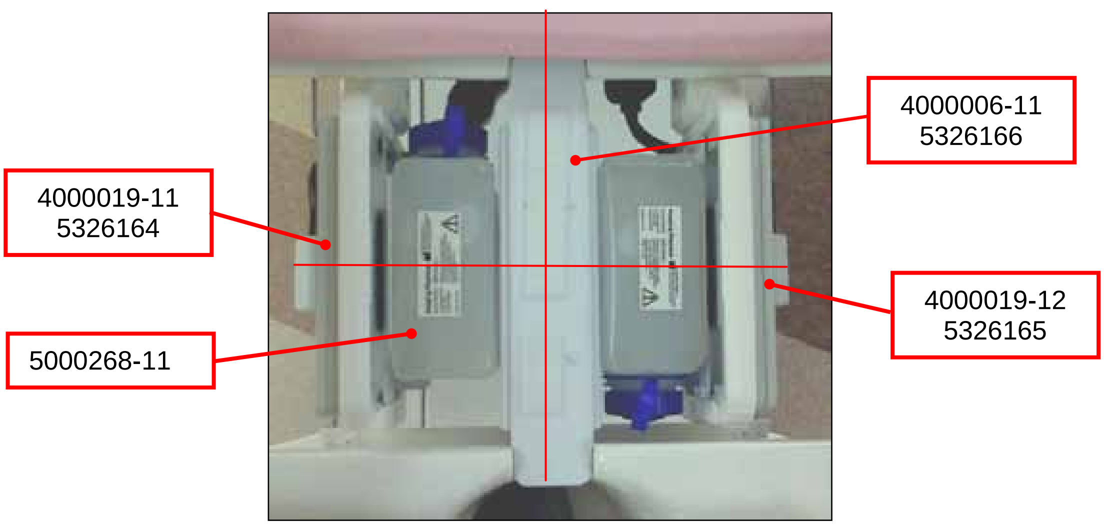


Figure 56: Positioning Phantoms in Frame

7.15.2 Prescribing a Scan

Click on **New Patient** and enter the following information:

<u>Patient Information</u>

Note: You must press enter to confirm data entered in the <u>Patient</u> ID and <u>Patient Name</u> fields. Select Accept in message box that appears while Patient information is being entered.

Patient ID: **geservice**
Patient Name: **SMI "date"** (eg. 21 NOV 2009)
Patient Weight: **120** lb

<u>Patient Position</u>

Patient Entry: **Feet First**
Patient Position: **Prone**
Coil: **... > currently connected > 8Ch Bilateral HD Breast Array by Sentinelle > 8Brst BL SMI**
Series Description: **8Ch SNR CHANNELS**

<u>Imaging Parameters</u>

Plane: **Coronal**
Mode: **2D**
Pulse Seq: **… > Spin Echo > Spin Echo > Accept**
The window will open to allow more scan information to be viewed and entered.

<u>Acquisition Timing</u>

Freq: **256**
Phase: **256**
Freq Dir: **S/I**
NEX: **1**
Phase FOV: **1**

<u>Scan Timing</u>

TE: **14**
TR: **450**
Receiver Bandwidth: **15** (.63 will auto-fill)

<u>Scan Range</u>

FOV: **32**
Slice Thickness: **5**
Slice Spacing: **0**

Start:**A40**
End: **A40**
# of Slices: **1** (will auto-fill)

- Select **Save Series >** right click **Research Operations >** select **Display CVs**
- In **CV Name** field type **saveinter**. Press enter and type **1** in **Current Value** field
- Click **Accept**, right click **Research Operations** and select **Download**.
- Select **Scan.**
- Scan is approximately 2:30. Images will preview in window on right of screen.

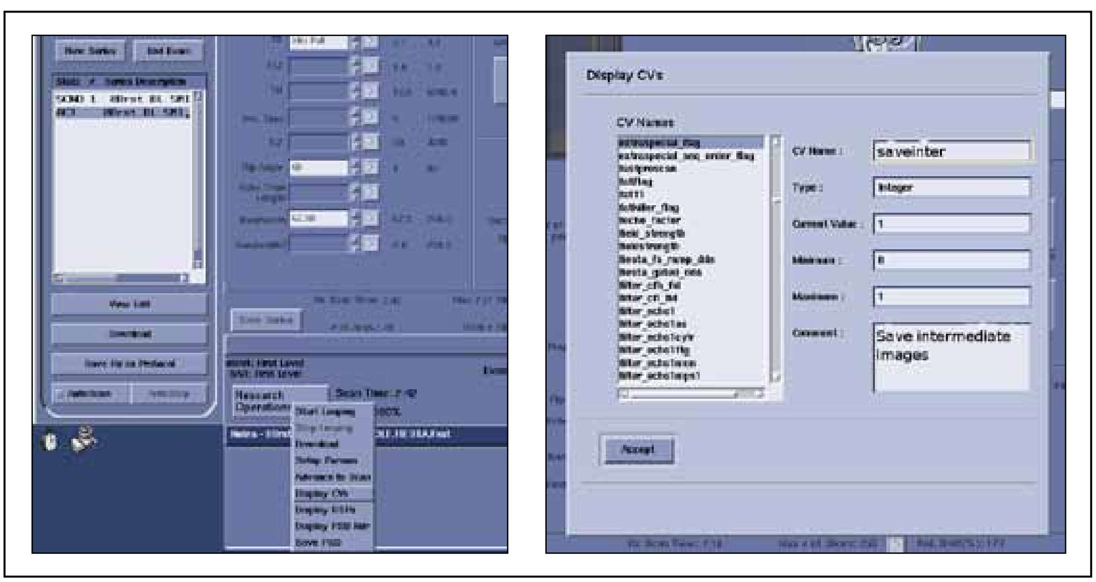
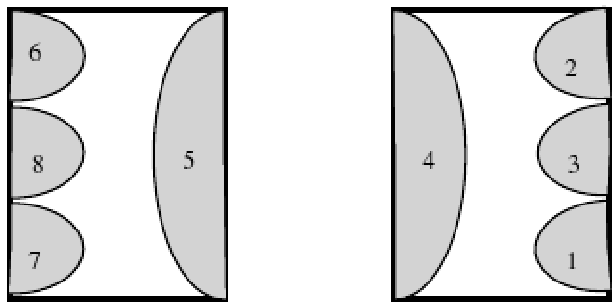


Figure 57: Setting the CV saveinter to 1, to obtain individual coil element images

7.15.3 Analysing Images

- On the top left of the screen click on the **Browser** button.
- Select examination **SMI "12 ABC 1234"** which was just performed and select **Viewer** from column of buttons on right side of the browser screen (if necessary you may return to the browser by selecting the **Browser** button at the middle left of the viewer screen). Select Format button to change to single image view format if necessary.
- By selecting the **Image –** and **+** buttons at middle left of screen, view images 1-8 to ensure that the signal for each channel is present and similar in intensity to the diagram below (all channels shown simultaneously for clarity).

Any black, grainy or low intensity signal areas indicate an image quality problem. Use the left and right phantoms to compare signal quality.



Left Phantom Right Phantom

SNR

- SNR will be calculated for the single image showing all channels activated, which is image 9.
- Select and hold the **Measure** button and select the **rectangle**. This will produce a rectangular Region of Interest (ROI) on the image.
- The rectangle may be positioned by selecting and holding any of the sides and may be resized by selecting the square in the upper right corner.
- Place a ROI over each phantom (1,2) which encompasses >90% of the signal producing area of the phantom. Ensure no part of the ROI is out of this area.
- Place a third ROI (3) of approximately 1100 mm^2 in the region above and between the phantoms in the image.

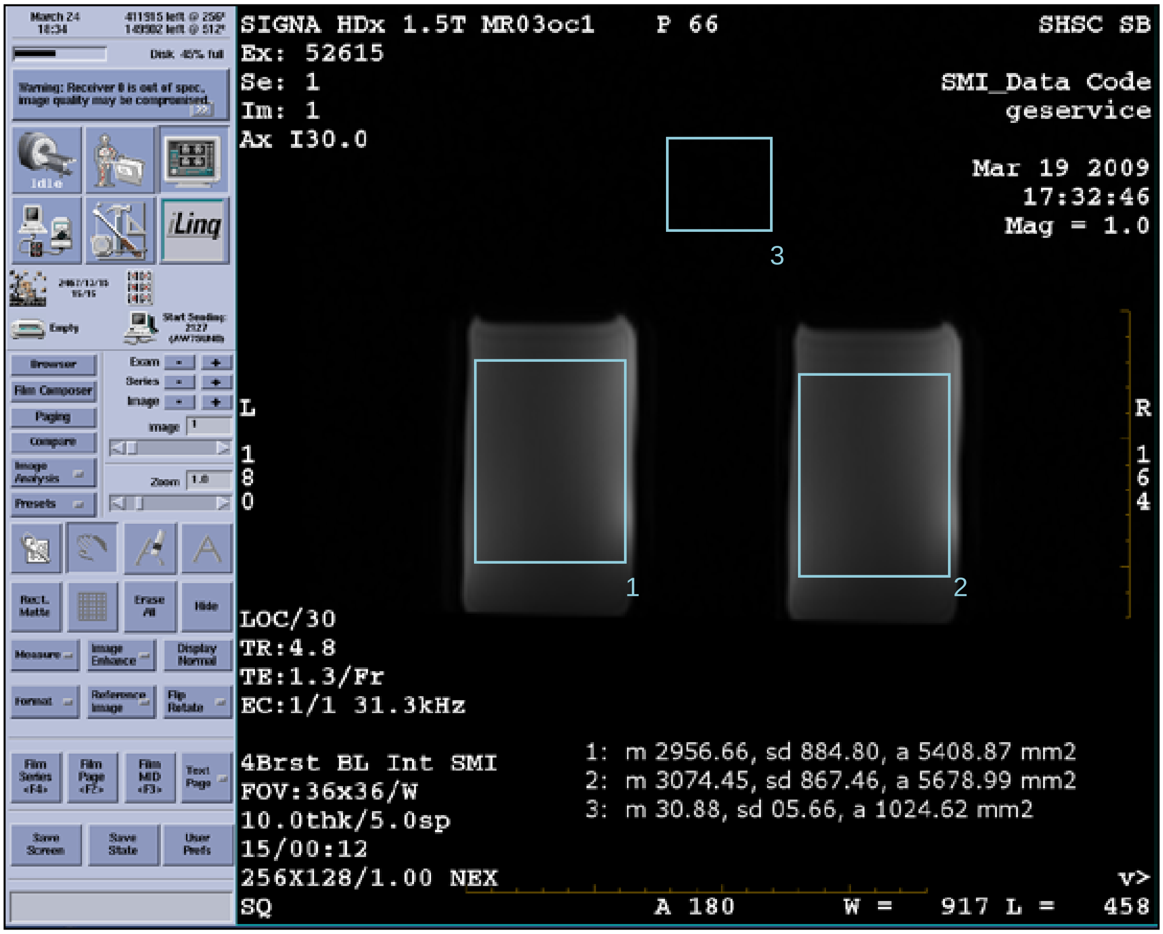


Figure 58: Placing ROI's in Image

The values for the **mean** (m), **standard deviation** (sd) and **area** (a) for each ROI are displayed at the bottom right of the image window. Use these values to calculate the signal-to-noise ratio (SNR) by dividing the average of the signal ROIs of the two phantoms by the standard deviation of the noise ROI.

$$\mathbf{SNR} = \frac{\text{(Right signal mean + Left signal mean) / 2}}{\text{Noise standard deviation}}$$

This table provides a format which can be copied for logging the values used to calculate SNR.

Config	Left Signal Mean	Right Signal Mean	Noise St'd Deviation	SNR
8ch				
4ch				
2ch R				
2ch L				

**Round all values to nearest whole number.*

7.16 RF Coil Testing on GE Vanguard 3T 8Ch Table System

Perform a QA scan of the GE 10 cm SiOil Spherical phantoms (P/N 2360034) to ensure all coils are working to specification. The QA scan must be completed for 8ch and 4ch configurations of the Vanguard.

*Note: The coil PASS/FAIL values should be determined using the procedure outlined in **7.16.1 Instructions for MR Imaging on Phantoms** and entered into the Quality Assurance SNR Section in the **Installation Verification Checklist** which can be found in Appendix A.*

7.16.1 Instructions for MR Imaging on Phantoms

Sentinelle Medical recommends that the coils be checked for imaging integrity on a regular basis, by performing the Multi-Coil Quality Assurance (MCQA) procedure. The phantoms used for the QA scan are part of GE's system phantoms. The instructions below only apply to the Sentinelle Vanguard system.

Note: If bubbles are apparent in the phantom, tap the outside of the phantom until these bubbles disappear.

Test 1: 8-channel Bilateral Breast SNR Verification

1. Dock the Vanguard to the MR scanner.
2. Install and connect the 8-channel coil set to the Vanguard table and insert the Vanguard cable into the receptacle. Close the catchment tray.
3. Move the lateral coils to their furthest outward position and slide them to their highest upward position. Locate the phantom positioners on either side of the medial coil as shown in Figure 59.

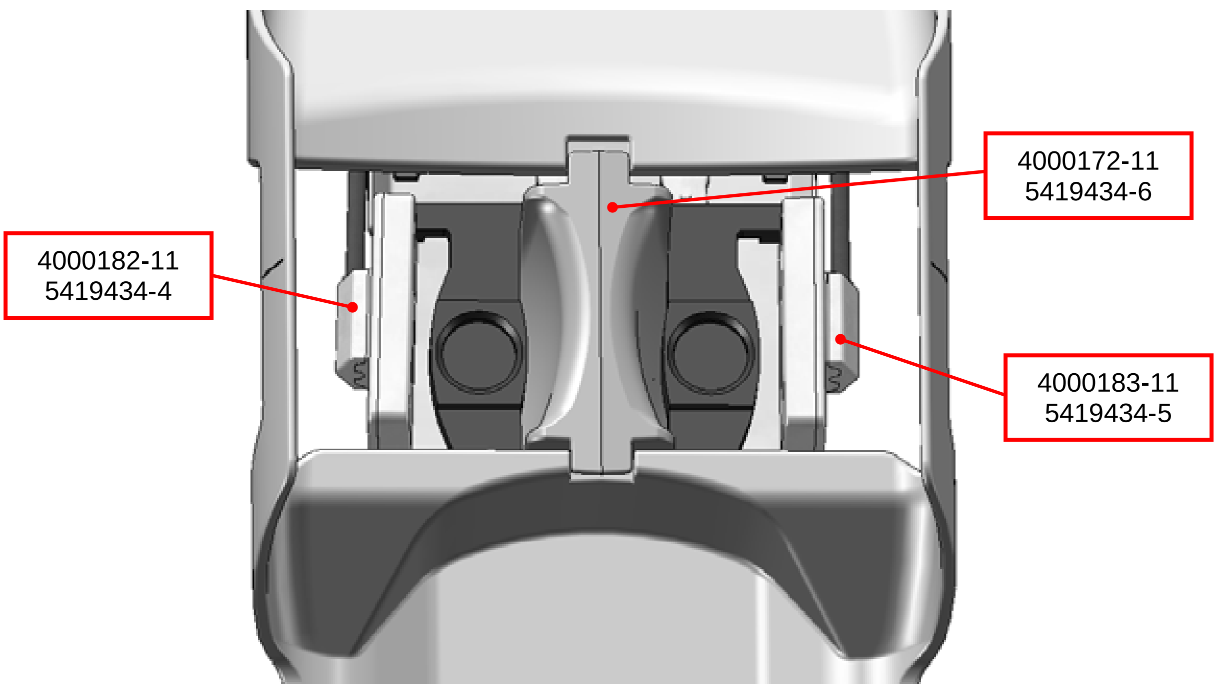


Figure 59: 8 Channel - Phantom Positioner Setup

4. Add spherical GE phantoms and move the lateral coils to their lowest position. Adjust lateral coils towards the phantom so that they lightly touch the phantoms as shown in Figure 60.

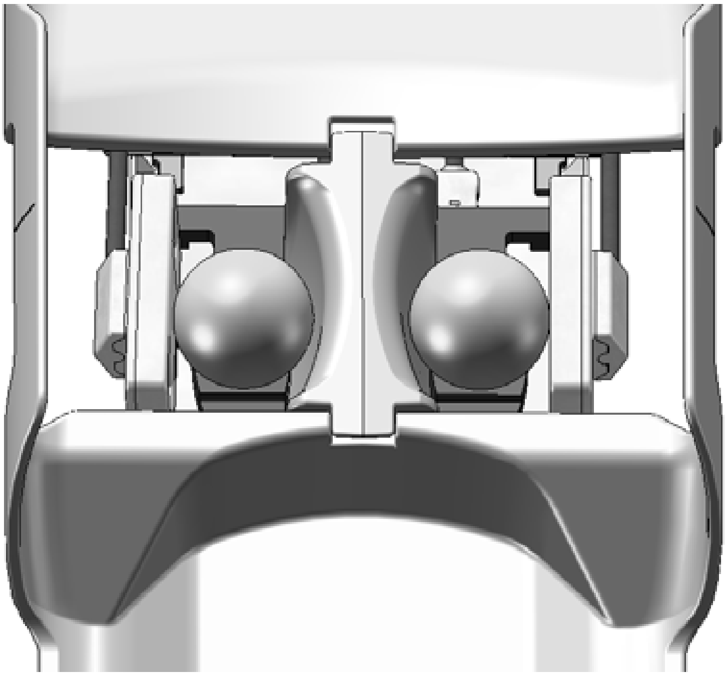


Figure 60: 8 Channel - Phantom Setup

5. Landmark using the notch on the edge of the patient support as shown in Figure 61.

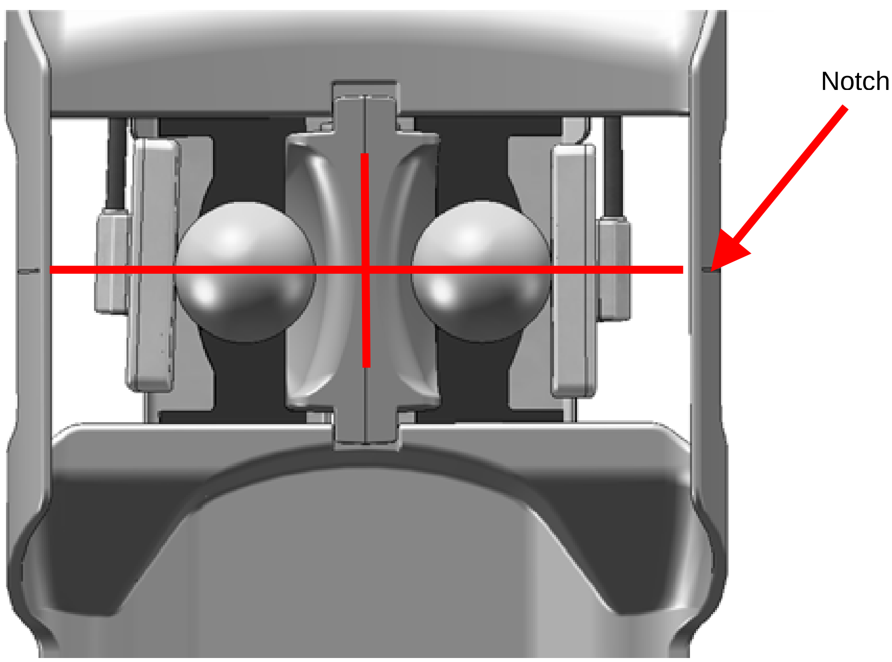


Figure 61: 8 Channel - Landmark Phantoms

6. Do not start a new patient; open the Tools box, and start the System Browser. Select the top tab marked 'Image Quality.' In the side menu, select 'Multi-Coil QA Tool', then 'Click here to start this tool.' The program shown in Figure 62 will start:

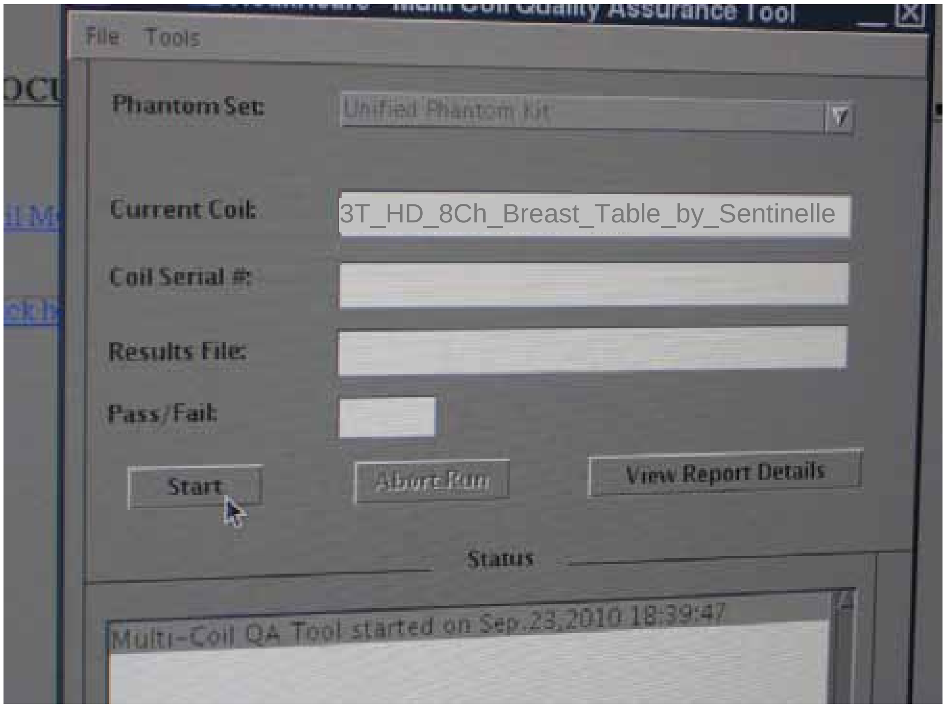


Figure 62: Screen Capture – MCQA Tool Start

7. Click ‘Start’, and answer ‘Yes’ to the dialog box that opens. The MCQA program will run for approximately 10 minutes, and test each channel of the system independently. When the program finishes, the screen capture shown in Figure 63 will appear.

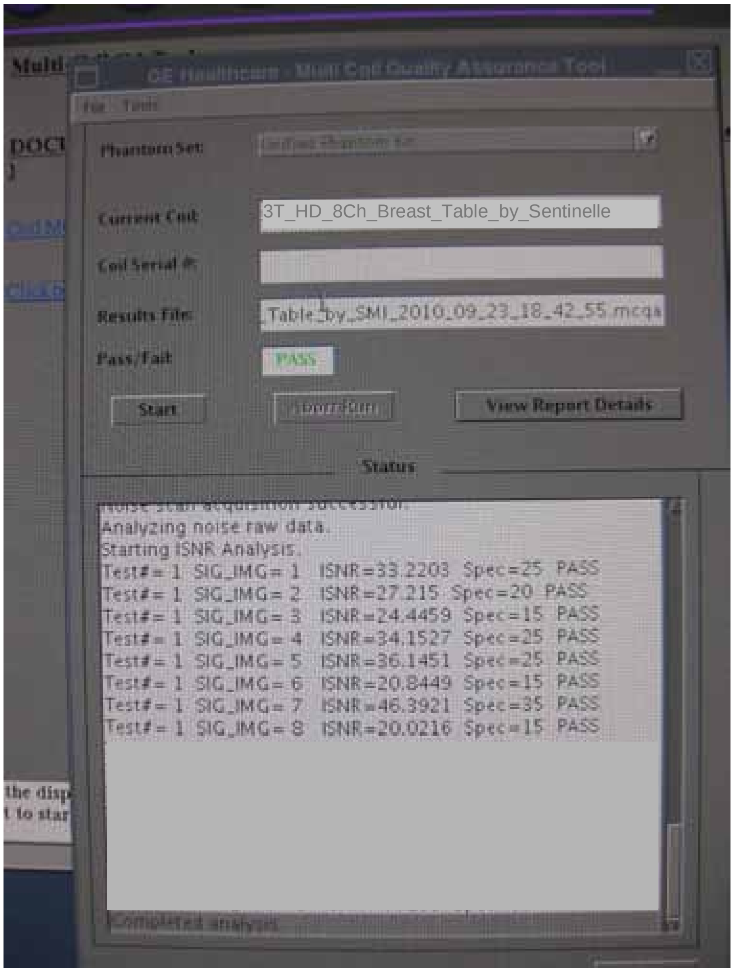


Figure 63: Screen Capture – MCQA Tool Finish

8. The MCQA program will display a PASS/FAIL indicator. If you receive a FAIL indication, check that all coils are connected properly, and re-run the test.

Test 2: 4-channel Bilateral Breast SNR Verification

1. Dock the Vanguard to the MR scanner.
2. Install and connect the 4-channel coil set to the Vanguard table and insert the Vanguard cable into the receptacle.
3. Close the catchment tray. Move the lateral coils to their furthest outward position and slide them to their highest upward position. Locate the phantom positioners on either side of the medial coil as shown in Figure 64.

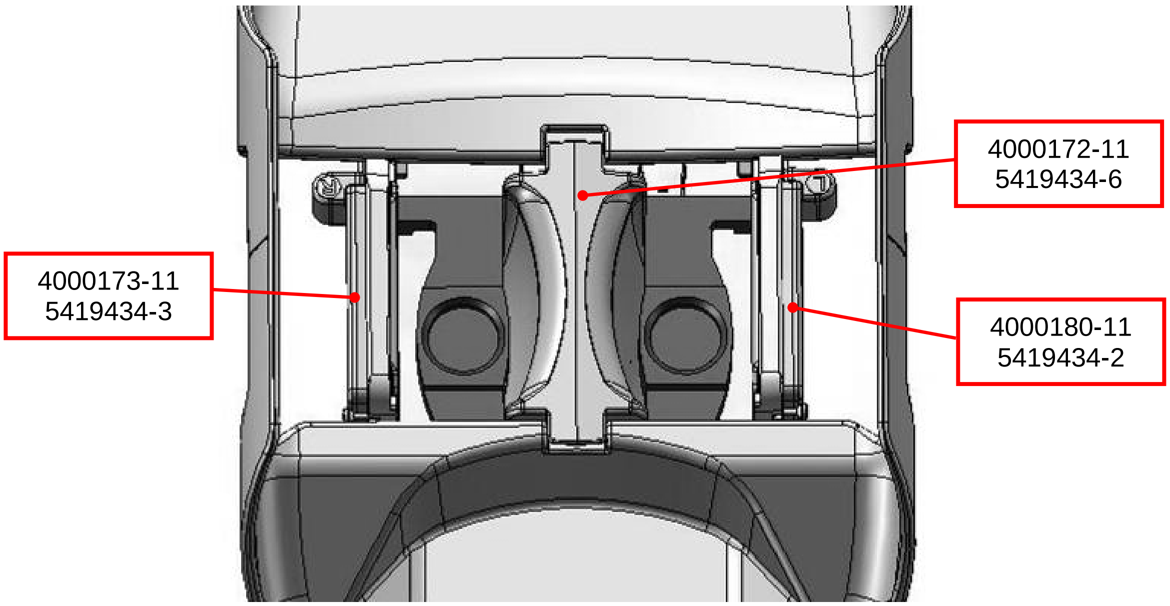


Figure 64: 4 Channel - Phantom Positioner Set-Up

4. Add spherical GE phantoms and move the lateral coils to their lowest possible position. Adjust lateral coils towards the phantom so that they lightly touch the phantoms as shown in Figure 65.

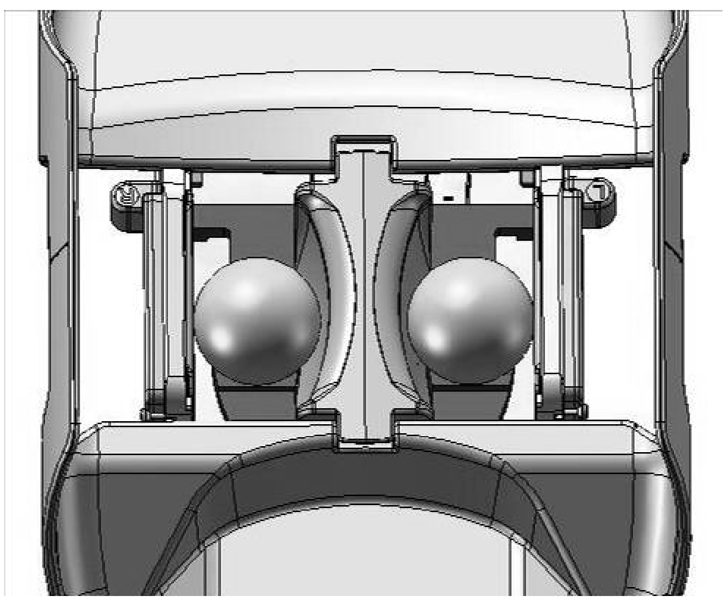


Figure 65: 4 Channel - Phantom Set-Up

5. Landmark using the notch on the edge of the patient support as shown in Figure 66.

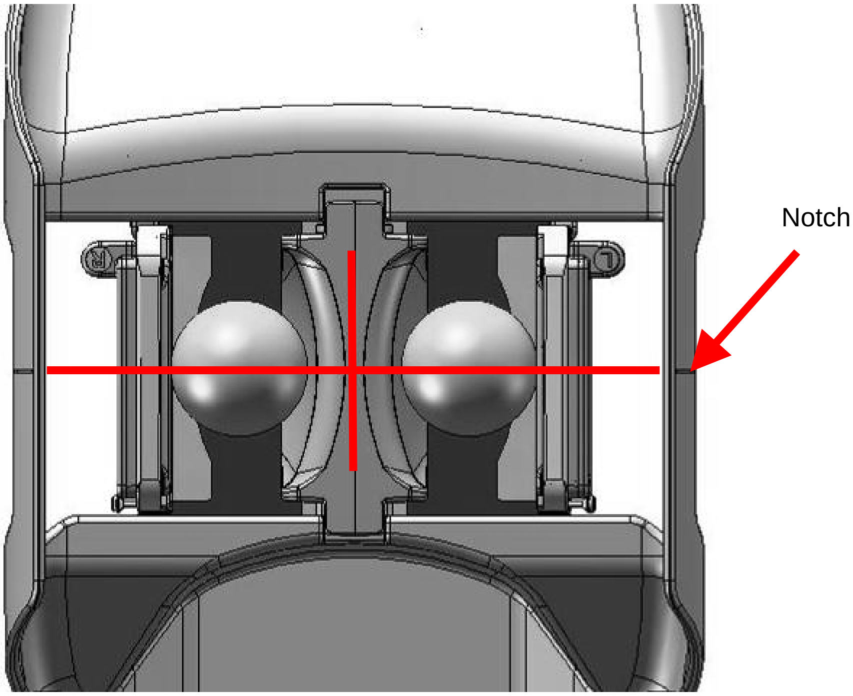


Figure 66: 4 Channel - Landmark Phantoms

6. Do not start a new patient; open the Tools box, and start the System Browser. Select the top tab marked 'Image Quality.' In the side menu, select 'Multi-Coil QA Tool', then 'Click here to start this tool.' The program shown in Figure 67 will start:

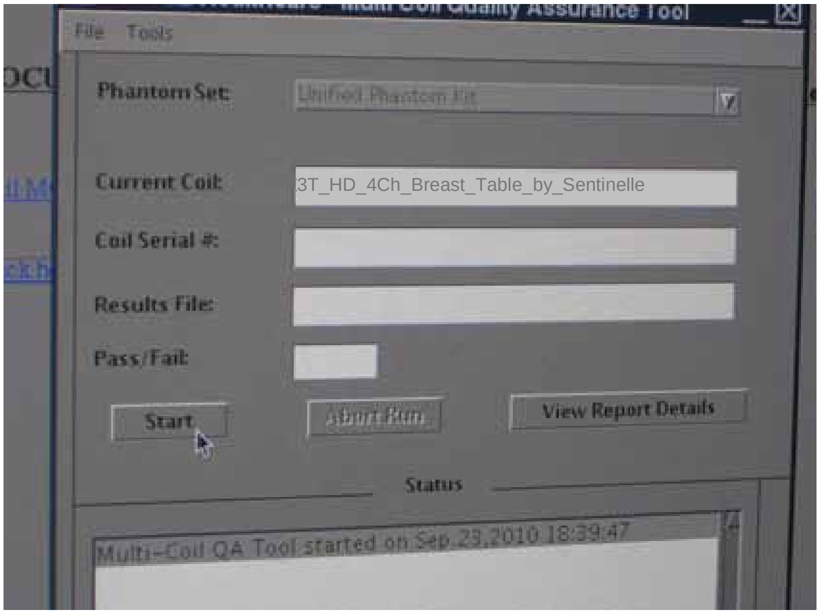


Figure 67: Screen Capture – MCQA Tool Start

7. Click ‘Start’, and answer ‘Yes’ to the dialog box that opens. The MCQA program will run for approximately 10 minutes, and test each channel of the system independently. When the program finishes, the screen capture shown in Figure 68 will display.

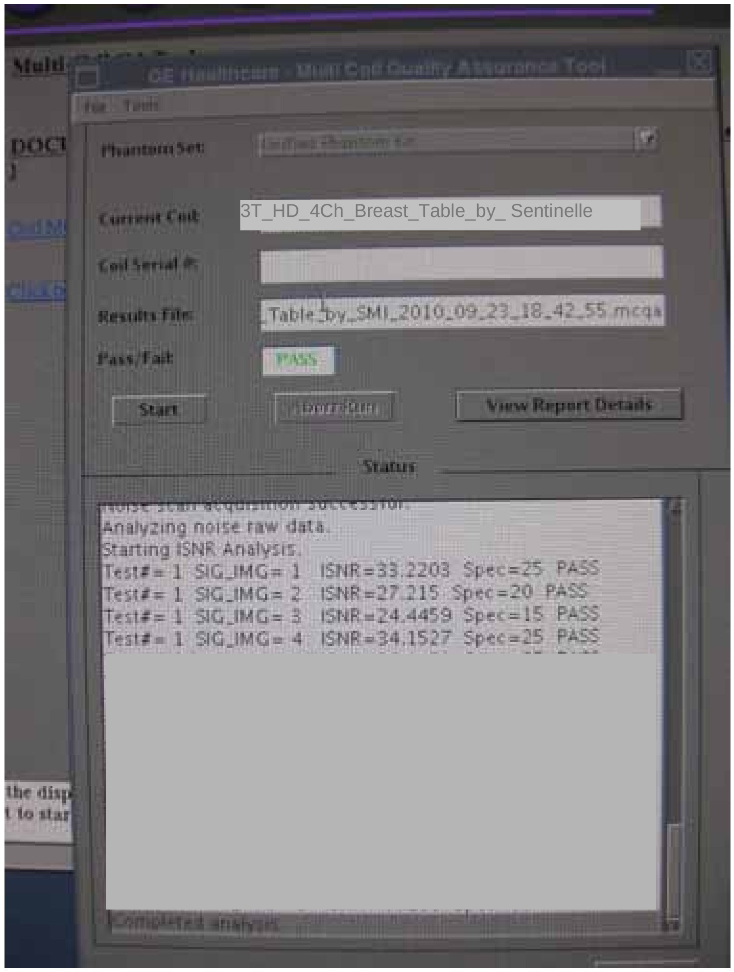


Figure 68: Screen Capture – MCQA Tool Finish

8. The MCQA program will display a PASS/FAIL indicator. If you receive a FAIL indication, check that all coils are connected properly, and re-run the test.

Troubleshooting tips

1. Ensure that no patient has been started before beginning the MCQA procedure.
2. Landmark the coils before starting the MCQA tool.
3. Only use the spherical GE phantoms (GE Model 46-317586G1), and the Sentinelle phantom positioners.
4. Ensure that all coils are touching the phantoms.

7.17 Coil Files

SMI ONLY	On GE Signa scanners with software version 16.0_V02 service pack 1 or later, the coil config files required to operate the Vanguard are pre-installed on the scanner. The first time each coil configuration is connected a message will appear prompting user to add coil files. Upon accepting this option the particular coil configuration can be used. Scanners with 12 and older revision software will require manual installation of the coil config files to operate the Vanguard. The coil files are installed by the **SMI Coil Config File Instructions**.

7.17.1 Coil Configuration File Names

The following coil configuration files are available:

- 8BrstBL SMI
- 8BrstLt SMI
- 8BrstRt SMI
- 4Brst BL Int SMI
- 2Brst Int SMI

7.18 Final Verification Checklist

Final verification of install is an important series of functional and visual checks which is undertaken to ensure that the installation procedure was executed correctly.

SMI ONLY	Complete the Installation Verification Checklist (section 14 Appendix A) and submit to SMI Service department.

GEHC ONLY	Ensure that System ID has been created for the Sentinelle table from the GON. Install table system ID label in a conspicuous location on the table visible to the customer.

7.19 System Storage

When installation is complete, store the system as directed by the customer or inside the scanner room in an area where it will not pose a hazard or impede access to any part of the room. The storage environment should be ***230cm x 100cm x 150cm*** or ***91in x 40in x 60in*** and meet the criteria listed in the table below. The weight of the system is ***162kg*** or ***357lb***.

Condition	Minimum	Maximum
Ambient Temperature (°C)	-30	+65
Relative Humidity (non-condensing)	10%	95%
Atmospheric Pressure	500 hPa	1060 hPa

8 Preventative Maintenance

WARNING **If any blood-bourne pathegens (BBPs) are present, stop work and notify customer that the Vanguard must be cleaned prior to performing service.**

Ensure that the person following this manual is the correct person to service this system.

GEHC ONLY	GE Field Engineers shall only service systems that have a service contract with GEHC for the table. Tables with a service contract will have a separate System ID.

If service on the scanner is performed involving the Dock, Bridge and any component of the LPCA, adjustments to the Vanguard must be confirmed if necessary:

CRITICAL ITEM

- If the Bridge is serviced, fine height adjustment of the stretcher may need to be performed.
- If the Dock is serviced, the bumper and/or bumper spring assemblies of the stretcher may need to be adjusted.
- If the LPCA is serviced, the LPCA interlock on the patient support may need to be adjusted.

8.1 Vanguard PM Procedure

This procedure will take approximately 1 hour to complete, provided that no major components are being replaced. This preventative maintenance procedure should be executed semi-annually for low use systems and quarterly for high use systems. A high use system is one that is used to perform >30 procedures weekly and a low use is one which is used for <30.

SMI ONLY	Any service activities performed on a system are always logged in the SMI service database. Before servicing a unit in the field, the service technician should confirm the date of last service for that particular unit. If the unit is, or will soon be due for a preventative maintenance call, one should be carried out while the service technician is on site, time permitting. Before servicing the Vanguard, the SMI service technician should inspect the scanner and its accessories to look for any obvious damage or signs of wear. Any perceived problems should be reported to the GE FE for that particular site. The SMI service technician should also speak with a site staff member who has personally used the system and ask if they have any functionality concerns to report. Use the PM checklist in Appendix A to document all checks and tests performed during PM procedure.

Dock the system and advance to scan; listen for any noises which could indicate broken wheels, interferences between the patient support and the scanner and stretcher, or within the patient support itself. It is sometimes helpful to clean the scanner's bridge of any debris or dust.

8.1.1 Verification of Installation Settings

In order to verify the settings of the Vanguard a series of functionality tests must be made. If any of these tests are failed, refer to the troubleshooting section in order to restore full functionality.

1) Dock stretcher and assess whether docking force required is appropriate. Listen for any noise or vibrations. Test docking force by pushing stretcher sharply toward scanner and by applying moderate left and right force to stretcher handle.
2) Check LPCA actuator sensor to make sure it is not pushing the pin on the docking fixture too far in and possibly overloading it.
3) Check gap between stretcher and scanner to ensure it is 1" to 1.5" (25.4mm to 38.1mm).
4) Advance LPCA to table and ensure LPCA mechanism is properly actuated.

5) Advance patient support completely into the magnet with weight applied and ensure it is not contacting bore or making any creaking, scratching, scraping, clicking, grinding, vibrating or unusual noises.
6) Ensure latch tab is completely retracted into LBS and is not protruding.
7) Verify UBS/LBS surfaces are parallel and level with scanner's bridge
8) Use tool to ensure that guide rails are collinear.
9) At the end of the LBS nearest the scanner apply heavy left right force to check LR centering of the stretcher.
10) Test patient support emergency release and re-latch patient support.
11) Check no-undock hook to ensure it is hanging freely and effectively preventing undock pedal from being pressed when the patient support is advanced into the scanner.
12) Return patient support to home position, again checking for unusual noises.
13) Check undock hook to ensure it is well clear of undock pedal arm.
14) Press undock pedal and leave system against scanner.
15) Ensure that the LPCA unlatches and moves into scanner.
16) Check docking hook height to make sure it is not contacting scanner dock's bronze hook when rolling the stretcher up to the scanner.
17) Perform the Phantom Scan Quality Assurance Procedure (as described in sections 7.15 and 7.16) and compare results with baseline figures entered from first scan on install.
18) Complete all image quality testing procedures issued for this product
19) Refer to all service bulletins for this product
20) Perform any Service Actions Pending for this product. If you do not have an up to date list of actions outstanding, telephone Sentinelle Service for an update.

8.1.2 Subsystem Tests and Inspection – Stretcher and Patient Support

1) Check the rolling and swivel brakes of all four casters
2) Actuate travel stop mechanism by hand, inspect cable at mount points
3) Check proper function of no dock spring and that there are no signs of damage.

PINCH HAZARD

CAUTION

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

4) Latch tab actuation and height, condition of cable and cable elbows
5) Function of bridge, and articulation of latch and stopper
6) Check function of lighting system, individual LEDs
7) By hand check articulation of the docking hook to ensure that it moves up and down freely without binding.
8) Check the undock spring to ensure it is not contacting any other parts of the linkage other than its mounting hardware.

PINCH HAZARD

CAUTION

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

9) Apply moderate left and right force to the dock and undock pedals to check if they have become worn and are contacting other components of the stretcher or docking mechanism.
10) While actuating the dock pedal by hand look at the drive link to make sure it is not rubbing against the tension coupling or center plate.
11) Dock the system and then, while undocking, check to ensure that the undock pedal arm does not contact the link wheel pin.
12) Remove drawers and inspect latch tab cable and cable elbows for wear or fatigue.
13) Check function of side handles to ensure they extend and retract smoothly and that gap covers are moving freely.
14) Check all fasteners on moving parts of docking mechanism for tightness.
15) Check drawers for smooth function and integrity of sliders.
16) Check all system labels are present and intact. Refer to section 6.2 for label locations.
17) Actuate and test emergency release with table in scanner several times
18) Check all wheels on the underside of the patient support for excessive noise (needing lubrication or replacement) or damage such as cracks.
19) Lift foot end of patient support and hang over end of LBS to expose the LPCA mechanism and check that mechanism functions properly and isn’t damaged or worn.
20) Ensure the nut on the LPCA mechanism plunger screw is tight using a small wrench.
21) Check the headrest for smooth operation and any cracks or damage.
22) Check the padding to ensure that is has no rips or damaged Velcro.

8.1.3 Subsystem Tests and Inspection – RF and Compression System

1) Inspect all of the coils to ensure that they are not cracked or otherwise damaged.
2) Check the coil cables for any rips, strains or loose connections.
3) Inspect the cable and balun at the end of the cable tray for damage.
4) Inspect the system plug by looking for bent pins and ensuring that it connects smoothly to the MRI with moderate force and good retention.
5) Check the sliders, frames and compression rails for damage or wear and that the slider locks lock firmly on the compression rail and frames.
6) Check contralateral support for cracking or damage.
7) Check all system labels are present and intact. Refer to section 6.2 for label locations.
8) Test the RF system by completing tests #2 and #3: SNR and Channel verification, from TP-151 - Simplified Imaging Tests and record results in SMI-0636 - VG PM Verification. If results from tests #2 and #3 do not pass, run all tests in TP-151 and forward results to SMI.

8.1.4 Cleaning

1) Clean all surfaces of the patient support and stretcher. Use an alcohol solution (70% isopropyl alcohol, 30% water) and a soft cloth or alcohol wipes, or other approved solution as described in the Vanguard User Manual.

GEHC ONLY	Use alcohol wipes (GE part # 46-183039P1) to clean the patient support and stretcher.

2) Make sure there is no tape or tape residue on any of the surfaces that would interfere with normal operation of the stretcher patient support or RF system

3) Flip over the patient support and inspect all of the wheels on the underside of the patient support and ensure that they are clean and rolling smoothly and quietly. Brush away any dust or debris.
4) Clean the mirrors and catchment tray if necessary using the same cleaning solution as for the patient support and stretcher.
5) Check the casters to ensure they have no debris caught in them which could inhibit their ability to roll or swivel.
6) Clean any dirt or debris from the stretcher drawers.
7) Check and clean contralateral support.
8) Clean any of the padding that may be dirty using a bleach solution (10% bleach containing 5.25% sodium hypochlorite, 90% water), or other approved solution as described in the Vanguard User Manual. **Do not use any alcohol-based cleaner on the Vanguard padding.**

9 Troubleshooting Guide for Mechanical Components

9.1 Troubleshooting Methodology

The following steps will guide you to identify and resolve the problem

1) ***Identify*** the malfunction.
2) ***Review the possible causes*** of the malfunction, considering User error and non-technical issues which may be impeding function of the Vanguard.
3) ***Execute the actions*** which correspond to the possible causes.
4) Damaged parts can be the cause of the reported problem. ***Repair or replace*** affected parts or hardware and then complete any associated adjustments
5) Every action includes a ***final "verification"*** step, including all interlocks and sub-assemblies. This full system check is performed for preventative maintenance.

9.2 Potential causes Triggering Service Call

- Damage caused during use - to parts and assemblies
- User error in operating the system
- Incorrect interpretation of the problem being reported
- Improper training/Improper use of system
- GE service related changes to the scanner (e.g. installed/upgraded software patches)

9.3 Mechanical Troubleshooting Guidance

The following table details potential problems, causes, checklist items and possible actions to resolve problems with the Vanguard.

PROBLEM	**POSSIBLE CAUSES**	**CHECKS to be Performed**	**REPAIR or REPLACE**
Unable to undock stretcher from scanner	· Damage to docking mechanism · Undock pedal jammed · Patient support not in home position · Caster(s) locked	· Check bridge is closed · Check for damage/ misalignment to parts of docking mechanism and foreign objects. · Review Procedures: ü Alignment ü 7.5 Bumper Adjustment ü 7.9 Gap Adjustment ü 7.6 Docking Force Adj. ü 7.7 Bumper Plunger Adj. ü 7.10 Latch Tab Adjustment ü 7.12 No-undock Hook Adj. ü 7.13 LPCA Mech. Adj. ü 7.16 Final Verification	Ø 13.9 Bridge Latch Assembly Ø 13.24 Latch Tab and Spring Ø 13.40 Cable Elbow 105 Ø 13.41 Cable Elbow 75 Ø 13.42 Cable Ø 13.20 LPCA Mechanism Ø 13.10 Swing Assembly Ø 13.2 Spring Bumper Assembly Ø 13.11 Docking Linkage Assembly Ø 13.28 Tension Coupling Ø 13.39 Caster

PROBLEM	POSSIBLE CAUSES	CHECKS to be Performed	REPAIR or REPLACE
Unable to dock stretcher to scanner	· Items interfering with docking · Docking hook misaligned · Docking pedal jammed · Damage to docking mechanism. · Stretcher-scanner gap not to specification · Caster(s) locked	· Investigate damage or misalignment to docking mechanism and · Foreign objects present · Review Procedures: ü 7.1 Alignment ü 7.5 Bumper Adjustment ü 7.9 Gap Adjustment ü 7.6 Docking Force Adj. ü 7.7 Bumper Plunger Adj. ü 7.16 Final Verification	Ø 13.10 Swing Assembly Ø 13.2 Spring Bumper Assembly Ø 13.11 Docking Linkage Assembly Ø 13.28 Tension Coupling Ø 13.39 Caster
Patient support will not advance from stretcher and/or in scanner	· Bridge not closed · Stretcher not docked · Damage to patient · support hardware · Items impeding motion of patient support · LPCA interlock mechanism not actuated · Scanner's bridge higher/lower · Latch tab protruding · Stretcher not centred	· Check for damage/misalignment to parts of patient support hardware and foreign objects. · Review Procedure: ü 7.9 Gap Adjustment ü 7.8 Centering Guide Adj. ü 7.13 LPCA Mech. Adj. ü 7.10 Latch Tab Adjustment ü 7.16 Final Verification	Ø 13.9 Bridge Latch Assembly Ø 13.24 Latch Tab and Spring Ø 13.40 Cable Elbow 105 Ø 13.41 Cable Elbow 75 Ø 13.42 Cable Ø 13.20 LPCA Mechanism Ø 13.10 Swing Assembly Ø 13.2 Spring Bumper Assembly Ø 13.11 Docking Linkage Assembly Ø 13.39 Caster
Patient support stuck in scanner or on stretcher and cannot advance or return to home position	Ø Bridge not closed Ø Items impeding motion of patient support. Ø Latch tab protruding Ø Docking sensor not actuated Ø Stretcher-scanner gap not in spec. Ø Stretcher not centred	· Check for damage/misalignment to parts of LPCA mechanism, docking mechanism and patient support hardware. · Review Procedures: ü 7.5 Bumper Adjustment ü 7.6 Docking Force Adj. ü 7.7 Bumper Plunger Adj. ü 7.8 Centering Guide Adj. ü 7.13 LPCA Mech. Adj. ü 7.10 Latch Tab Adjustment ü 7.16 Final Verification	Ø 13.10 Swing Assembly Ø 13.2 Spring Bumper Assembly Ø 13.11 Docking Linkage Assembly Ø 13.28 Tension Coupling Docking Linkage Ø 13.9 Bridge Latch Assembly Ø 13.24 Latch Tab and Spring Ø 13.40 Cable Elbow 105 Ø 13.41 Cable Elbow 75 Ø 13.42 Cable

PROBLEM	POSSIBLE CAUSES	CHECKS to be Performed	REPAIR or REPLACE
Patient support comes unlatched from LPCA when advancing or returning	· Items preventing latching of LPCA to patient support. · Emergency release being actuated · Stretcher not centred · Stretcher not docked straight · Stretcher not level	· Check for damage/misalignment to parts of LPCA mechanism, docking mechanism and patient support hardware. · Review Procedures: ü 7.3 Fine Height Adj. ü 7.5 Bumper Adjustment ü 7.6 Docking Force Adj. ü 7.7 Bumper Plunger Adj. ü 7.8 Centering Guide Adj. ü 7.13 LPCA Mech. Adj. ü 7.10 Latch Tab Adjustment ü 7.16 Final Verification	Ø 13.10 Swing Assembly Ø 13.2 Spring Bumper Assembly Ø 13.9 Bridge Latch Assembly Ø 13.24 Latch Tab and Spring Ø 13.40 Cable Elbow 105 Ø 13.41 Cable Elbow 75 Ø 13.42 Cable
Rough or noisy motion of patient support in scanner or on stretcher	· Bridge not closed · Latch tab protruding · LPCA mechanism not fully actuated · Items impeding motion of patient support · Damage to wheels and tabletop hardware	· Check for damage/misalignment to parts of patient support hardware and foreign objects. · Review Procedure: ü 7.1 Alignment ü 7.3 Fine Height Adj. ü 7.5 Bumper Adjustment ü 7.16 Final Verification	Ø 13.9 Bridge Latch Assembly Ø 13.24 Latch Tab and Spring Ø 13.40 Cable Elbow 105 Ø 13.41 Cable Elbow 75 Ø 13.42 Cable Ø 13.2 Spring Bumper Assembly Ø 13.37 Patient Support Wheel

10 Electrical Troubleshooting Guidance

The following section details basic electrical troubleshooting. An important aspect of the RF system of this product is that the preamplifiers are inside the coil enclosure. When troubleshooting an open circuit or bias error in the cable tray, specific channels can be checked for continuity with a multi-meter. This will help to determine if fault is located in the coil or cable tray. The majority of errors can be addressed with coil replacement.

10.1 Electrical Troubleshooting

PROBLEM	**POSSIBLE CAUSES**	**CHECKS to be Performed**	**REPAIR or REPLACE**
Poor Image Quality (Signal Drops, Noise; Artefacts, low SNR etc.)	· Coil Configurations are incorrectly entered or detected after upgrade · Discrete Pins for preamp inside HYP plug are bent or loose · Coil or Cable Tray Failure (pre-amps or cables)	ü Run Manual Prescan and check for number of receivers (2, 4, or 8) ü Check that pins are not damaged inside plug ü Run Manual Prescan to check peaks on all receivers. (flat or no peak indicate failures)	Ø Re-enter/Repair Coil Configurations Ø Replace Cable tray Ø Replace Coil(s) or Cable Tray
Pre-Scan Failure Error: “Open Circuit”	· Coil not properly connected to cable tray · (HYP) Cable Plug not properly connected to scanner · Discrete and/or Coax Pins inside plug are bent · Intermittent/broken electrical connection	ü Ensure coil is properly connected ü Ensure HYP Cable connector is fully seated ü Center pin of coax and discrete pins are not damaged inside plug ü Check all cables for electrical continuity	Ø Connect coil Ø Connect Cable Plug Ø Replace coil or cable

PROBLEM	POSSIBLE CAUSES	CHECKS to be Performed	REPAIR or REPLACE
Pre-Scan Failure Error: "Signal too Small"	· Discrete Pins for preamp inside HYP plug are bent · (HYP) Cable Plug not properly connected to scanner- Valid coil set not connected to system · Medial Terminator Plug is not plugged into system for interventional procedures only · Discrete pins are bent/damaged on Coils or HYP Cable	ü Pins are not damaged inside plug ü Ensure HYP Cable Plug is fully seated ü Check physical coil connection (HYP Cable is connected) ü Center pin of coax and discrete pins are not damaged inside plug	Ø Replace Cable Tray Ø Connect to Scanner Ø Attach appropriate coil array
Vanguard is not recognized by the scanner	· Not all coils are properly connected to cable tray · Medial Terminator Plug is not plugged into cable tray for interventional procedures only · Valid coil set not connected to system · Discrete pins are bent/damaged on Coils or HYP Cable Plug	ü Ensure HYP Cable Plug is fully seated ü Check physical coil connection to the HYP Cable Tray ü Ensure the Medial Terminator Plug is connected to HYP cable tray ü Ensure valid coil sets are connected to cable tray ü Ensure discrete pins are not damaged inside HYP Plug or other side of Cable Tray where coils are connected ü Ensure discrete pins are not damaged inside coil connectors	Ø Repair Connection Ø Connect Coils properly Ø Connect Medial Plug Ø Replace coil or cable
Lighting system is inoperative	· Loose connectors · Damaged cables · Dead batteries · Blown fuse	ü Check all connectors ü Check all cables for damage ü Check batteries ü Check fuse	Ø Replace batteries Ø Replace fuse Ø Lighting Wiring Harness

10.2 1.5T RF System Coil Configurations

Imaging Configuration	***Connect to Left Cable Plug***	***Connect to Middle Receptacle***	***Connect to Right Cable Plug***	****Configuration to Select on Scanner***	***Channels Used by Coils***
Bilateral 8-channel	Lateral Array Coil Left	Medial Array Coil	Lateral Array Coil Right	8Brst BL SMI	**1 – 8**
Unilateral 8-channel Left	Lateral Array Coil Left	Medial Array Coil	Lateral Array Coil Right	8Brst Lt SMI	**1 – 4**
Unilateral 8-channel Right	Lateral Array Coil Left	Medial Array Coil	Lateral Array Coil Right	8Brst Rt SMI	**5 – 8**
Bilateral 4-channel	Single Loop Coil Left	Medial Array Coil	Single Loop Coil Right	4Brst BL Int SMI	**3 – 6**
Unilateral 2-channel Right	Medial Single Loop Coil Left	Medial Plug	Lateral Single Loop Coil Right	2Brst Int SMI	**3,6**
Unilateral 2-channel Left	Lateral Single Loop Coil Left	Medial Plug	Medial Single Loop Coil Right	2Brst Int SMI	**3,6**

*The “currently connected” option should correctly identify SMI Coils

10.3 3T RF System Coil Configurations

Imaging Configuration	***Connect to Left Cable Plug***	***Connect to Middle Receptacle***	***Connect to Right Cable Plug***	****Configuration to Select on Scanner***	***Channels Used by Coils***
Bilateral 8-channel	Lateral Array Coil Left	Medial Array Coil	Lateral Array Coil Right	8Brst BL SMI	**1 – 8**
Unilateral 8-channel Left	Lateral Array coil Left	Medial Array Coil	Lateral Array Coil Right	8Brst Lt SMI	**1 – 4**
Unilateral 8-channel Right	Lateral Array coil Left	Medial Array Coil	Lateral Array Coil Right	8Brst Rt SMI	**5 – 8**
Bilateral 4-channel	Single Loop Coil Left	Medial Array Coil	Single Loop Coil Right	4Brst BL Int SMI	**3 – 6**
Unilateral 2-channel (Right or Left)	Lateral Single Loop Coil Left	Medial Plug	Lateral Single Loop Coil Right	2Brst Int SMI	**3,6**

*The “currently connected” option should correctly identify SMI Coils

10.4 Final Verification and checks of Electrical Components/Assemblies

After the repair has been completed on any/all electrical components or assemblies (RF Components) perform a Quality Assurance Check as detailed in sections 7.15 and 7.16.

SMI ONLY	Use Appendix A to record the results obtained and return this documentation to appropriate Sentinelle Medical employees.

10.5 Follow-up

Upon completion of the PM, report condition of Vanguard to site manager and leave a completed calling card. If any wear or damage has been detected due to misuse, consider making site manager aware of what has occurred and suggest actions to take to prevent future damage or premature wear to system.

11 FRU List with GE Part Numbers

The following tables list all Field Replaceable Units (FRUs) available for order from Sentinelle or GEHC.

11.1 Stretcher FRUs

GEHC Part #	***Sentinelle Part #***	***Part Description***	***Section #***
5326119	SVC-00341	Drawer assembly	12.1, 13.1
5326120	SVC-00345	Bumper spring plunger assembly	12.1, 13.2
5326121	SVC-00348	Access door assembly	12.1, 13.3
5326123	SVC-00427	Lighting wiring harness	12.1, 13.5
5326124	SVC-00351	Upper body support assembly	12.1, 13.6
5326125	SVC-00380	Lower body support assembly	12.1, 13.7
5326126	SVC-00356	Travel stop pivot assembly	12.1, 13.8
5326127	SVC-00321	Swing assembly	12.1, 13.10
5326129	SVC-00323	Docking linkage assembly	12.1, 13.11
5326130	SVC-00324	Travel stop plunger assembly	12.1, 13.12
5326131	SVC-00428	Stopper assembly	12.1, 13.26
5326132	SVC-00332	No-undock hook assembly	12.1, 13.14
5326133	SVC-00429	Pedal block	12.1, 13.15
5326134	SVC-00430	Pedal arm assembly	12.1, 13.16
5326135	SVC-00369	Light enclosure	12.1, 13.4
5326136	SVC-00333	Shroud stretcher	12.1, 13.27
5326137	SVC-00335	Tension coupling docking linkage	12.1, 13.28
5326138	SVC-00358	Cable elbow 75 service assembly	12.1, 13.41
5326139	SVC-00336	Upper body support handle	12.1, 13.30
5326135	SVC-00369	Light Enclosure	12.1, 13.4
5326140	SVC-00378	Bumper	12.1, 13.31
5326141	SVC-00338	Spacer pin bumper	12.1, 13.32
5326142	SVC-00339	L-R centering bumper	12.1, 13.38
5326143	SVC-00340	Pedal spring	12.1, 13.34
5326144	SVC-00342	Undock spring	12.1, 13.35
5326145	SVC-00343	Bridge spring block	12.1, 13.33
5326146	SVC-00349	Bridge latch assembly	12.1, 13.24
5326148	SVC-00350	LBS cable elbow guard	12.1, 13.29
5326149	SVC-00352	Tray table (starboard parts + hinge)	12.1, 13.13
5326150	SVC-00353	Caster assembly	12.1, 13.39
5408273	SVC-00355	Caster boot	12.1, 13.39
5408274	SVC-00357	Mirror assembly (stretcher)	12.1
5408275	SVC-00359	Cable elbow 105 service assembly	12.1, 13.40
5408276	SVC-00360	Cable for stretcher	12.1
5408277	SVC-00361	Sentinelle table battery	12.1

11.2 Patient Support FRUs

GEHC Part #	***Sentinelle Part #***	***Part Description***	***Section #***
N/A	5000268-11	Rectangular Imaging phantom	-
5326151	SVC-00362	Catchment latch assembly	6.1.1, 12.2, 13.17
5326152	SVC-00363	Catchment assembly	12.2, 13.36
5326153	SVC-00387	Compression slider	6.1.1, 12.2, 13.18
5326154	SVC-00364	Emergency release handle assembly	12.2, 13.19
5326155	SVC-00381	LPCA mechanism assembly	12.2, 13.20
5326156	SVC-00365	Emergency release spring assembly	12.2, 13.21
5326157	SVC-00366	L-R centering wheel assembly	
5326159	SVC-00368	Wheel (bearing) patient support	12.2, 13.37
5326168	SVC-00432	Patient support for GE Signa	6.1.1, 12.2
5399108	SVC-00389	Compression frame assembly left	6.1.1, 12.2
5399109	SVC-00390	Compression frame assembly right	6.1.1, 12.2
5408237	SVC-00388	Headrest mirror assembly	12.2
5408239	SVC-00307	Compression plate	6.1.1, 12.2
5408279	SVC-00306	Sternum support	6.1.1, 12.2
5408280	SVC-00301	Portable head rest assembly (with mirror)	6.1.1, 12.2
5408283	SVC-00367	Compression rail assembly	6.1.1, 12.2, 13.18
5408284	SVC-00302	Shoulder bridge	6.1.1, 12.2
5419434-9	SVC-00401	GE spherical phantom positioner	6.1.1, 12.2

11.3 1.5T RF System FRUs

GEHC Part #	***Sentinelle Part #***	***Part Description***	***Section #***
5326160	4000012-11	Lateral single loop coil right VG	6.1.2
5326161	4000012-12	Lateral single loop coil left VG	6.1.2
5326162	4000013-11	Medial single loop coil right VG	6.1.2
5326163	4000013-12	Medial single loop coil left VG	6.1.2
5326164	4000019-11	Lateral array coil right VG	6.1.2
5326165	4000019-12	Lateral array coil left VG	6.1.2
5326166	4000006-11	Medial array coil VG	6.1.2
5326167	4000058-11	Medial plug	6.1.2
5326169	4000015-11	Vanguard HYP cable system	6.1.2, 13.23

11.4 3T RF System FRUs

GEHC Part #	***Sentinelle Part #***	***Part Description***	***Section #***
5419434-3	4000173-11	Coil assembly single loop right VG 3T	6.1.3
5419434-2	4000180-11	Coil assembly single loop left VG 3T	6.1.3
5419434-4	4000182-11	Coil assembly lateral array right VG 3T	6.1.3
5419434-5	4000183-11	Coil assembly lateral array left VG 3T	6.1.3
5419434-6	4000172-11	Coil assembly medial array VG 3T	6.1.3
5419434-7	4000184-11	Assembly medial plug 3T	6.1.3
5419434-8	4000181-11	Vanguard HYP cable system 3T	6.1.3, 13.24

11.5 Padding Kit FRUs

GEHC Part #	***Sentinelle Part #***	***Part Description***	***Section #***
5408293	SVC-00304	Pad arm support	6.1.3
5408294	SVC-00305	Pad body	6.1.3
5408287	SVC-00308	Pad headrest thin	6.1.3
5408288	SVC-00309	Pad wedge	6.1.3
5408292	SVC-00440	Padding kit stretcher (7 Pads)	-
Contained in 5408292 Kit	SVC-00311	Pad shoulder wedge right	6.1.3
Contained in 5408292 Kit	SVC-00312	Pad shoulder wedge left	6.1.3
Contained in 5408292 Kit	SVC-00317	Pad medial array VG	6.1.3
Contained in 5408292 Kit	SVC-00315	Pad PS guard right	6.1.3
Contained in 5408292 Kit	SVC-00316	Pad PS guard left	6.1.3
Contained in 5408292 Kit	SVC-00314	Pad body wedge	6.1.3
Contained in 5408292 Kit	SVC-00313	Pad foot rest	6.1.3

GEHC ONLY	Pads and straps are designed as customer consumable components. They are not intended to be covered by any GE service contract. If pads and straps are replaced they should be billed to the customer.

12 FRU and Fastener Location Diagrams

This section shows the locations of the FRUs in the overall table layout and labels the fasteners used to mount the FRUs.

The following legend can be used when referring to the type of fasteners in the tables and diagrams below.

Legend

SS	Stainless Steel
SHCS	Square Head Cap Screw
FHCS	Flat Head Cap Screw
BHCS	Button Head Cap Screw
HHMS	Hex-Head Machine Screw
RHMS	Round Head Machine Screw
PHMS	Pan Head Machine Screw
FHMS	Flat Head Machine Screw

12.1 Stretcher FRUs

Item	GEHC FRU #	SMI FRU #	FRU Description	Fasteners	
				Type	Qty
(A)	5326149	SVC-00352	Tray table	(1) SHCS 1/4 – 20 x 3/8L 316 SS	4
(B)	5326147	SVC-00379	Latch tab ass'y	(2) FHCS 10 – 24 x 3/4L 316 SS	2
(C)	5326148	SVC-00350	LBS cable elbow guard	(3) BHCS 10 – 32 x 3/8L 316 SS	4
(D)	5326150	SVC-00353	Caster	(4) HEX NUT 1 – 18L 316 SS (24) BHCS 1/4-20 x 1/2L 316 SS (Loosen, Do Not Remove)	4 8
(E)	5326127	SVC-00321	Swing ass'y	(5) SHCS 5/16 – 18 x 3/4L 316 SS (17) SHCS 5/16 – 18 x 3/4L 316 SS	2 2
(F)	5326143	SVC-00340	Pedal spring	(6) SHCS 5/16 – 18 x 1/2L 316 SS (25) SHCS 5/16 – 18 x 3/4L 316 SS	2 2
(G)	5326129	SVC-00323	Docking linkage ass'y	(7) 10 – 24 x 1 1/2L 316 SS SHCS (16) 5/16 – 18 x 3/8D x 5/8L 316 SS Shoulder Bolt	4 1
(H)	5326132	SVC-00332	No-undock hook ass'y	(8) Screw 1/4 – 20 x 5/8L 316 SS	2
(I)	5326130	SVC-00324	Travel stop plunger ass'y	(9) SHCS 1/4 – 20 x 1/2L 316 SS	4
(J)	5326135	SVC-00369	Light enclosure	(10) FHMS 8 – 32 x 1/2L Brass	8
(K)	5326134	SVC-00430	Pedal arm ass'y	(11) BHCS 10 – 32 x 3/8L 316 SS	1
(L)	5326133	SVC-00429	Pedal block	(12) Screw 5/16 – 18 x 3/4L	4
(M)	5326140	SVC-00378	Bumper stretcher	(13) SHCS 5/16 – 18 x 1 1/3L 316 SS (18) SHCS 5/16 – 18 x 1.5L 316 SS	4 2
(N)	5326145	SVC-00343	Bridge spring block	(14) SHCS 1/4 – 20 x 2 1/4L 316 SS	2
(O)	5326139	SVC-00336	UBS handle	(15) Screw 5/16 – 18 x 2L	2
(P)	5326137	SVC-00335	Tension coupling docking linkage	(7) SHCS 10 – 24 x 1 1/2L 316 SS (16) Shoulder Bolt 5/16 – 18 x 3/8D x 5/8L 316 SS (17) SHCS 5/16 – 18 x 3/4L 316 SS	4 1 2
(Q)	5326142	SVC-00339	L-R centering bumper	(18) SHCS 5/16 – 18 x 1.5L 316 SS	2
(R)	5326125	SVC-00380	Lower body support ass'y	(19) HHCS 3/8 – 16 x 1L 316 SS	4
(S)	5326126	SVC-00356	Travel stop pivot ass'y	(20) SHCS 10-32 x 1/2L 316 SS	2
(T)	5326146	SVC-00349	Bridge latch ass'y	(21) SHCS 10-24 x 1 1/2L 316 SS	4
(U)	5326124	SVC-00351	Upper body support ass'y	(20) SHCS 10-32 x 1/2L 316 SS (21) SHCS 10-24 x 1 1/2L 316 SS (22) HHCS 3/8 – 16 x 1L 316 SS	2 4 4
(V)	5326121	SVC-00348	Access door ass'y	(23) Screw Captive Panel 10-32 x .563L	1
(W)	5326131	SVC-00428	Stopper ass'y	(26) FHCS 10-24 x 3/4L 316 SS	4
(X)	5326119	SVC-00341	Drawer ass'y	(27) FHCS 8-32 x 1L SS316	6
(Y)	5326136	SVC-00333	Shroud stretcher	(28) SHCS ¼-20x3/4" 316SS (29) 1/4" 316SS Flat Washer	4 4
(Z)	5326120	SVC-00345	Bumper spring plunger ass'y	-	
(AA)	5326144	SVC-00342	Undock spring	(6) SHCS 5/16 – 18 x 1/2L 316 SS (25) SHCS 5/16 – 18 x 3/4L 316 SS	2 2
(AB)	5326138	SVC-00358	Cable elbow 75 service ass'y	-	-
(AC)	5326123	SVC-00427	Lighting wiring harness (not shown)	-	-
(AD)	5408274	SVC-00357	Mirror ass'y	-	-
(AE)	5326141	SVC-00338	Spacer pin bumper	-	-
(AF)	5408273	SVC-00355	Caster boot	-	-
(AG)	5408276	SVC-00360	Cable for stretcher (not shown)	-	-
(AH)	5408275	SVC-00359	Cable elbow 105 service ass'y	-	-
(AI)	5408277	SVC-00361	Sentinelle table battery (not shown)	-	-

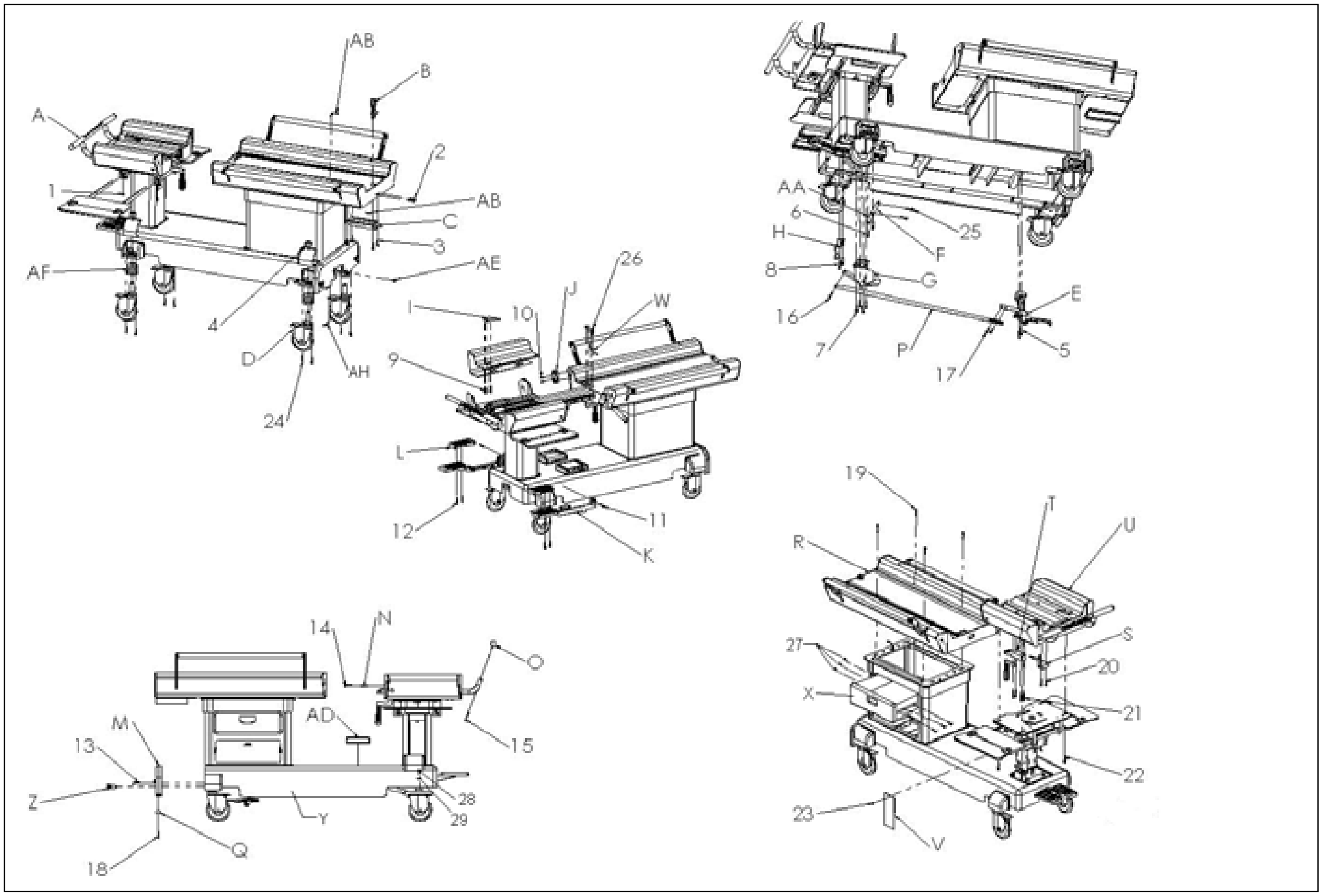


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Figure 69

12.2 Patient Support FRUs

Item	GEHC FRU #	SMI FRU #	FRU Description	Fasteners	
				Type	Qty
(A)	5326155	SVC-00381	LPCA mechanism assembly	(1) RHMS 10-32 x 1L Brass	2
(B)	5326156	SVC-00365	Emergency release spring assembly	(2) ER Spring Pin	1
(C)	5326152	SVC-00363	Catchment assembly	-	-
(D)	5326151	SVC-00362	Catchment latch assembly	(3) RHMS 10-32 x 1L Brass	4
(E)	5326154	SVC-00364	Emergency release handle assembly	(4) 5/16–18 x 1.5L Brass 260/270 (6) PHMS Slot 8-32 x 1/2L Brass	2 1
(F)	5408283	SVC-00367	Compression rail assembly	(5) FHMS 1/4 – 20 x 2.5L Brass	5
(G)	5326159	SVC-00368	Wheel (bearing) patient support	(7) RHMS 10-32 x 1L Brass	8
(H)	5326157	SVC-00366	L-R centering wheel ass'y	(6) BHCS 10-32 1.25L	16
(I)	5326168	SVC-00432	Patient support ass'y for GE Signa	-	-
(J)	5326153	SVC-00387	Compression slider (not shown)	-	-
(K)	5408280	SVC-00301	Portable head rest ass'y (not shown)	-	-
(L)	5399108	4000049-11	Compression frame ass'y left (not shown)	-	-
(M)	5399109	4000050-11	Compression frame ass'y right (not shown)	-	-
(N)	5408284	SVC-00302	Shoulder bridge (not shown)	-	-
(O)	5408239	SVC-00307	Compression plate (not shown)	-	-
(p)	5407765-9	SVC-00329	Spherical Phantom Positioner (1.5T not shown)	-	-
(Q)	5407764-9	SVC-00401	Spherical Phantom Positioner (3T not shown)	-	-
(R)	N/A	5000268-11	Rectangular Imaging Phantom (1.5T only; not shown)	-	-
(S)	5408237	SVC-00337	Headrest Mirror Assembly (not shown)	-	-
(T)	5408279	SVC-00306	Sternum Support (not shown)	-	-

FASTENERS:
1. 2 x RHMS 10-32x1L BRS
2. 1 x ER SPRING PIN
3. 4 x RHMS 10-32x1L BRS
4. 2 x 5/16-18x1.5L BRS 260/270
5. 5 x FHMS 1/4-20x2.5L BRS
6. 16 x BHCS 10-32x1.25L
7. 1 x PHMS SLOT 8/32 x 1/2L BRS
8. 8 x RHMS 10-32 x 1L BRS

REPLACEMENT PARTS:
1. 4000040-11 – LPCA MECHANISM
2. 4000104-11 – EMERGENCY RELEASE SPRING ASS'Y
3. 4000500-51 – CATCHMENT ASS'Y
4. 4000062-11 – CATCHMENT LATCH ASS'Y
5. 4000031-11 – EMERGENCY RELEASE HANDLE ASS'Y
6. 4000068-11 – COMPRESSION RAIL ASS'Y
7. AR10BDR-GL-1/2 – WHEEL (DOUBLE ROW BEARING)
8. 4000017-11 – L-R CENTERING WHEEL ASSEMBLY
9. 4000008-11 – PATIENT SUPPORT ASSEMBLY FOR GE SIGNA

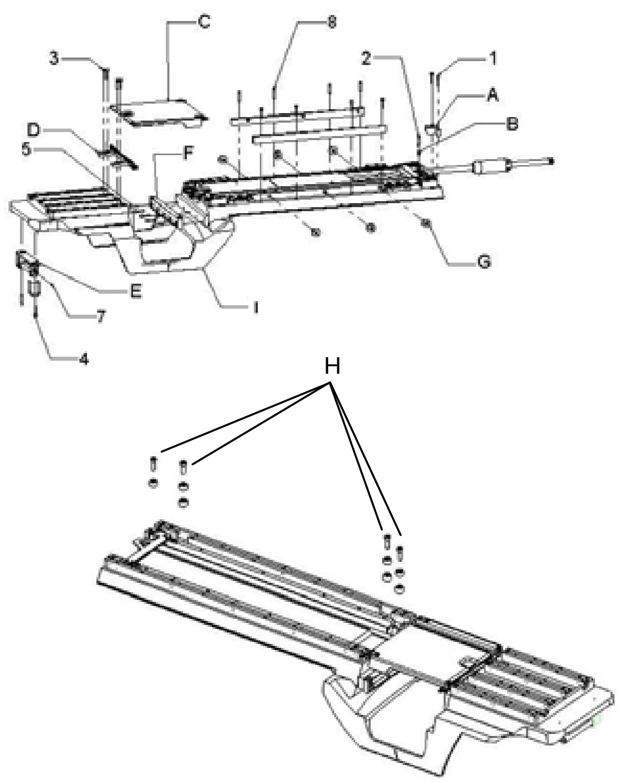


Figure 70

12.3 1.5T RF System FRUs

Item	GEHC FRU #	SMI FRU #	FRU Description	Fasteners	
				Type	Qty
(A)	5326169	4000015-11	Vanguard HYP Cable System	(1) FHMS Slot 1/4-20 x 1.5L Brass	2
(B)	5326161	4000012-12	Lateral Single Loop Coil Left VG (not shown)	-	-
(C)	5326160	4000012-11	Lateral Single Loop Coil Right VG (not shown)	-	-
(D)	5326162	4000013-11	Medial Single Loop Coil Right VG	-	-
(E)	5326163	4000013-12	Medial Single Loop Coil Left VG	-	-
(F)	5326164	4000019-11	Lateral Array Coil Right VG	-	-
(G)	5326165	4000019-12	Lateral Array Coil Left VG	-	-
(H)	5326166	4000006-11	Medial Array Coil VG	-	-
(I)	5326167	4000058-11	Medial Plug	-	-

FASTENERS:
1. 2 x FHMS SLOT 1/4-20x1.5L BRS

REPLACEMENT PARTS:
A. 4000015-11 – VANGUARD HYP CABLE SYSTEM

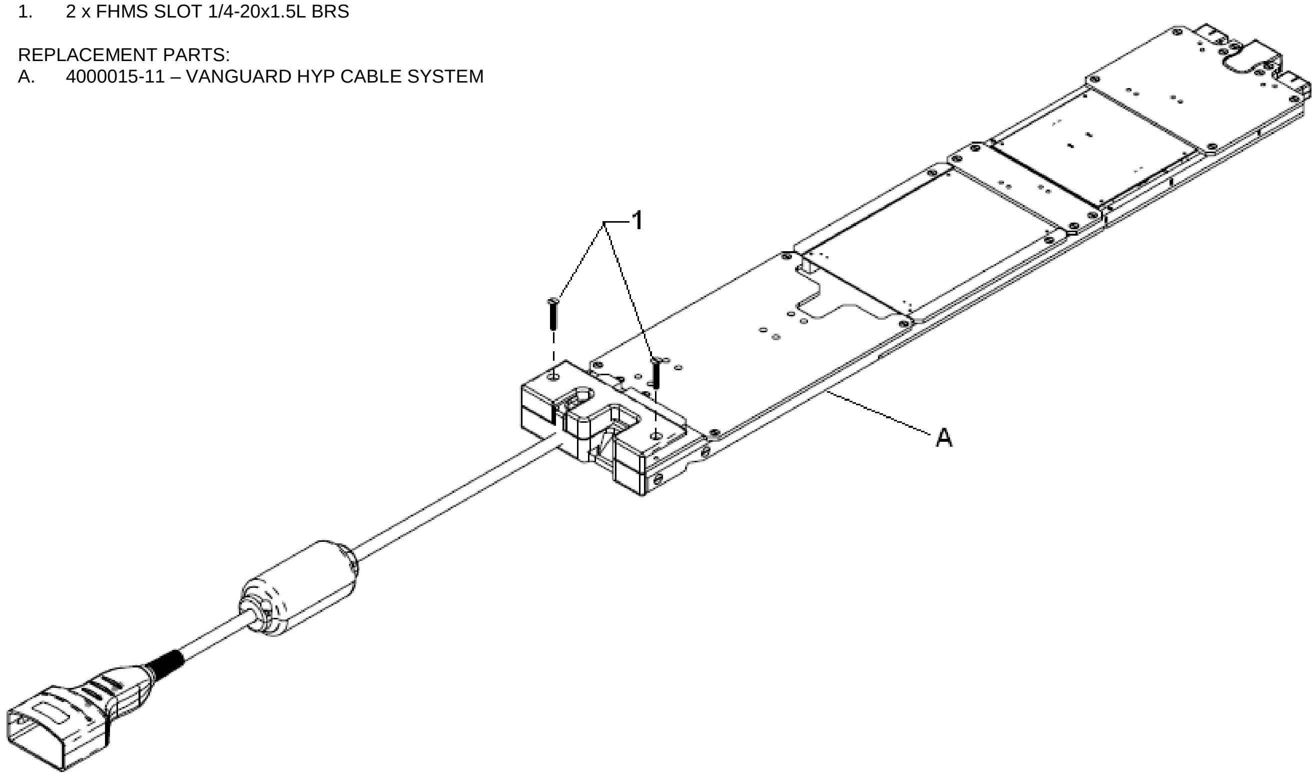


Figure 71

12.4 3T RF System FRUs

Item	GEHC FRU #	SMI FRU #	FRU Description	Fasteners	
				Type	Qty
(A)	5419434-8	4000181-11	Vanguard HYP Cable System 3T	(1) FHMS Slot 1/4-20 x 1.5L Brass	2
(B)	5419434-6	4000172-11	Coil Assy Medial Array VG 3T (not shown)	-	-
(C)	5419434-3	4000173-11	Coil Assy Single Loop Right VG 3T (not shown)	-	-
(D)	5419434-2	4000180-11	Coil Assy Single Loop Left VG 3T (not shown)	-	-
(E)	5419434-4	4000182-11	Coil Assy Lateral Array Right VG 3T (not shown)	-	-
(F)	5419434-5	4000183-11	Coil Assy Lateral Array Left VG 3T (not shown)	-	-
(G)	5419434-7	4000184-11	Assy Medial Plug 3T (not shown)	-	-

FASTENERS:
1. 2 x FHMS SLOT 1/4-20x1.5L BRS

REPLACEMENT PARTS:
A. 4000181-11 – VANGUARD HYP CABLE SYSTEM 3T

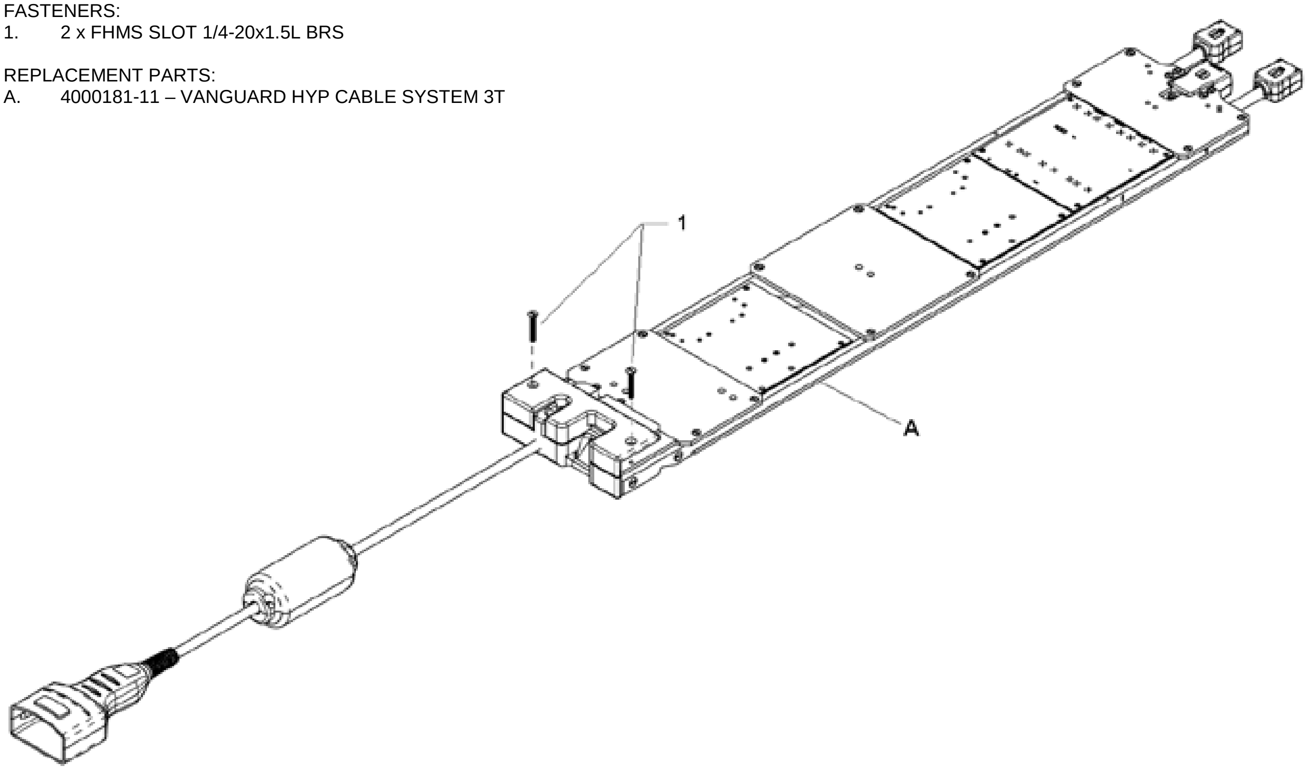


Figure 72

13 Replacing Mechanical Parts and Assemblies

Most parts and assembles require disassembly, reassembly, adjustment and testing. With all parts and assemblies, unless otherwise noted, the procedure for re-assembly is the reverse order of the steps described for disassembly. Parts and assemblies which are simple to replace will have no steps described.

⚠ WARNING **If any BBPs are present, stop work and notify customer that the Vanguard must be cleaned prior to performing service.**

⚠ WARNING **The battery is slightly-ferrous and must be removed/replaced outside of the scan room. Personal Injury may result if the battery is removed in close proximity to the magnet. Once the batteries have been replaced, re-install the retaining plate by tightening the screw and secure the battery door with its screw.**

⚠ CAUTION	GEHC ONLY	***Note: The Step Stool is not to be used as a service tool.***

13.1 Drawer assembly

SMI Part #: SVC-00341
GEHC Part #: 5326119
Time required: 40 minutes
Tools required: 1/8" Non-magnetic Allen wrench
Personnel required: 1

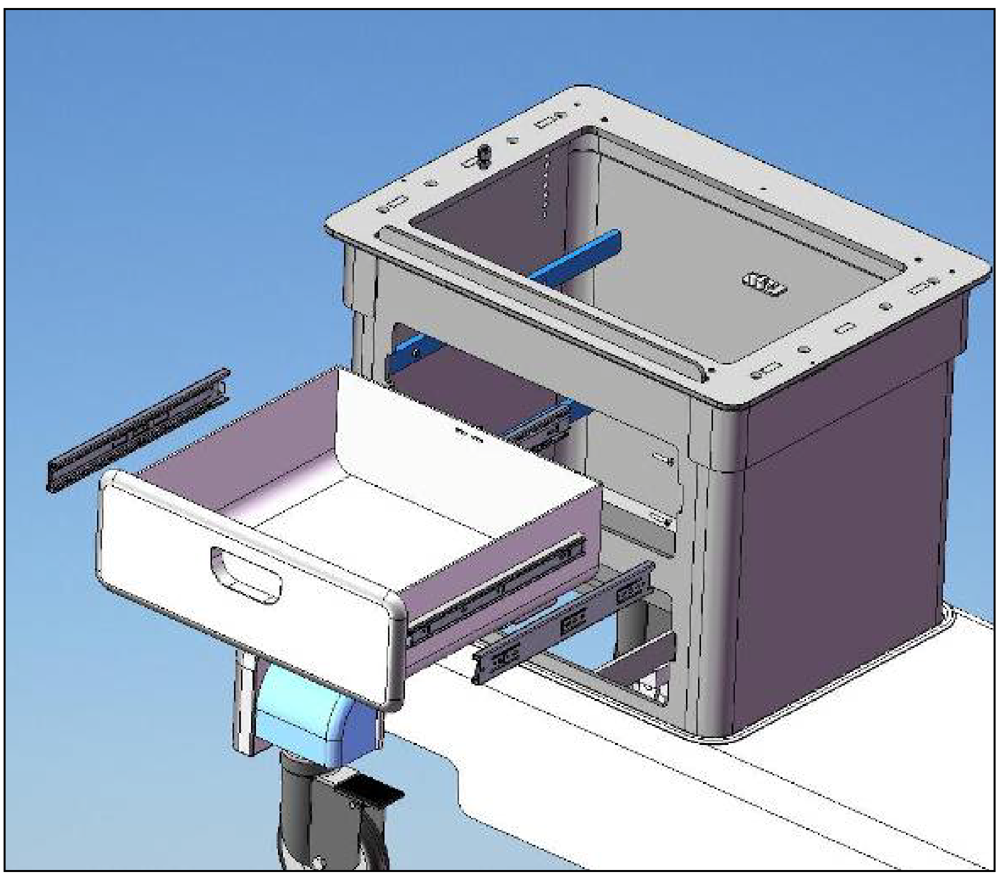


Figure 73

1. Remove drawer from RSA.
2. Remove drawer sliders from RSA.
3. Fasten new drawer sliders to RSA.
4. Insert new drawer to RSA.

Finalization Steps

i. Test drawer for smooth motion by opening and closing it 5 times.

13.2 Bumper Spring Plunger Assembly

SMI Part #: SVC-00345
GEHC Part #: 5326120
Time required: 50 minutes
Tools required: Non-magnetic adjustable wrench
Personnel required: 1

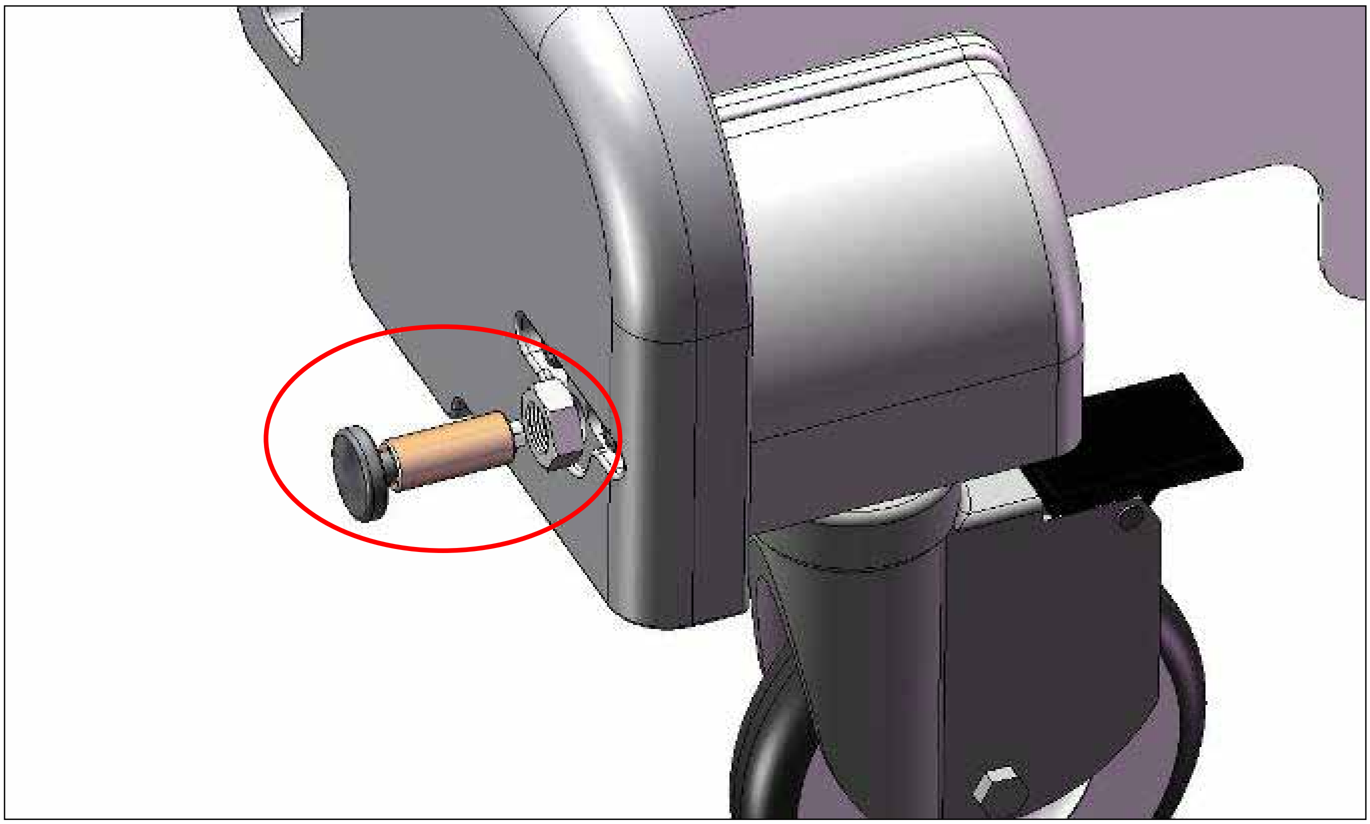


Figure 74

1. Note position of Spring Bumper Assembly from bumper face.
2. Unscrew Spring Bumper Assembly from Bumper.
3. Clean hole and let dry. Alcohol wipes suggested for cleaning.
4. Screw in new Spacer Pin Bumper to the correct amount to preserve the setting and tighten white plastic nut.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times.

13.3 Access Door Assembly

SMI Part #: SVC-00348
GEHC Part #: 5326121
Time required: 50 minutes
Tools required: 5/32" non-magnetic Allen wrench
Personnel required: 1

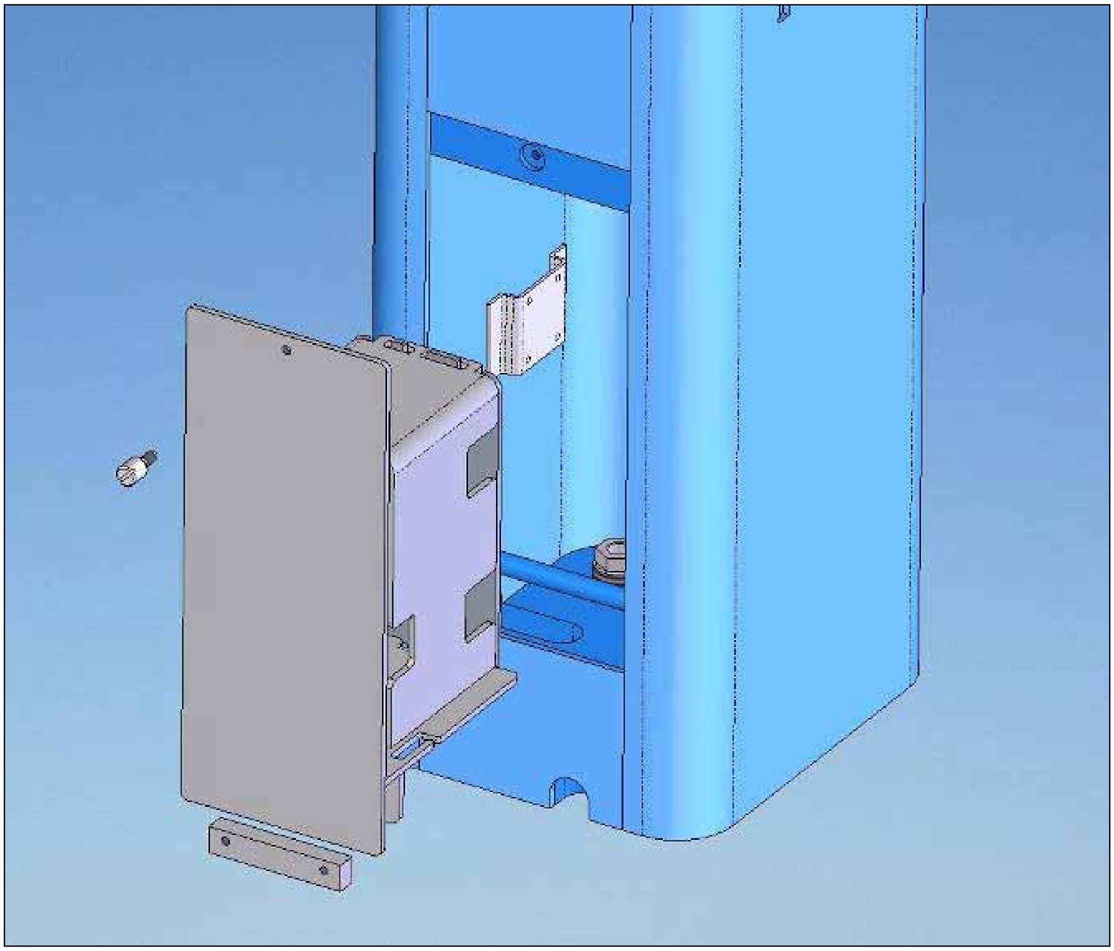


Figure 75

1. Remove fuse from Battery PCB.
2. Unplug Access Door Assembly from lighting system.
3. Unscrew block from bottom of enclosure assembly.

4. Insert new assembly on to mounting rod weldment and fasten block.
5. Plug into lighting system and insert fuse.

Finalization Steps

i. Turn the lighting system on and off 3 times leaving it on for 30 seconds each time.
ii. Repeat step i) for the lighting system on both sides of the stretcher.

13.4 Light Enclosure

SMI Part #: SVC-00369
GEHC Part #: 5326135
Time required: 40 minutes
Tools required: Non-magnetic small slot screwdriver
Personnel required: 1

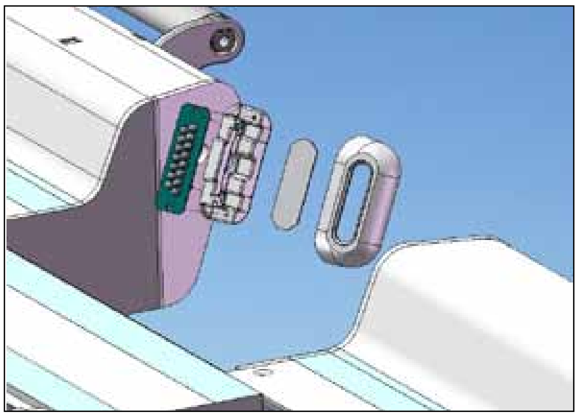


Figure 76

1. Remove Light Enclosure cover.

Figure 77

2. Remove Light Enclosure by unplugging the connector plug.

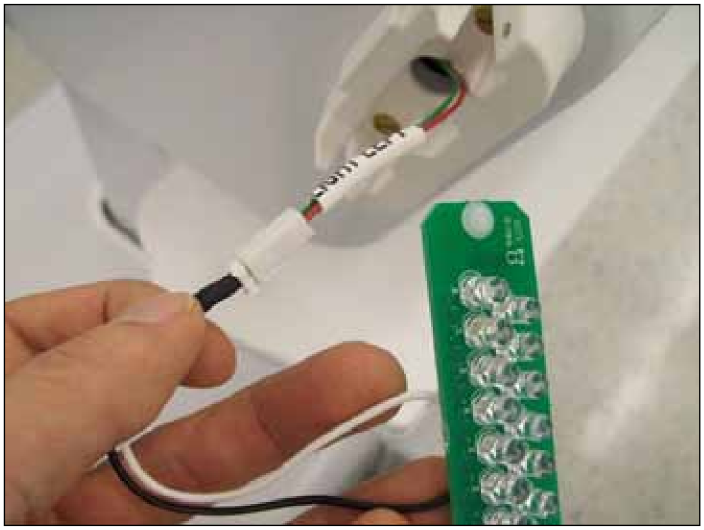


Figure 78

3. Attach connector of new Light Enclosure. Place the Light Enclosure into the slot feature and tap it slightly (with a rubber tool) to ensure that it is fixed.

Figure 79

4. Attach new Light Enclosure Cover and mount with two plastic white screws.

Figure 80

Finalization Steps

i. Turn the lighting system on and off 3 times leaving it on for 30 seconds each time.

ii. Repeat step i) for the lighting system on both sides of the stretcher.

13.5 Lighting Wiring Harness

SMI Part #: SVC-00427
GEHC Part #: 5326123
Time required: 100 minutes
Tools required: None
Personnel required: 1

1. Open Access Door Assembly (Figure 81 and Figure 82) to gain access to Lighting Wiring Harness.

Figure 81

Figure 82

2. Disconnect Lighting Wiring Harness from Light Enclosures in UBS and LBS.
3. Remove Lighting Wiring Harness from mounting tabs in FSA and RSA.
4. Remove Lighting Wiring Harness from conduit below Shroud.
5. Detach distribution PCB and remove Lighting Wiring Harness from FSA.
6. Attach distribution PCB and new Lighting Wiring Harness to FSA.
7. Route through conduit and attach new harness to tabs in FSA and RSA.
8. Connect Lighting Wiring Harness to Light Enclosures in UBS and LBS.

Finalization Steps

i. Turn the lighting system on and off 3 times leaving it on for 30 seconds each time.

ii. Repeat step i) for the lighting system on both sides of the stretcher.

13.6 Upper Body Support Assembly

SMI Part #: SVC-00351
GEHC Part #: 5326124
Time required: 90 minutes
Tools required: 9/16” non-magnetic wrench, 5/32” non-magnetic Allen key
Non-magnetic pliers, needle nose, 6";
GE Only: Non-magnetic pliers, side cutter, titanium
Note: This FRU contains one cable and two cable ends
Personnel required: 2

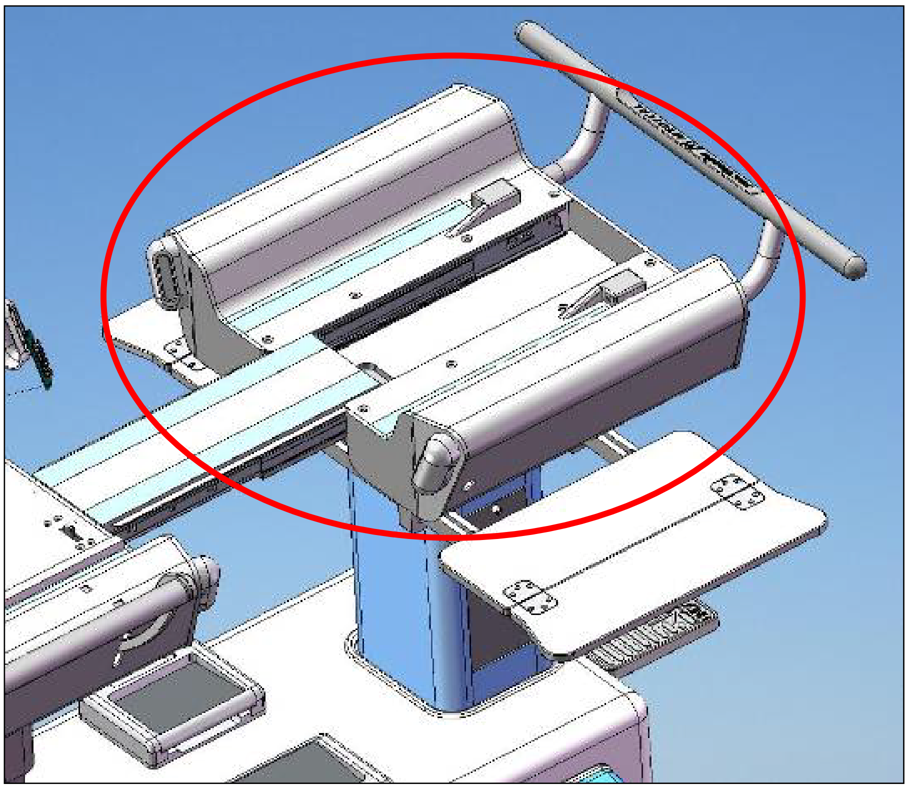


Figure 83

1. Remove the Patient Support from the foot end by following the steps outlined in *Section 7.1 Initial Alignment, steps 3 to 9.*

HEAVY ITEM

WARNING

Use proper heavy lifting techniques and be careful when lifting the patient support. Ensure that two people are lifting, because the patient support weighs approximately 55lbs.

2. Detach Travel Stop Cable from No-Undock Hook.
3. Unplug Lighting Wiring Harness from Light Enclosures. Refer to Section 13.5 for instructions on how to gain access to the Lighting Wiring Harness. Refer to *Section 13.4 Light Enclosure* for instructions on how to replace the Light Enclosure.
4. Remove four mounting bolts from inside of UBS.
5. Carefully lift off the UBS.

HEAVY ITEM

WARNING

The UBS weighs over 35lbs. Use proper heavy lifting techniques when lifting to prevent back injury. At least two people are required when lifting.

6. Lower new LBS into position.
7. Pass Travel Stop Cable through FSA and let hang freely.
8. With bridge Guide Block aligned to LBS hole, fasten UBS with four mounting bolts from inside of UBS.
9. Attach Travel Stop Cable to No-Undock Hook and set hook height. The side of the no-undock hook should be ¼” (6.35mm) away from the Undock pedal. Once this has been set, tighten the screw to fix the cable, cut the cable if it is too long and crimp on a new cable end using non-magnetic side cutter pliers.
10. Plug Lighting Wiring Harness into Light Enclosures.
11. Replace the patient support by following the steps outlined in *Section 7.1 Initial Alignment, steps 2 to 9* in reverse.

PINCH HAZARD

CAUTION

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

Finalization Steps

i. Dock the stretcher onto the scanner.
ii. Advance the patient support to scan.
iii. Return the patient support into the home position.
iv. Undock the stretcher from the scanner.
v. Repeat finalization steps five times.

13.7 Lower Body Support Assembly

SMI Part #: SVC-00380
GEHC Part #: 5326125
Time required: 90 minutes
Tools required: 9/16” non-magnetic wrench
Non-magnetic pliers, needle nose, 6"
GE Only: Non-magnetic pliers, side cutter, titanium
Note: This FRU contains one cable and two cable ends
Personnel required: 2

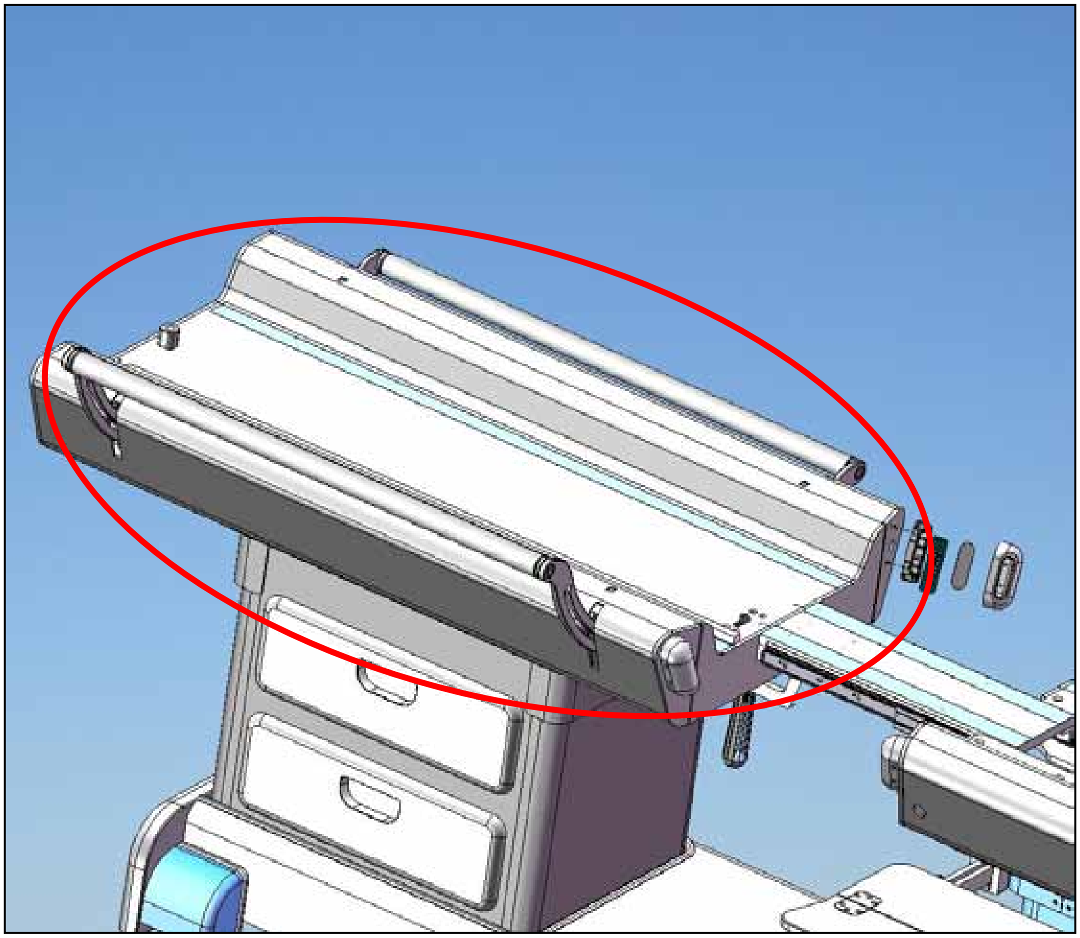


Figure 84

1. Remove the Patient Support from the foot end by following the steps outlined in *Section 7.1 Initial Alignment, steps 3 to 9.*

HEAVY ITEM

WARNING

Use proper heavy lifting techniques and be careful when lifting the patient support. Ensure that two people are lifting, because the patient support weighs approximately 55lbs.

2. Remove drawers from RSA.
3. Remove the cable end from the Latch Tab Cable and detach the cable from the Swing Assembly (Figure 85).
4. Unplug Lighting Wiring Harness from Light Enclosures. Refer to Section 13.5 for instructions on how to gain access to the Lighting Wiring Harness. Refer to Section 13.4 for instructions on how to replace the Light Enclosure.
5. Mark position of LBS on base and remove four mounting bolts from inside of LBS. Ensure that the four bolts removed are not those used for coarse height adjustment. Refer to Figure 47 for the location of the correct bolts.
6. Carefully lift off LBS.

HEAVY ITEM

WARNING

The LBS weighs over 65lbs. Use proper heavy lifting techniques when lifting to prevent back injury. At least two people are required when lifting.

7. Lower new LBS into position.
8. Insert Latch Tab Cable through cable elbows (Figure 85). An old or a new cable can be used depending on the condition of the cable.

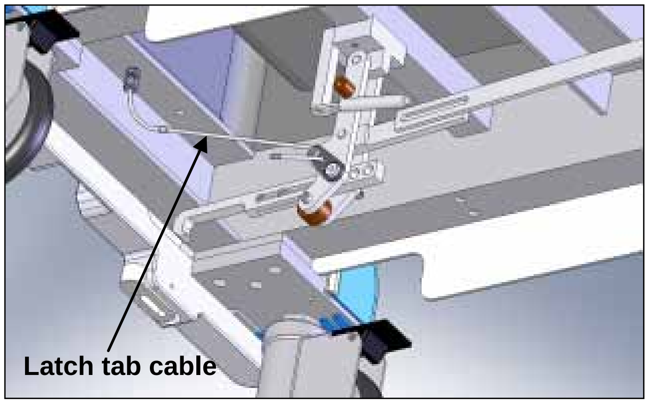


Figure 85

9. After the cable is through the cable elbows, cut the cable if it is too long using non-magnetic side cutter pliers.
10. Terminate the new cable with a new cable end using non-magnetic needle nose pliers.
11. Fasten LBS with four mounting bolts from inside of LBS in previously marked position.
12. Plug Lighting Wiring Harness into Light Enclosures.
13. Attach Latch Tab Cable and set tab height. Refer to section 7.10.
14. Replace drawers in RSA.
15. Replace the patient support by following the steps outlined in *Section 7.1 Initial Alignment, steps 3 to 9* in reverse.

PINCH HAZARD

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

Finalization Steps

i. Dock the stretcher onto the scanner.
ii. Advance the patient support to scan.
iii. Return the patient support into the home position.
iv. Undock the stretcher from the scanner.
v. Repeat finalization steps five times.

13.8 Travel Stop Pivot Assembly

SMI Part #: SVC-00356
GEHC Part #: 5326126
Time required: 50 minutes
Tools required: 3/16” non-magnetic Allen wrench,
Non-magnetic pliers, needle nose, 6"
GE Only: Non-magnetic pliers, side cutter, titanium
Note: This FRU contains one cable and two cable ends
Personnel required: 1

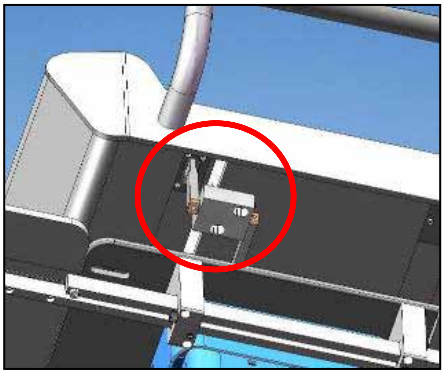


Figure 86

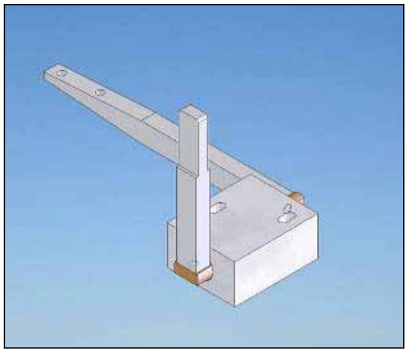


Figure 87

1. Detach Latch Travel Stop Cable from no undock hook.
2. Remove 2 mounting screws.
3. Carefully remove Travel Stop Pivot Assembly.
4. Thread new cable into pivot arm.
5. Mount Travel Stop Pivot Assembly and pass Stop Cable through FSA and let hang freely.
6. Fasten Travel Stop Pivot Assembly with 2 screws and align rocker arm with Travel Stop Plunger Assembly
7. Attach Travel Stop Cable to No-Undock Hook and set hook height. The side of the no-undock hook should be ¼” (6.35mm) away from the Undock pedal. Once this has been set, tighten the screw to fix the cable and crimp on a new cable end using non-magnetic needle nose pliers.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times.

13.9 Bridge Latch Assembly

SMI Part #: SVC-00349
GEHC Part #: 5326146
Time required: 20 minutes
Tools required: 3/16" non-magnetic Allen wrench
Personnel required: 1

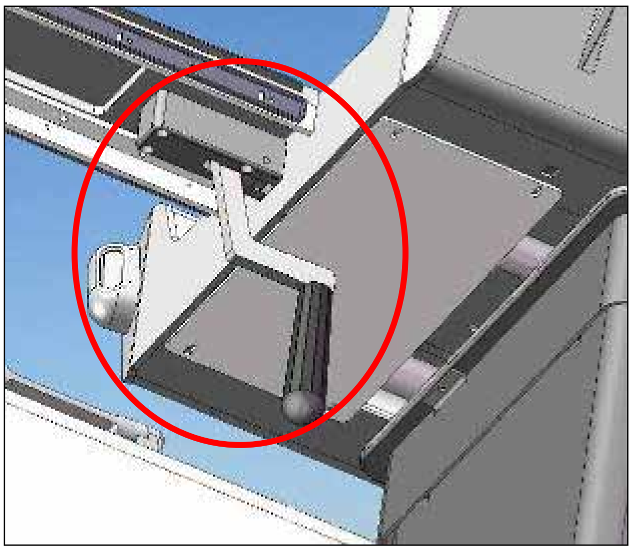


Figure 88

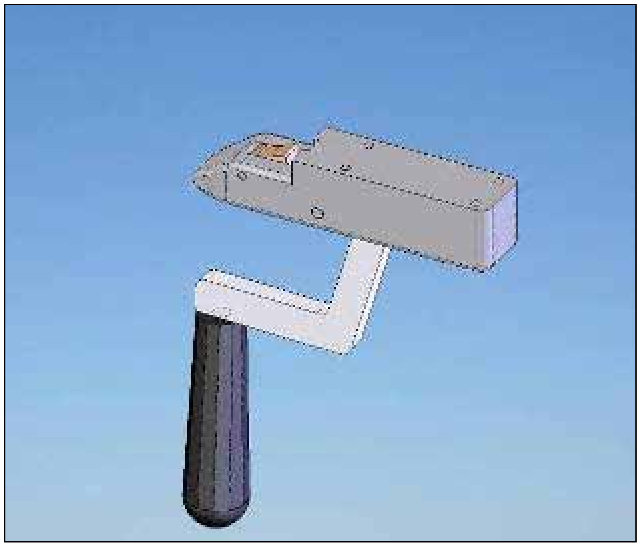


Figure 89

1. Remove Bridge Latch Assembly by removing four mounting screws.
2. Replace Bridge Latch Assembly and fasten with four mounting screws.

Finalization Steps

i. Open and close the bridge, ensuring that the motion is smooth and there is no noise while sliding.

ii. Ensure that the bridge latches on when closed and that it cannot be moved unless the handle is pressed.

iii. Repeat steps i) and ii) ten times.

13.10 Swing Assembly

SMI Part #: SVC-00321
GEHC Part #: 5326127
Time required: 70 minutes
Tools required: ¼” non-magnetic Allen wrench
Non-magnetic pliers, needle nose, 6"
GE Only: Non-magnetic pliers, side cutter, titanium
Note: This FRU contains one cable and two cable ends
Personnel required: 1

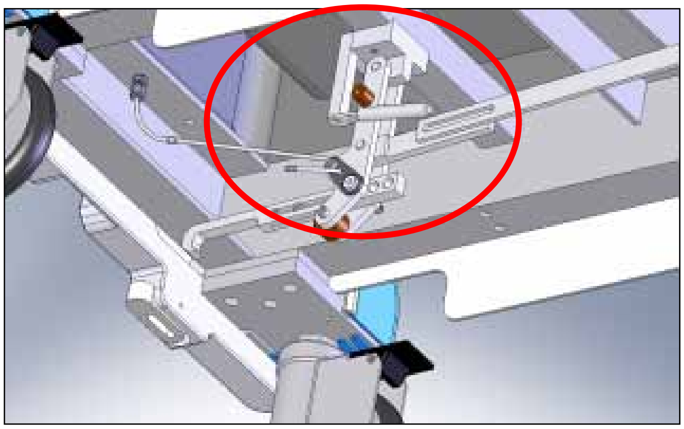


Figure 90

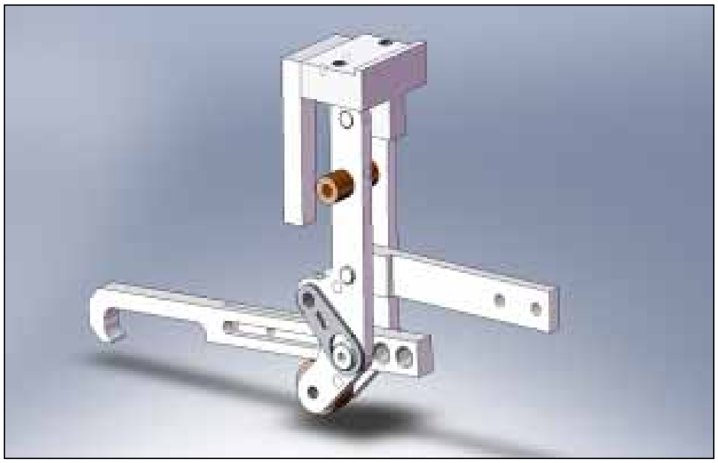


Figure 91

1. Remove drawers from RSA.
2. Remove cable end from Latch Tab Cable and detach the cable from the Swing Assembly.
3. Detach Tension Coupling Docking Linkage from Swing Assembly by removing two bolts.
4. Detach Swing Assembly from Frame by removing mounting bolts.
5. Mount Swing Assembly to Frame with mounting bolts.
6. Attach Tension Coupling Docking Linkage to Swing Assembly by inserting two bolts while mounting Undock Spring.

PINCH
HAZARD

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

7. Attach Latch Tab Cable to Swing Assembly and refer to 7.10 to set Latch Tab Height. After the cable is adjusted, cut the cable if it is too long and terminate it with a new cable end using non-magnetic side cutter pliers.
8. Replace Drawers.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times.

13.11 Docking Linkage Assembly

SMI Part #:	SVC-00323
GEHC Part #:	5326129
Note:	Order consumables FRU kit (P/N 4001158-11, GEHC P/N 5408278) for Loctite 290
Time required:	90 minutes
Tools required:	¼" non-magnetic Allen wrench
Personnel required:	1

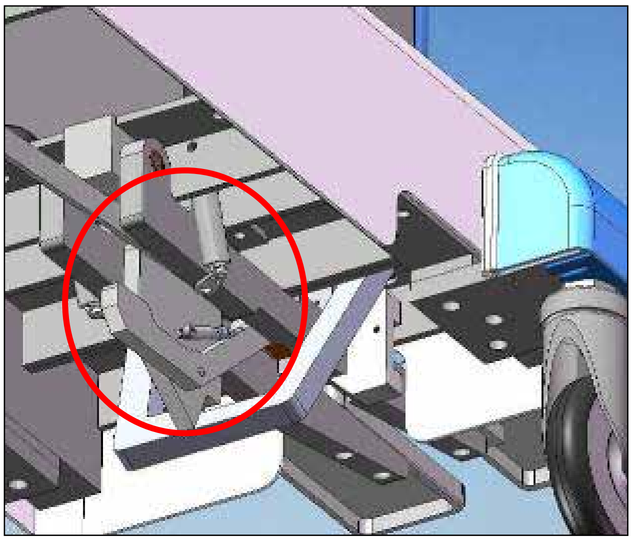


Figure 92

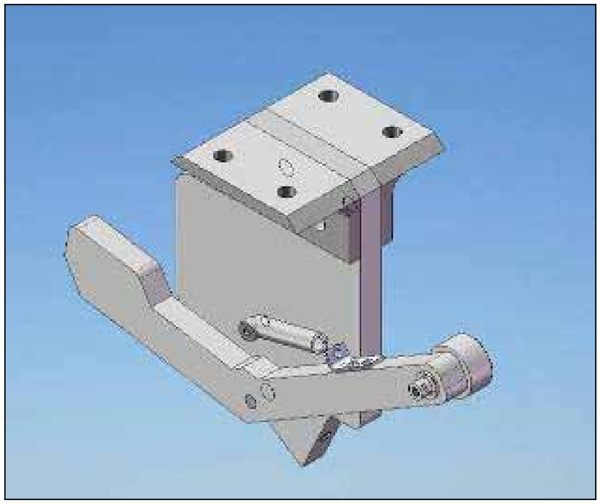


Figure 93

- Detach Tension Coupling Docking Linkage from Link Wheel Pin by removing screw.
- Remove Docking Linkage Assembly by removing four screws from mounting brackets.
- Insert new Docking Linkage Assembly and fasten by inserting four screws into mounting brackets.

PINCH HAZARD

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

- Attach Tension Coupling Docking Linkage to Link Wheel Pin with screw and Loctite 290.

Finalization Steps

i. Dock the stretcher onto the scanner.
ii. Advance the patient support to scan.
iii. Return the patient support into the home position.
iv. Undock the stretcher from the scanner.
v. Repeat finalization steps five times.

13.12 Travel Stop Plunger Assembly

SMI Part #: SVC-00324
GEHC Part #: 5326130
Time required: 50 minutes
Tools required: 3/16" non-magnetic Allen wrench
Personnel required: 1

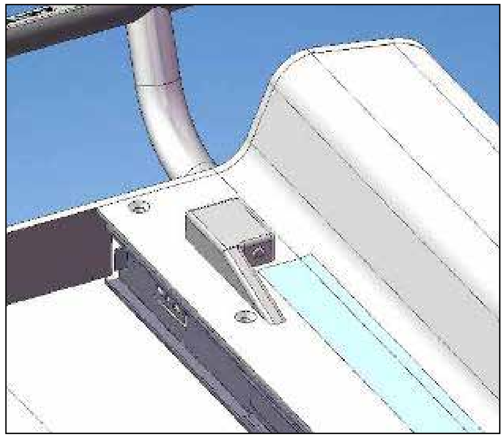


Figure 94

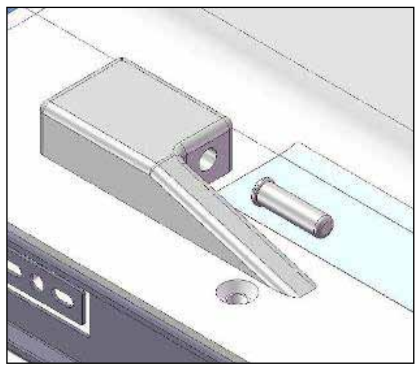


Figure 95

1. Remove Travel Stop Plunger Housing by removing four mounting screws from inside UBS.
2. Place new Travel Stop Plunger Assembly in position with plunger protruding on UBS and fasten with four mounting screws from inside UBS.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Ensure that the Travel Stop Plunger Assembly stops the Patient Support as it returns to the home position.

v. Undock the stretcher from the scanner.

vi. Repeat finalization steps five times.

13.13 Tray Table

SMI Part #: SVC-00352
GEHC Part #: 5326149
Time required: 30 minutes
Tools required: 5/32” non-magnetic Allen wrench
Personnel required: 1

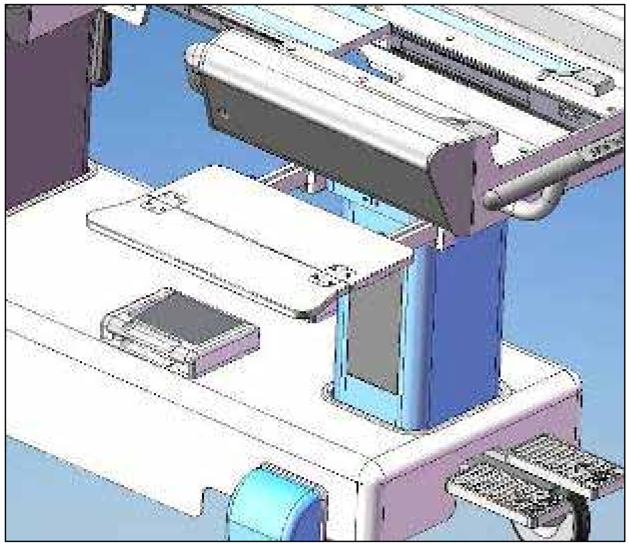


Figure 96

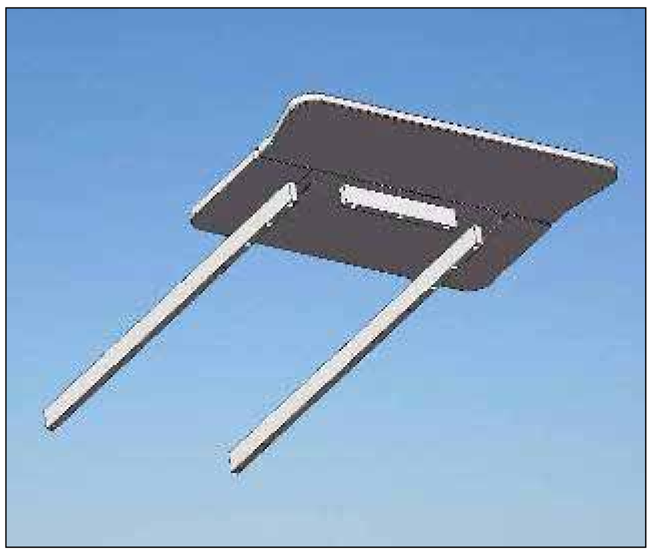


Figure 97

1. Remove Tray Assembly by removing 6 mounting screws from the underside.
2. Carefully position new Tray Assembly and fasten with 6 mounting screws from the underside using light force.

Finalization Steps

i. Pull the tray table out and open it.

ii. Close the tray table and push it back in towards the FSA.

iii. Repeat steps i) and ii) 5 times.

iv. Repeat steps i) to iii) for both tray tables.

13.14 No-Undock Hook Assembly

SMI Part #: SVC-00332
GEHC Part #: 5326132
Time required: 70 minutes
Tools required: 5/32” and 3/16” non-magnetic Allen wrench
Personnel required: 1

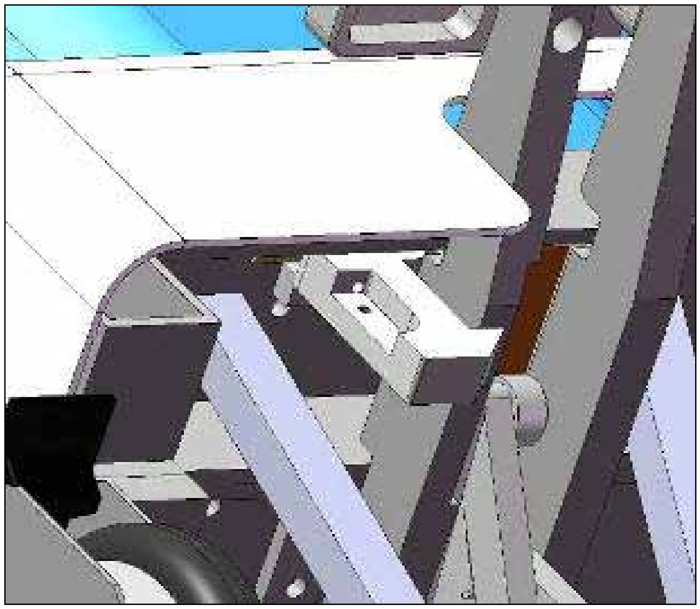


Figure 98

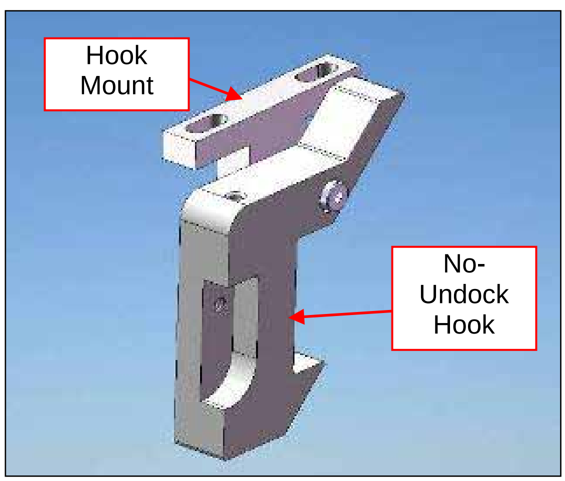


Figure 99

1. Remove two mounting screws from the No Undock Hook Mount and remove assembly.
2. Position new No-Undock Hook Assembly so the hook is positioned correctly with the pedal arm and fasten with two mounting screws in the No-Undock Hook Mount.

PINCH HAZARD

CAUTION

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Try pressing the undock pedal with the patient support in the scanner. Ensure that the stretcher does not undock. If the stretcher does undock, this step has failed. Carefully dock to the scanner again, return the patient support to the home position and refer to Section 7.12 to readjust the no-undock hook.

iv. If the hook passed step iii), return the patient support into the home position.

v. Undock the stretcher from the scanner.

vi. Repeat finalization steps five times.

13.15 Pedal Block Assembly

SMI Part #: SVC-00429
GEHC Part #: 5326133
Time required: 90 minutes
Tools required: ¼”, 3/16”, 1/8” non-magnetic Allen wrench
Personnel required: 1

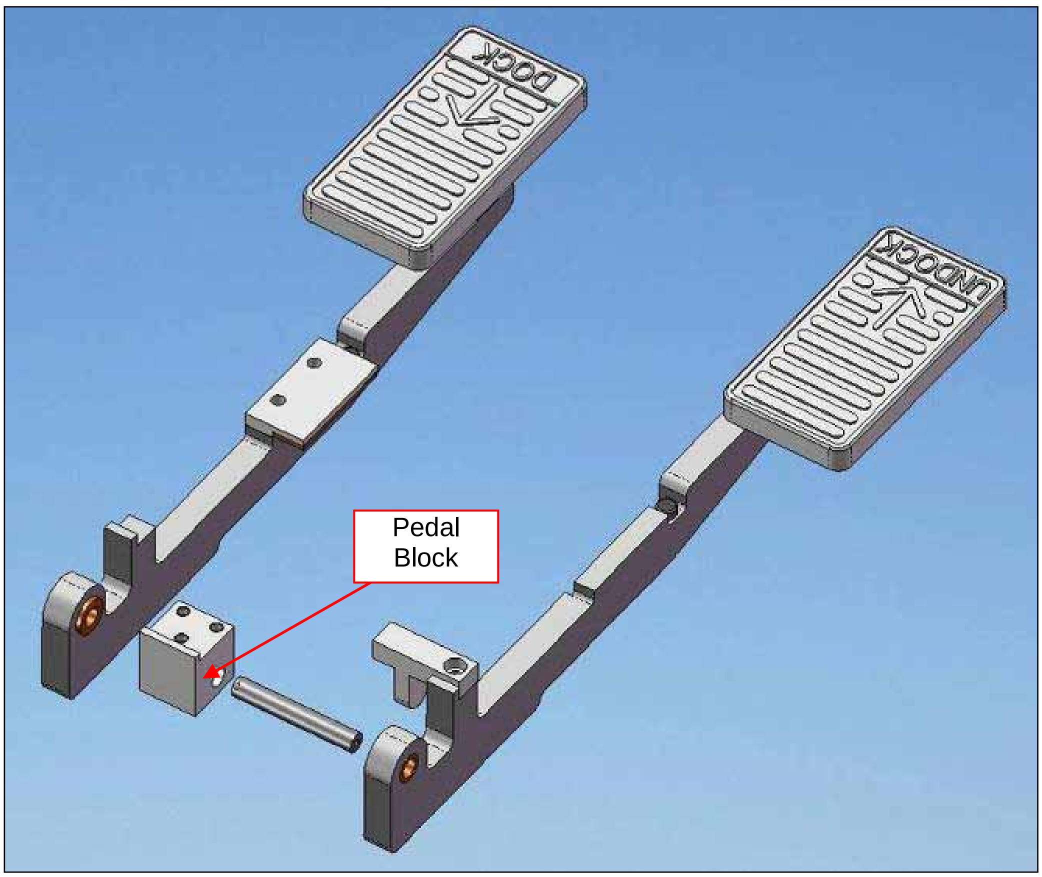


Figure 100

1. Remove Wheel Covers.
2. Remove Shroud mounting screws.

3. Raise Shroud to gain access to Pedal Block Assembly and detach from frame by removing three mounting screws.
4. Remove Pedal Arms from Pedal Block Assembly by removing screws from each end of shaft.
5. Position new Pedal Block Assembly and fasten Pedal Arms to shaft with screws.
6. Fasten new Pedal Block Assembly to Frame with three mounting screws.
7. Lower Shroud into position and replace Shroud mounting screws.
8. Replace Wheel Covers.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times.

13.16 Pedal Arm Assembly

SMI Part #:	SVC-00430
GEHC Part #:	5326134
Note:	Order consumables FRU kit (P/N 4001158-11, GEHC P/N 5408278) for Loctite 290
Time required:	60 minutes
Tools required:	1/8” non-magnetic Allen wrench
Personnel required:	1

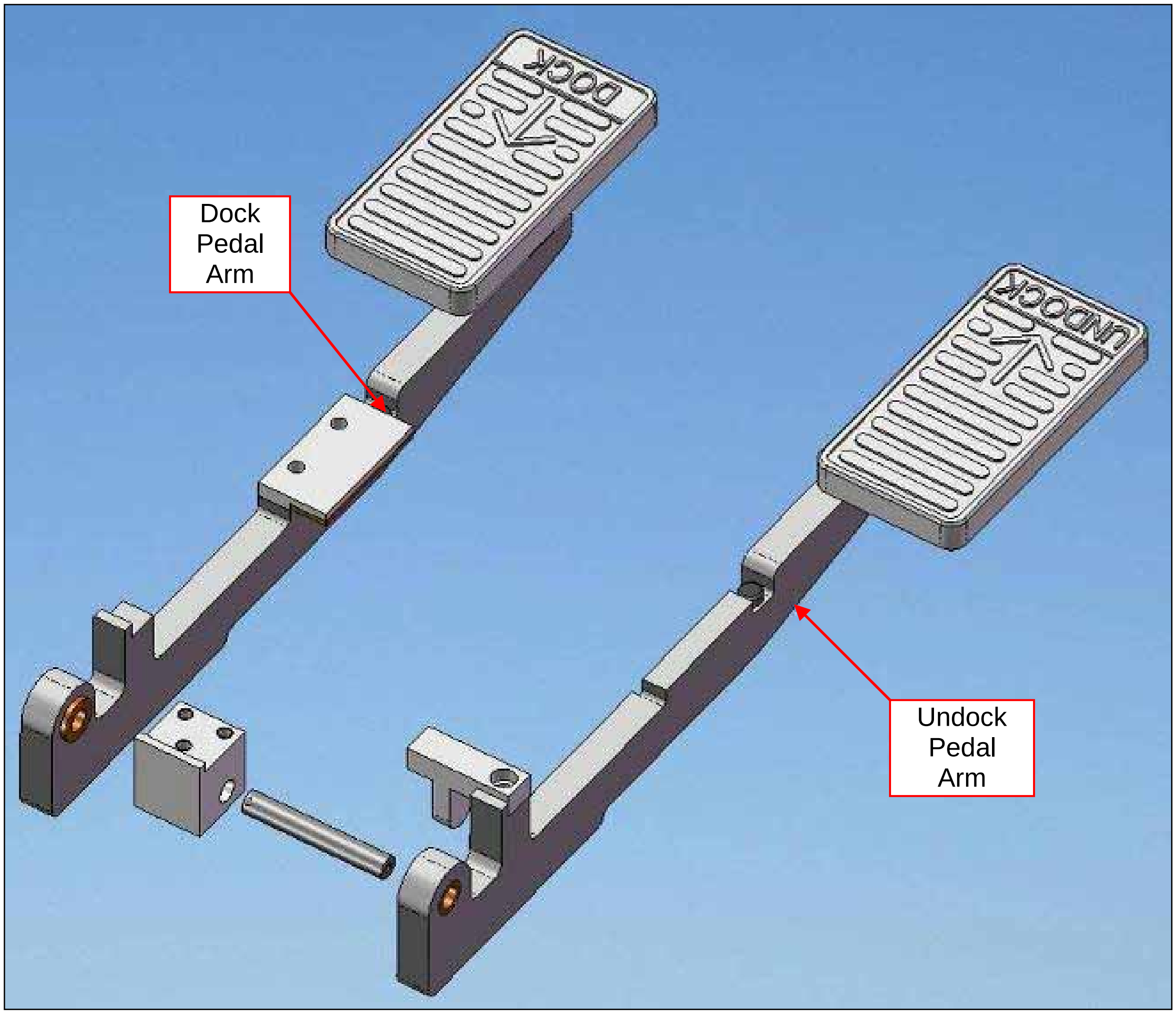


Figure 101

1. Remove Pedal Arm from Pedal Block Assembly by removing screw from end of shaft.
2. Place new Pedal Arm on shaft and fasten with screw in end of shaft with Loctite 290.
3. Test the function of the new Pedal Arm Assembly by pressing it several times.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Ensure that the Travel Stop Plunger Assembly stops the Patient Support as it returns to the home position.

v. Undock the stretcher from the scanner.

vi. Repeat finalization steps five times.

13.17 Catchment Latch Assembly

SMI Part #: SVC-00362
GEHC Part #: 5326151
Time required: 60 minutes
Tools required: Medium non-magnetic slot screwdriver
Personnel required: 1

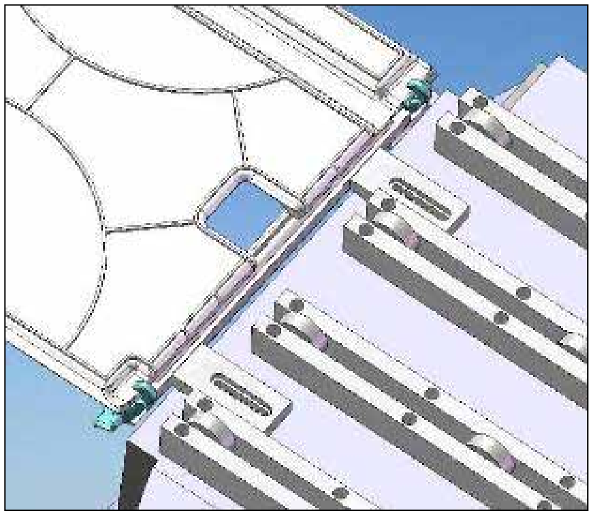


Figure 102

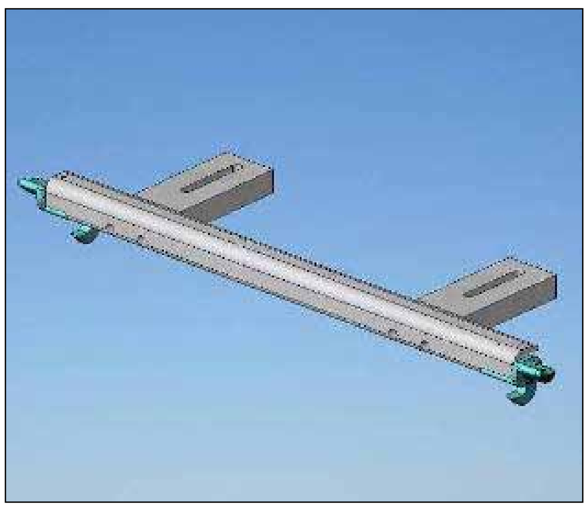


Figure 103

1. Access underside of Patient Support and loosen Door Catch Mount screws to remove Catchments Latch Assembly.
2. Replace Catchments Latch Assembly and tighten Door Catch Mount screws while aligning Catchments Latch Assembly to Catchments Tray.

Finalization Steps

i. Turn the small handle on the side of the catchment latch assembly to release the catchment.

ii. Ensure that not too much force is required to turn the handle, in order to release the catchment.

iii. Close the catchment and ensure that it catches onto the catchment latch assembly.

iv. Repeat steps i) to iii) five times.

13.18 Compression Rail Assembly and Compression Slider

SMI Part #:	Rail: SVC-00367
	Slider: 4000095-11
GEHC Part #:	Rail: 5408283
	Slider: 5326153
Time required:	Rail: 40 minutes
	Slider: 10 minutes
Tools required:	Large slot screwdriver
Personnel required:	1

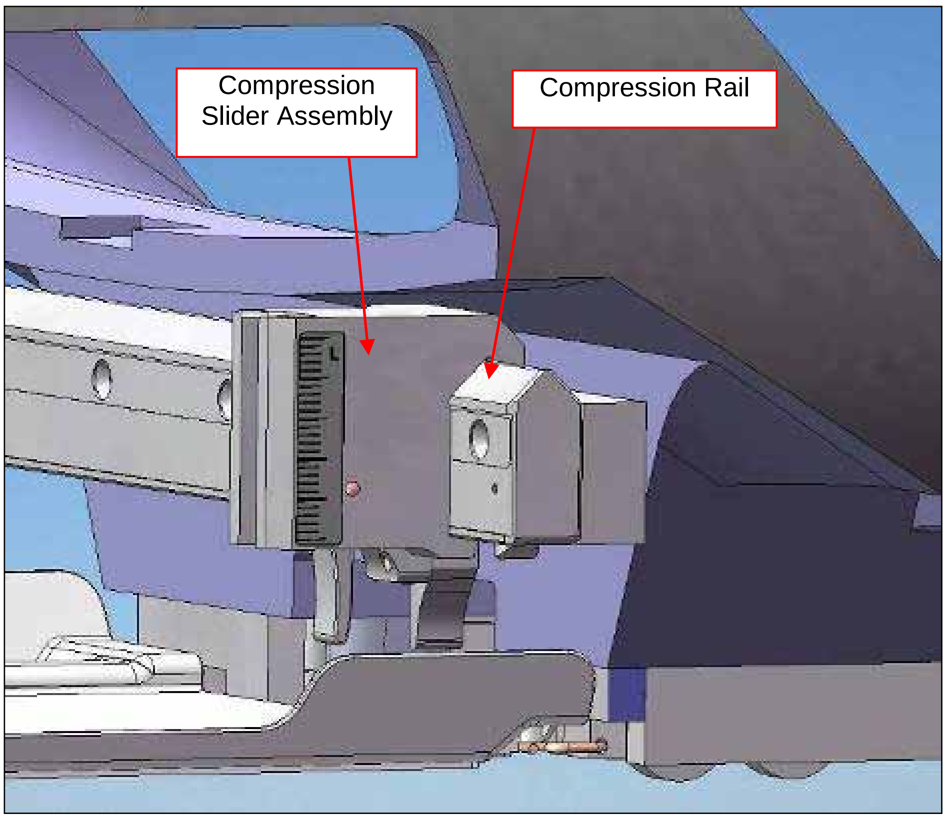


Figure 104

1. Depress retaining feature on end of Compression Rail and remove Slider Assembly(s).
2. Remove five ¼-20 FHMS and remove compression Rail.
3. Assemble replacement Compression rail by fastening five ¼-20 FHMS very tight.
4. Replace Slider Assembly and slide across entire length of rail to ensure functionality.

Finalization Steps

i. Slide both compression sliders on the compression rail and check for smooth movement.

ii. Lock the compression sliders, and then apply a lateral force (along the rail), ensuring that the sliders do not move when locked.

13.19 Emergency Release Handle Assembly

SMI Part #: SVC-00364, Kevlar cord (10 feet)
GEHC Part #: 5326154
Time required: 90 minutes
Tools required: 3/16" non-magnetic Allen wrench
Personnel required: 1

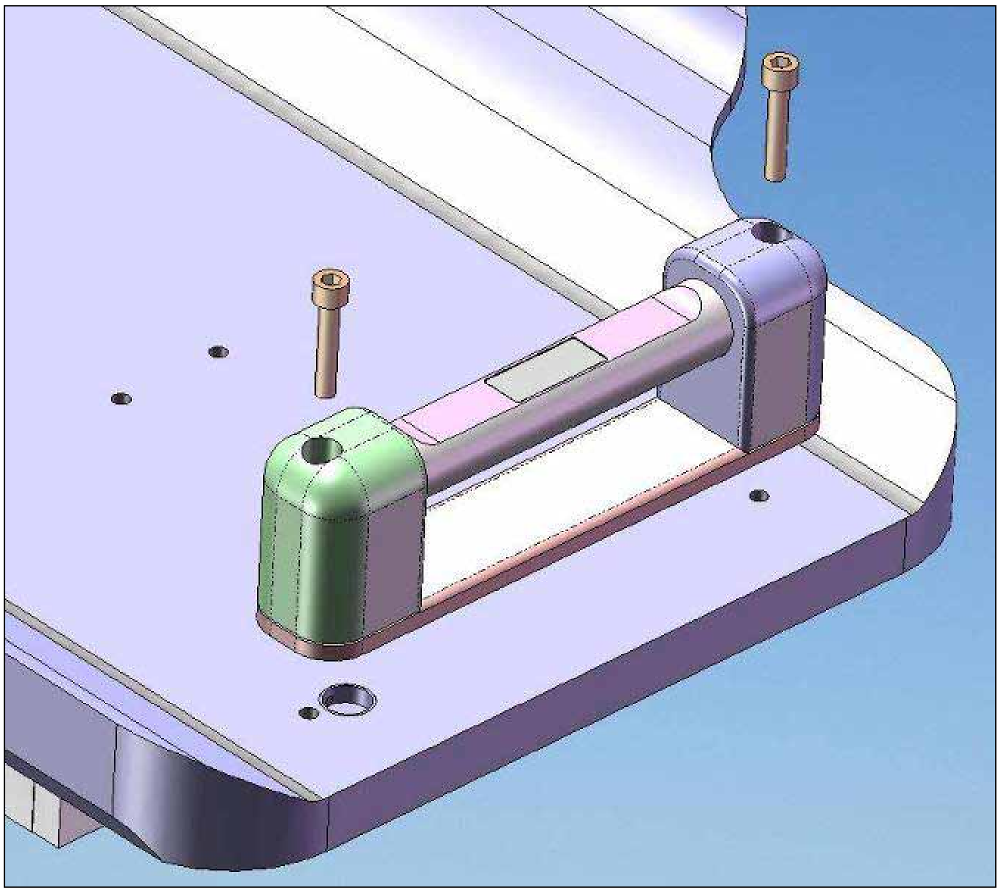


Figure 105

1. Remove Emergency Release Handle Assembly by removing two mounting screws.
2. Cut Kevlar cable as close to Emergency Release Handle Assembly as possible to leave excess.
3. Firmly knot Kevlar cable from inside patient support to Kevlar cable from new Emergency Release Handle Assembly and place Emergency Release Handle Assembly in position on patient support.
4. Access underside of Patient Support and pull Kevlar line through until taught.
5. Mount Emergency Release Handle Assembly on Patient Support with two screws.
6. Remove old Kevlar cable from Emergency Release Actuator and firmly knot new cable ensuring adequate tension and seal with tape.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Move to the side of the stretcher, turn the emergency handle and pull the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times.

13.20 LPCA Mechanism Assembly

SMI Part #: SVC-00381
GEHC Part #: 5326155
Time required: 70 minutes
Tools required: Non-magnetic adjustable wrench, non-magnetic large and medium slot screwdriver
Personnel required: 1

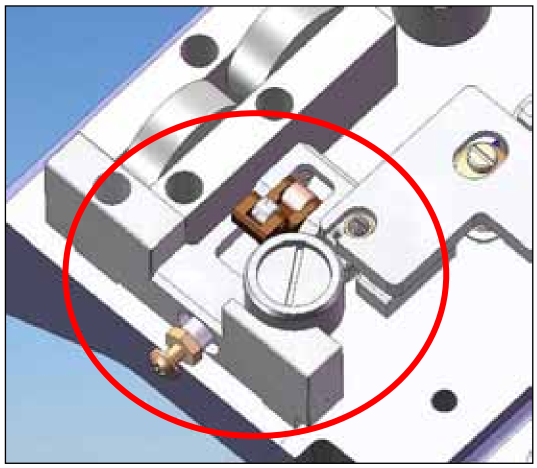


Figure 106

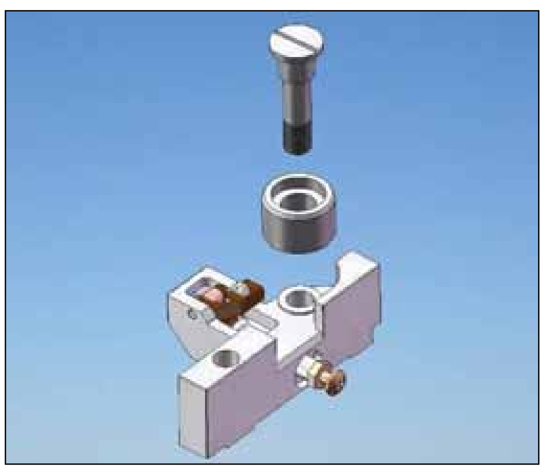


Figure 107

1. Measure protrusion distance of screw from plunger.
2. Access underside of Patient Support and remove L-R Centering Wheel Assembly and mounting screw and

remove LPCA Mechanism Assembly

3. Replace LPCA Mechanism Assembly by fastening with L-R Centering Wheel Assembly and mounting screw.
4. Re-set LPCA Mechanism by adjusting screw to previously measured setting and tighten locking nut firmly.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan. If the scanner hook does not engage the latch fixture base, adjust the protrusion distance by screwing in the knurled screw head and restart at step i).

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times.

13.21 Emergency Release Spring Assembly

SMI Part #: SVC-00365
GEHC Part #: 5326156
Time required: 70 minutes
Tools required: Non-magnetic small slot screwdriver
Personnel required: 1

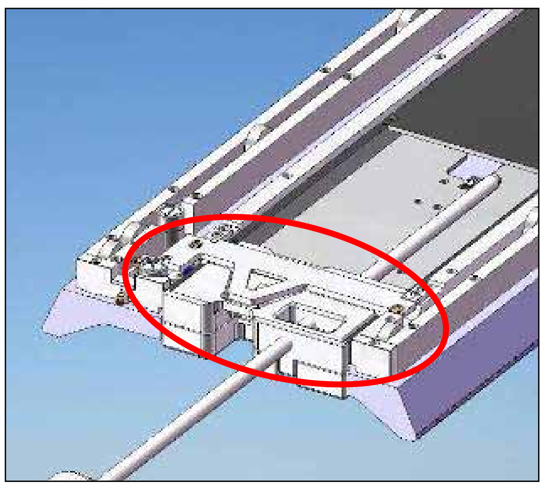


Figure 108

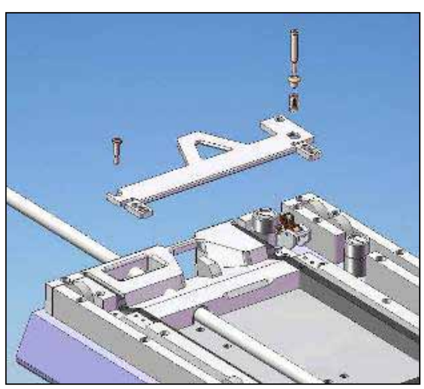


Figure 109

1. Access underside of Patient Support and remove mounting screws of Emergency Release Spring Assembly.
2. Remove tape and untie knotted Kevlar cable from Emergency Release Spring Assembly.
3. Position new Emergency Release Spring Assembly and fasten to Patient Support with mounting screws.
4. Firmly knot Kevlar cable to Emergency Release Actuator ensuring adequate tension and seal with tape.

Finalization Steps

i. Dock the stretcher onto the scanner.
ii. Advance the patient support to scan.
iii. Move to the side of the stretcher, turn the emergency handle and pull the patient support into the home position.
iv. Undock the stretcher from the scanner.
v. Repeat finalization steps five times.

13.22 L-R Centering Wheel Assembly

SMI Part #: SVC-00366
GEHC Part #: 5326157
Time required: 40 minutes
Tools required: Non-magnetic large slot screwdriver

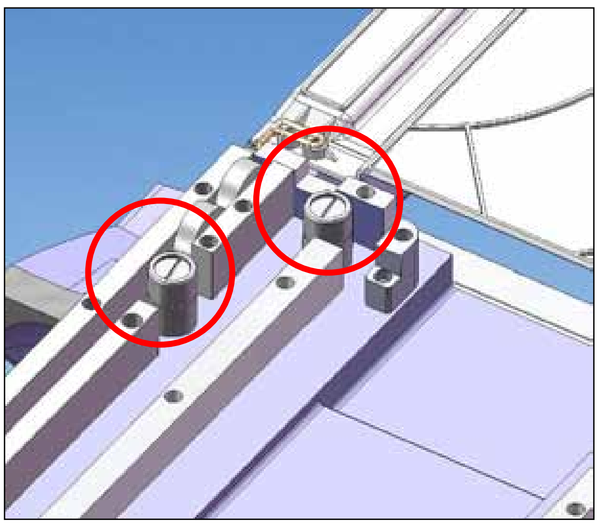


Figure 110

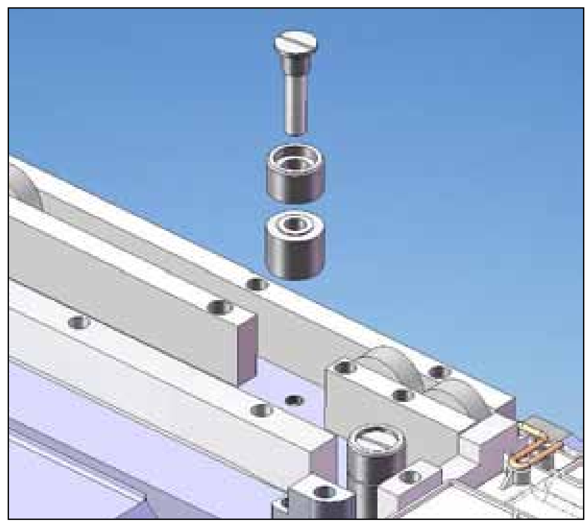


Figure 111

1. Access underside of Patient Support and unscrew mounting screw to remove L-R Centering Wheel Assembly.
2. Position replacement assembly and fasten carefully to underside of Patient Support.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times

13.23 VG HYP 1.5T Cable Tray

SMI Part #: 4000015-11
GEHC Part #: 5326169
Time required: 60 minutes
Tools required: Non-magnetic large slot screwdriver
Personnel required: 1

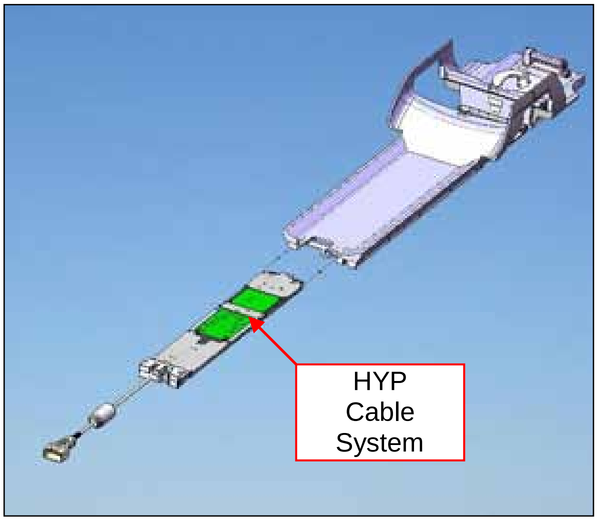


Figure 112

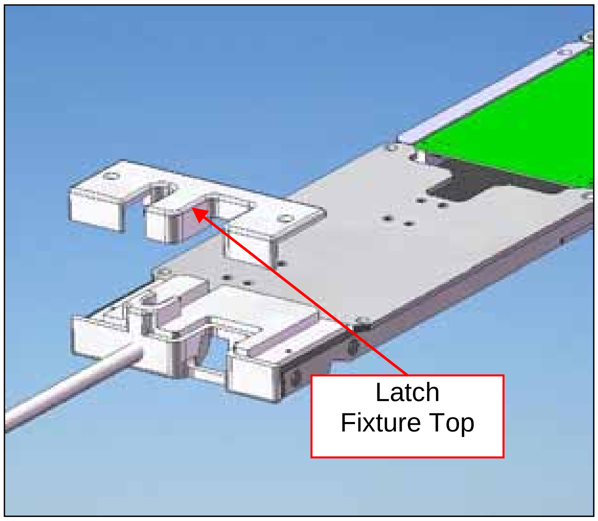


Figure 113

1. Remove latch fixture top by unscrewing two screws.
2. Remove Vanguard HYP Cable System from Patient Support by pulling on the bottom of the latch fixture.
3. Insert new Cable Tray into end of Patient Support and carefully slide completely in, positioning connectors at other end as required. If resistance is felt, stop and try to position the connectors such that they move freely, allowing the entire Cable Tray to slide under the patient support.
4. Position latch fixture top, align the holes and fasten with two screws.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan. If the scanner hook does not engage the latch fixture base, adjust the protrusion distance by screwing in the knurled screw head on the Cable Tray and restart at step i).

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times.

13.24 Vanguard HYP Cable System 3T

SMI Part #:	4000181-11
GEHC Part #:	5419434-8
Time required:	60 minutes
Tools required:	Non-magnetic large slot screwdriver

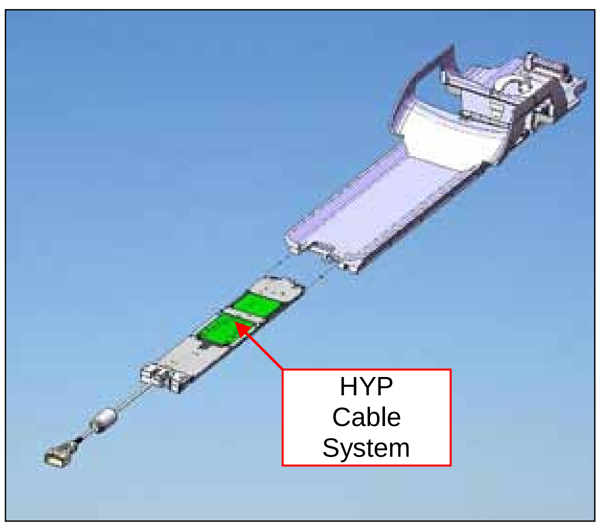


Figure 114

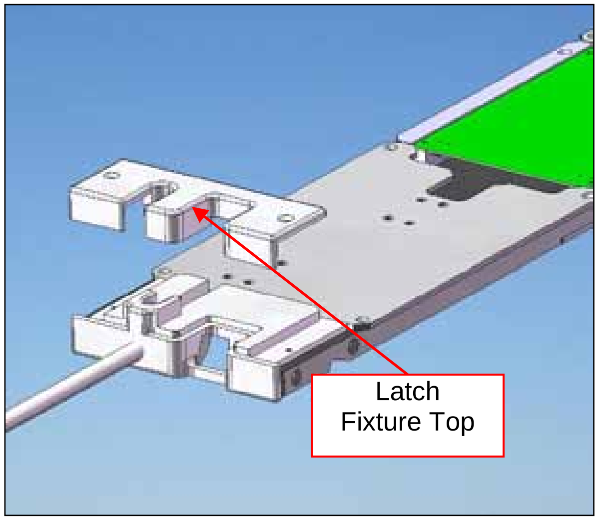


Figure 115

1. Remove latch fixture top by unscrewing two screws.
2. Remove Vanguard HYP Cable System 3T from Patient Support by pulling on the bottom of the latch fixture.
3. Insert new Cable Tray into end of Patient Support and carefully slide completely in, positioning connectors at other end as required.
4. Position latch fixture top and fasten with two screws.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan. If the scanner hook does not engage the latch fixture base, adjust the protrusion distance by screwing in the knurled screw head on the Cable Tray and restart at step i).

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times.

13.25 Latch Tab Assembly

SMI Part #: SVC-00379
GEHC Part #: 5326147
Time required: 40 minutes
Tools required: Non-magnetic large slot screwdriver, 3/16” and 1/8” Allen wrench
Non-magnetic pliers, needle nose, 6"
GE Only: Non-magnetic pliers, side cutter, titanium
Note: This FRU contains one cable and two cable ends
Personnel required: 2

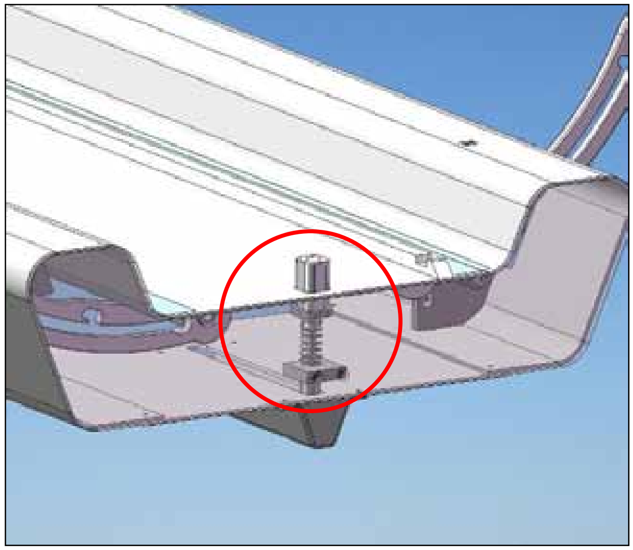


Figure 116

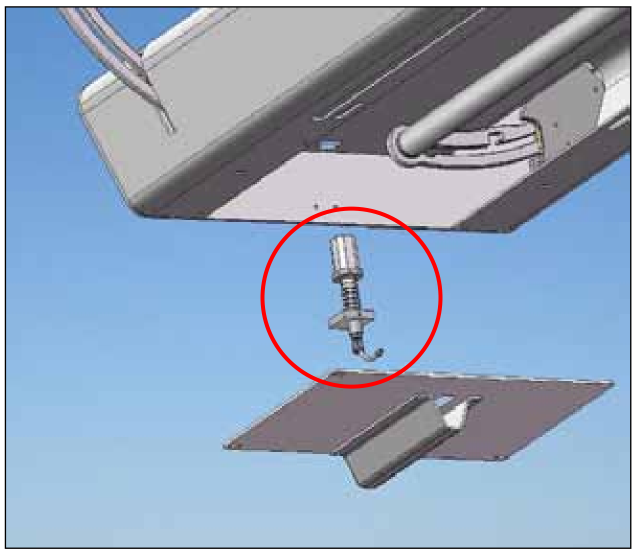


Figure 117

1. Remove cover by unscrewing four mounting screws while supporting cover.
2. Detach Latch Tab Cable from Swing Assembly.
3. Remove screw from top of Latch Tab and withdraw cable.
4. Insert the new cable into new Latch Tab Assembly and through the cable elbows.
5. Mount Latch Tab Assembly to LBS by screwing in two fasteners.
6. Position cover by screwing in four mounting screw while supporting cover.
7. Refer to Section 7.10 to set the Latch Tab.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times.

13.26 Stopper Assembly

SMI Part #: SVC-00428
GEHC Part #: 5326131
Time required: 40 minutes
Tools required: 5/32” non-magnetic Allen wrench
Personnel required: 1

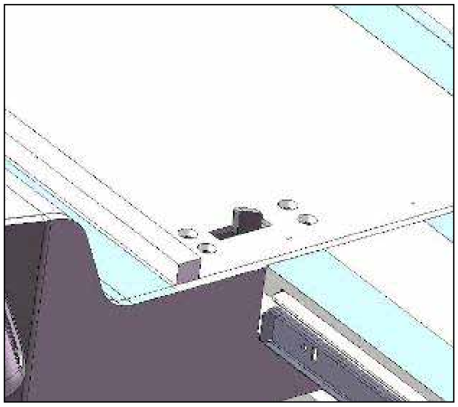


Figure 118

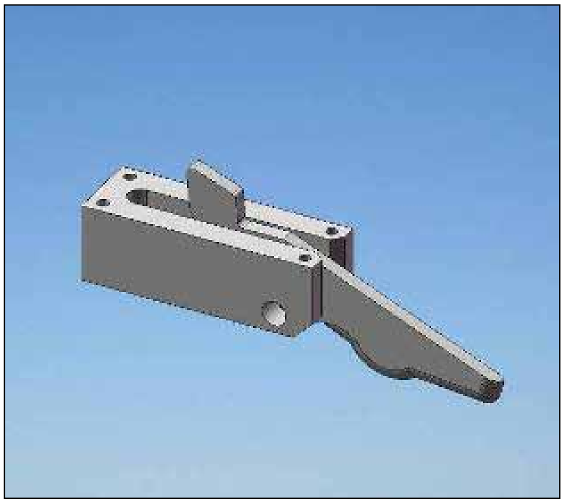


Figure 119

1. Remove four FHMS mounting screws accessed from the top surface of the LBS.
2. Remove Stopper Assembly from the underside of the top surface of the LBS
3. Align replacement stopper assembly with mounting holes and fasten with four FHMS mounting screws.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Open the bridge. Apply a force on the patient support into and out of the scanner and confirm that the patient support does not move at all. Close the bridge.

iii. Advance the patient support to scan.

iv. Return the patient support into the home position.

v. Open the bridge. Apply a force on the patient support into and out of the scanner and confirm that the patient support does not move at all. Close the bridge.

vi. Undock the stretcher from the scanner.

vii. Repeat finalization steps three times.

13.27 Shroud Stretcher

SMI Part #: SVC-00333
GEHC Part #: 5326136
Time required: 120 minutes
Tools required: 9/16” non-magnetic wrench, 5/32” and 3/16” non-magnetic Allen key
Personnel required: 2

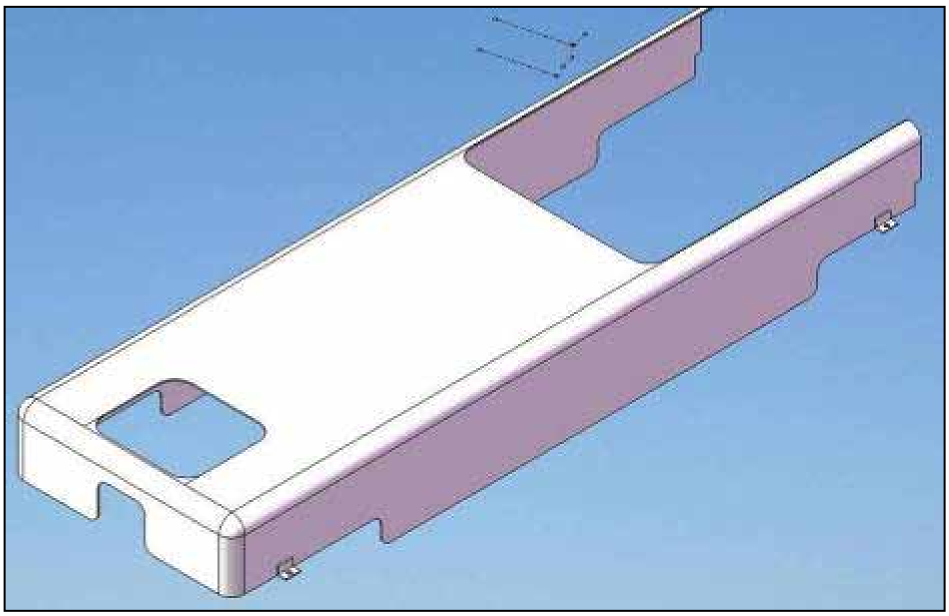


Figure 120

1. Loosen mounting screws and remove all 4 Wheel Covers.
2. Remove Shroud mounting screws.
3. Remove the FSA with the UBS still on it. To remove the FSA, open the battery door, disconnect the wiring circuit from the battery, loosen the four hex head bolt inside the bottom of the FSA. Two people should lift the FSA and UBS (still attached) at least 5cm and slide the shroud out towards the head of the stretcher.

HEAVY ITEM

WARNING ***Use proper heavy lifting techniques and use two people when lifting the FSA and UBS assembly.***

4. With the FSA and UBS assembly still in the air, slide in the new shroud from the head end of the stretcher.
5. Place the bottom of the FSA back down and tighten the four hex head bolts.
6. Fasten Shroud mounting screws.
7. Reconnect the wiring circuit to the battery.

8. Replace Wheel Covers and tighten mounting screws.

Finalization Steps

i. Dock the stretcher onto the scanner. Ensure that the shroud is not interfering with any part of the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps two times.

13.28 Tension Coupling Docking Linkage

SMI Part #: SVC-00335
GEHC Part #: 5326137
Note: Order consumables FRU kit (P/N 4001158-11, GEHC P/N 5408278) for Loctite 290
Time required: 50 minutes
Tools required: ¼", 5/32" and 1/8" non-magnetic Allen wrench
Personnel required: 1

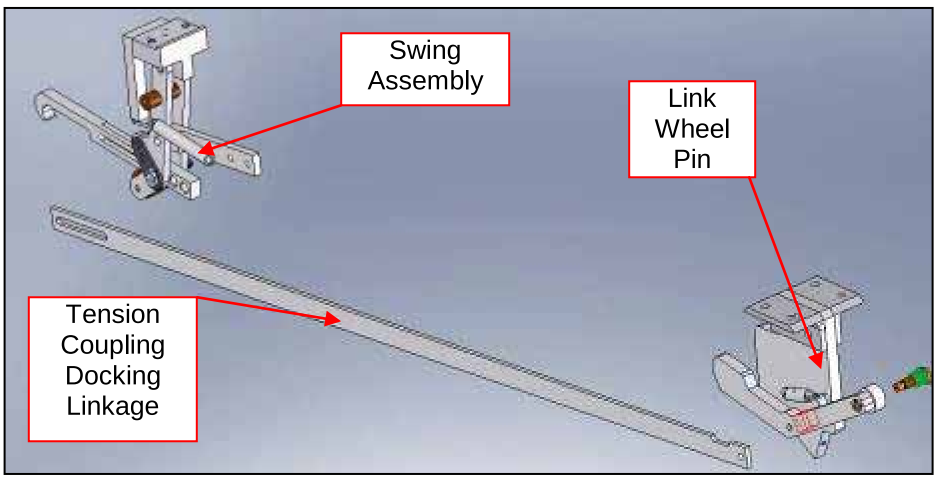


Figure 121

1. Unscrew mounting screw and remove Tension Coupling Docking Linkage from Link Wheel Pin.
2. Remove Tension Coupling Docking Linkage from Swing Assembly by removing the two mounting/adjustment screws.
3. Attach the new Tension Coupling Docking Linkage to the Swing Assembly by screwing in the mounting/adjustment screws.

PINCH HAZARD

⚠ CAUTION

Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.

4. Attach the Tension Coupling Docking Linkage to the Link Wheel Pin with the mounting screw and Loctite 290.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times.

13.29 LBS Cable Elbow Guard

SMI Part #: SVC-00350
GEHC Part #: 5326148
Time required: 15 minutes
Tools required: 1/8” non-magnetic Allen wrench
Personnel required: 1

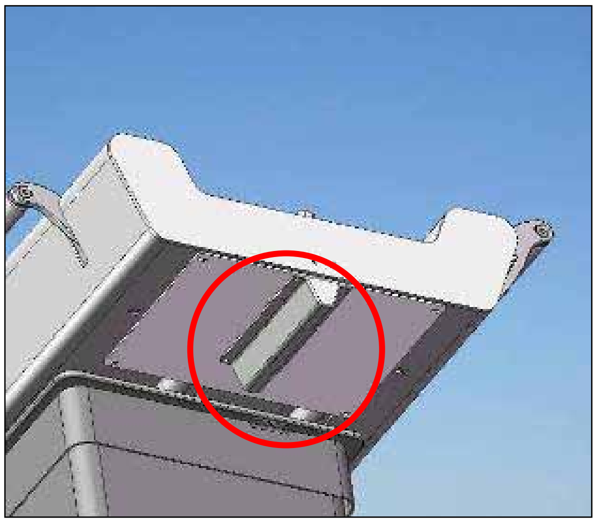


Figure 122

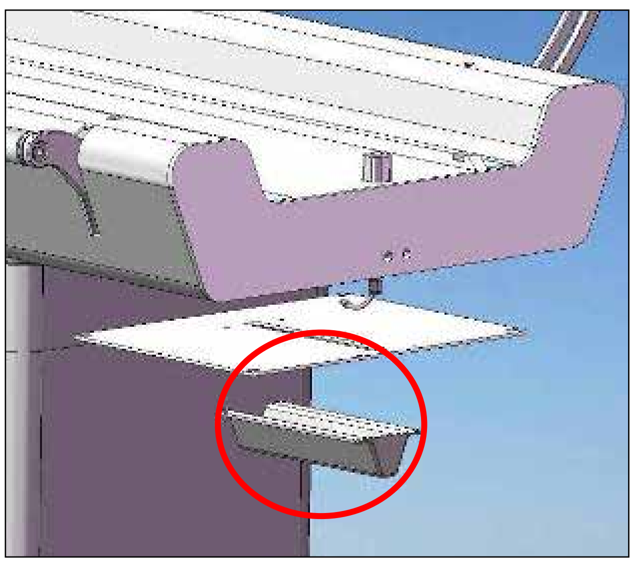


Figure 123

1. Unscrew four mounting screws to remove Cable Elbow Guard.
2. Position new Cable Elbow Guard and fasten with four mounting screws.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

13.30 UBS Handle

SMI Part #: SVC-00336
GEHC Part #: 5326139
Time required: 20 minutes
Tools required: ½" non-magnetic adjustable wrench
Personnel required: 1

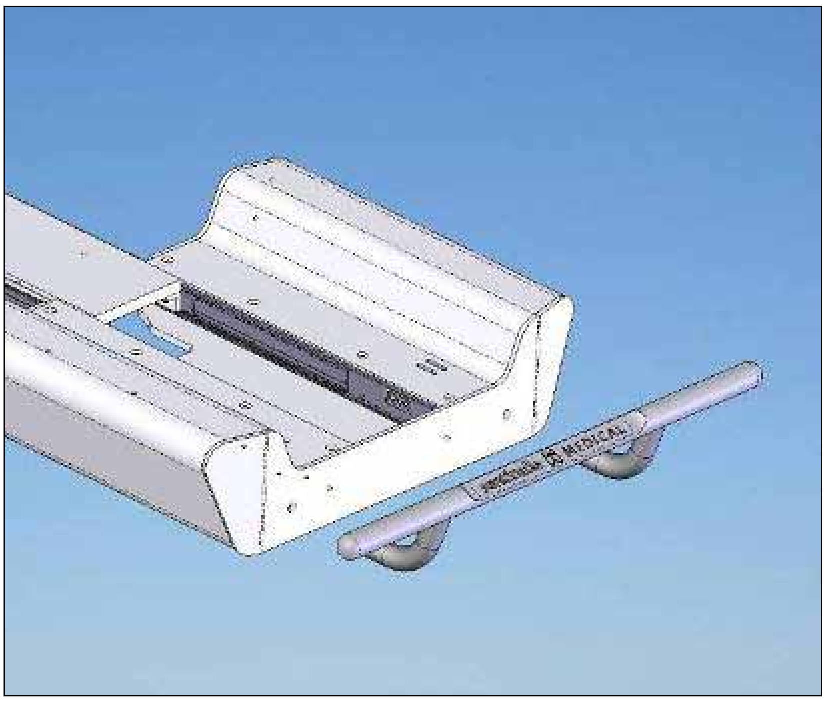


Figure 124

1. Support UBS Handle and remove two mounting screws from inside of UBS.
2. Position and support new UBS Handle and firmly tighten mounting screws from inside of patient support.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Pull the stretcher at least 3ft (91cm) away from the scanner using the UBS handle.

vi. Push the stretcher back onto the scanner using the UBS handle.

vii. Repeat steps i) to vi) five times.

13.31 Bumper

SMI Part #: SVC-00378
GEHC Part #: 5326140
Time required: 35 minutes
Tools required: ¼” non-magnetic Allen key, 5/32” non-magnetic Allen key
Personnel required: 1

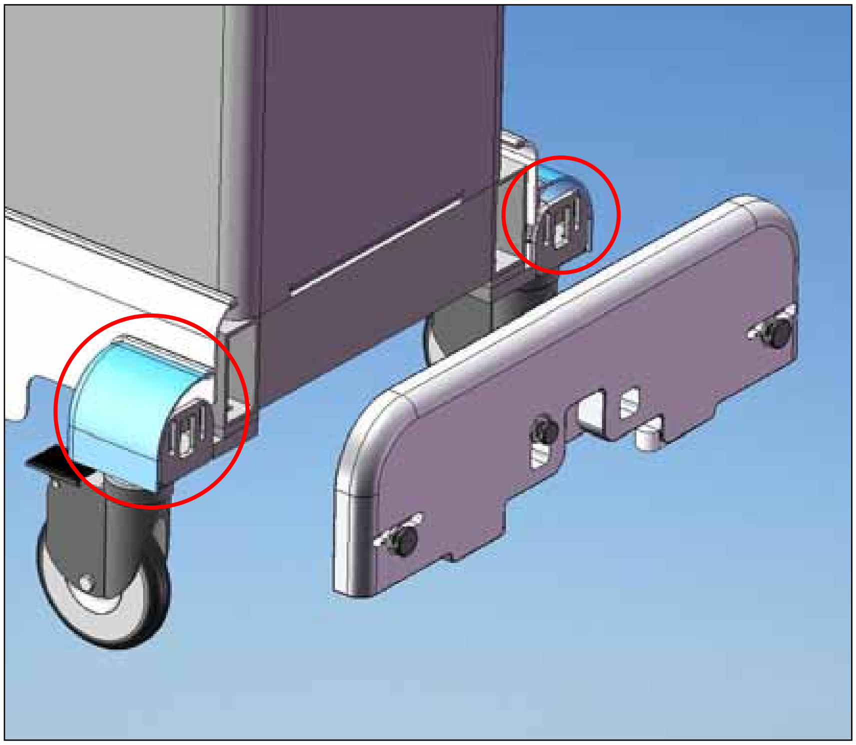


Figure 125

1. Loosen mounting screws and remove the 2 Wheel Covers next to the Bumper.
2. While supporting the bumper carefully unscrew the four mounting screws and remove the bumper.
3. Take measure the amount of protrusion of the Bumper Spring Assemblies and Spacer pin bumpers and screw them into the new bumper the same amount and apply a small amount of Loctite 290.
4. Attach new bumper to Frame by inserting mounting screws into bumper and screwing into tightening blocks.
5. Complete *Section 7.5 Bumper Adjustment* after installing the new bumper to ensure correct positioning.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times.

13.32 Spacer Pin Bumper

SMI Part #: SVC-00338
GEHC Part #: 5326141
Time required: 15 minutes
Tools required: Non-magnetic adjustable wrench
Personnel required: 1

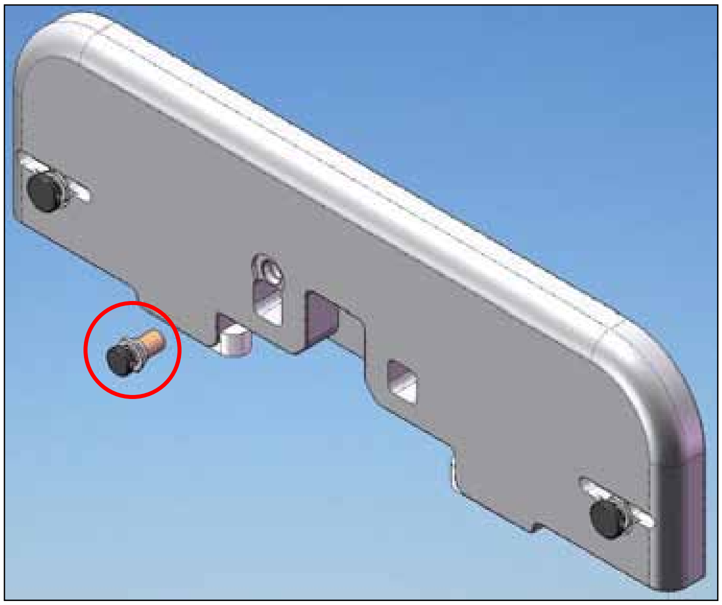


Figure 126

1. Measure protrusion distance of Spacer Pin Bumper from Bumper and remove.
2. Screw in new Spacer pin bumper to the correct amount to preserve the setting and apply a small amount of Loctite 290.

Finalization Steps

i. Dock the stretcher onto the scanner.
ii. Advance the patient support to scan.
iii. Return the patient support into the home position.
iv. Undock the stretcher from the scanner.
v. Repeat finalization steps five times.

13.33 Bridge Spring Block

SMI Part #: SVC-00343
GEHC Part #: 5326145
Time required: 40 minutes
Tools required: 3/16” non-magnetic Allen wrench
Personnel required: 1

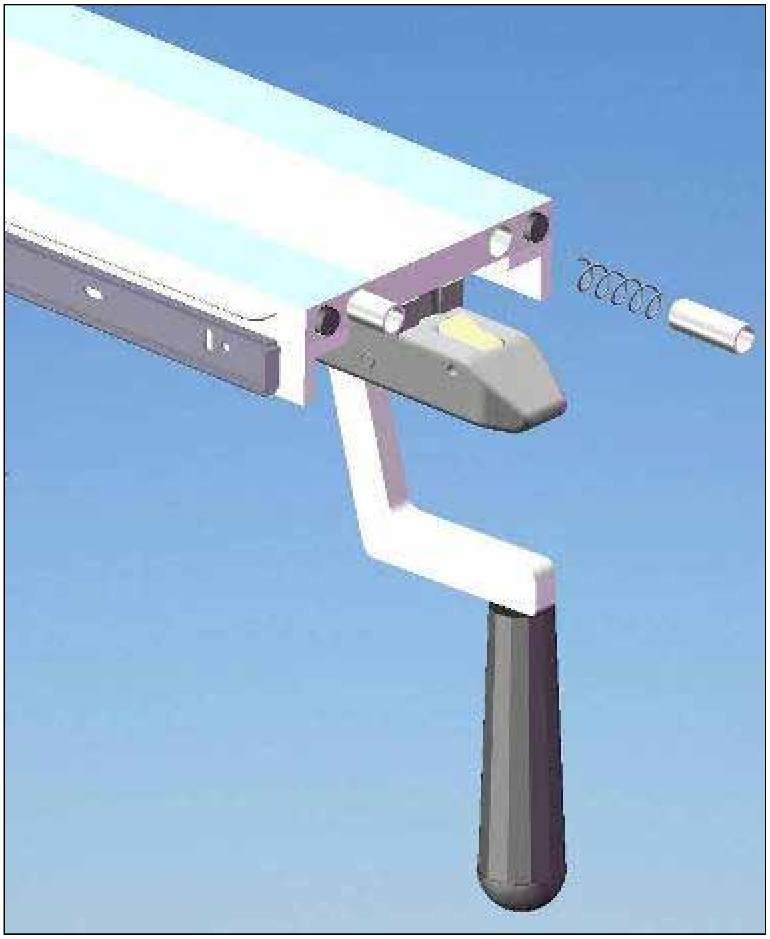


Figure 127

1. With bridge in open position unscrew countersunk screw inside Spring Block and remove screw, spring and Spring Block.
2. Insert new spring, spring block and screw and tighten screw.

Finalization Steps

i. Open and close the bridge, ensuring that the motion is smooth and there is no noise while sliding.

ii. Ensure that the bridge latches on when closed and that it cannot be moved unless the handle is pressed.

iii. Repeat steps i) and ii) ten times.

13.34 Pedal Spring

SMI Part #: SVC-00340
GEHC Part #: 5326143
Time required: 80 minutes
Tools required: 1/4” non-magnetic Allen wrench
Personnel required: 1

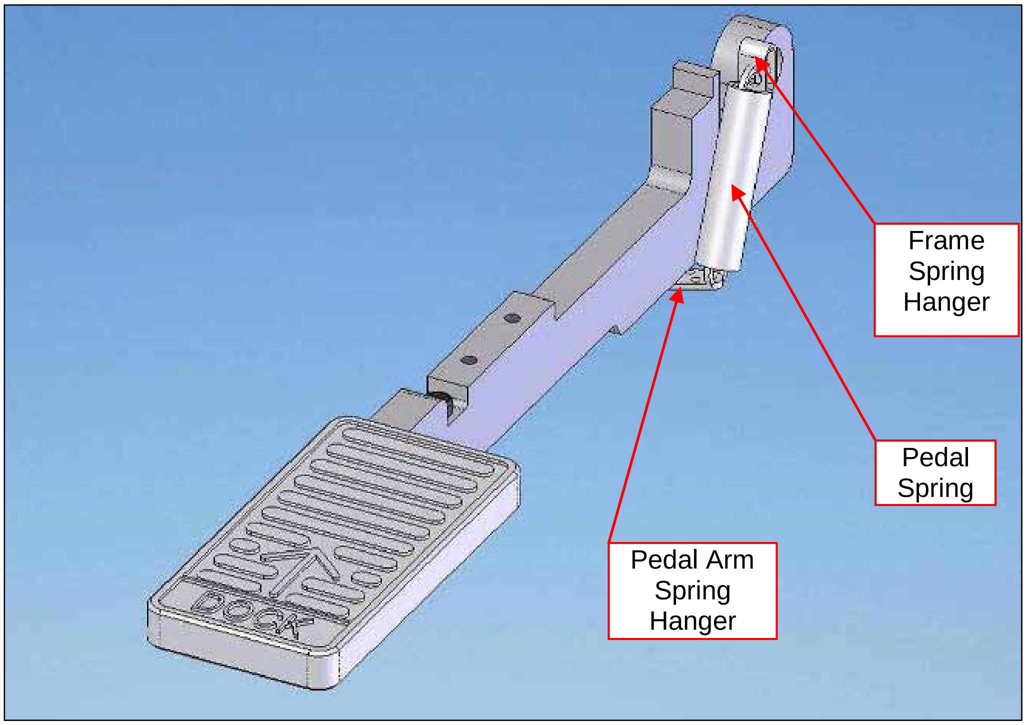
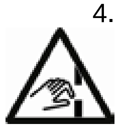


Figure 128

1. Remove spring hanger by unscrewing screw on bottom of pedal arm.
2. Remove spring hanger on frame by unscrewing screw on frame.
3. Place new spring on hanger and fasten to frame with mounting screw.
4. Attach spring hanger to new spring and stretch spring to position hanger on Pedal Arm.



PINCH HAZARD

CAUTION **Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.**

5. Fasten to Pedal Arm by screwing in mounting screw.

Finalization Steps

i. Dock the stretcher onto the scanner.
ii. Advance the patient support to scan.
iii. Return the patient support into the home position.
iv. Undock the stretcher from the scanner.
v. Repeat finalization steps five times.

13.35 Undock Spring

SMI Part #: SVC-00342
GEHC Part #: 5326144
Time required: 40 minutes
Tools required: ¼" and 5/32" non-magnetic Allen wrench
Personnel required: 1

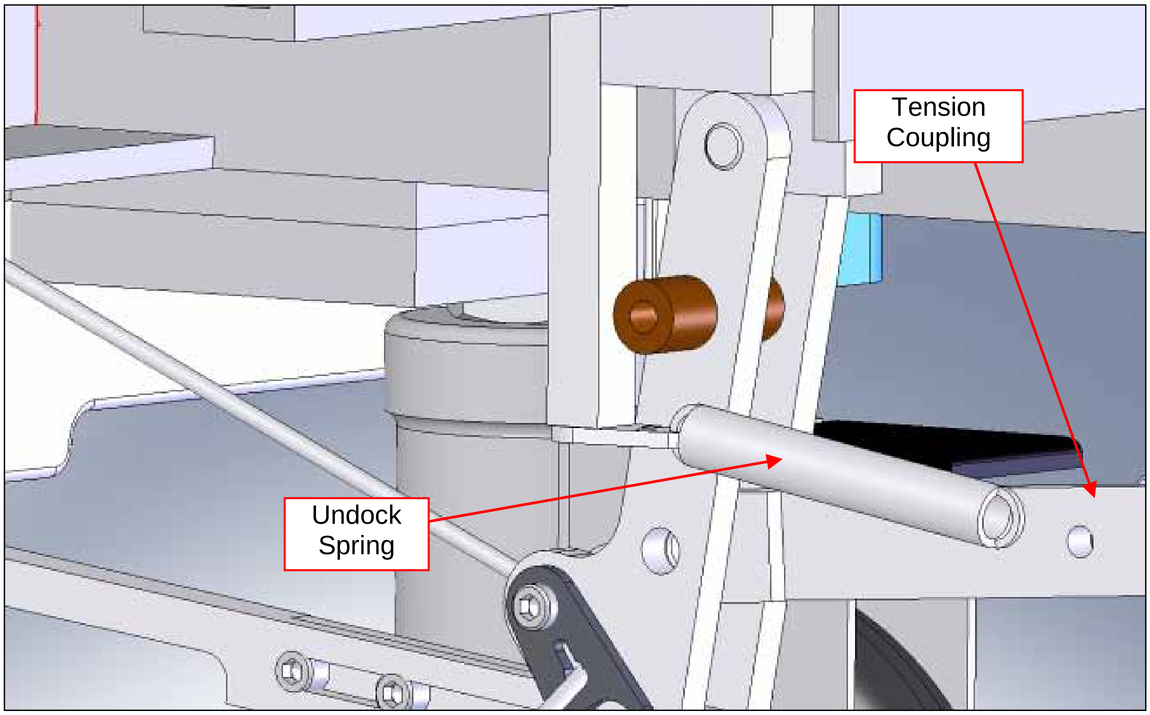


Figure 129

1. Remove Undock Spring Hanger from Swing Assembly by unscrewing mounting screw.
2. Remove Undock Spring from Tension Coupling Docking Linkage by unscrewing mounting screw and removing nut.
3. Attach new Undock Spring to Spring Hanger and attach to Swing Assembly by firmly screwing in mounting screw.


PINCH HAZARD

CAUTION **Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.**

4. Insert mounting screw into other end of Undock Spring, screw nut on to halfway point of mounting screw and screw into Tension Coupling Docking Linkage so that end of screw is flush with far side of Tension Coupling Docking Linkage.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Apply a small axial force (towards and away from the scanner) and a small lateral force (left and right) to the stretcher. Ensure that it does not undock due to this applied force.

iv. Return the patient support into the home position.

v. Undock the stretcher from the scanner.

vi. Repeat finalization steps three times.

13.36 Catchment

SMI Part #: SVC-00363
GEHC Part #: 5326152
Time required: 40 minutes
Tools required: Non-magnetic small and medium slot screwdriver
Personnel required: 1

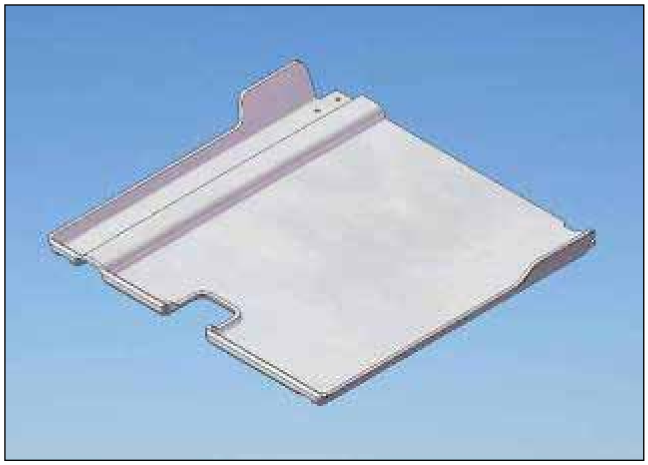


Figure 130

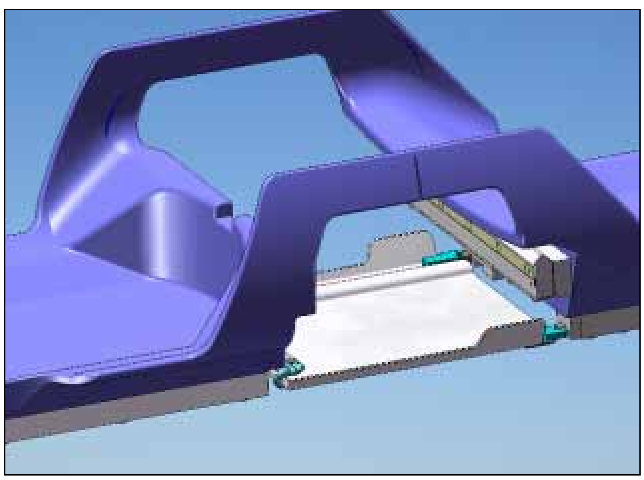


Figure 131

1. Separate Catchment from Catchment Hinges by removing mounting fasteners from underside of Catchment.
2. Remove Catchment Hinge Mounts from Patient Support by unscrewing mounting screws.
3. Remove Catchment Hinges from Catchment Hinge Mounts by unscrewing mounting screws.
4. Attach new Catchment Hinges on Catchment Hinge Mounts by screwing in mounting screws.
5. Attach Catchment Hinge Mounts to Patient Support by screwing in mounting screws.
6. Hold Catchment in position while fastening Catchment Hinges to Catchment.

Finalization Steps

i. Open the catchment handle and let the catchment drop open.

ii. Close the catchment.

iii. Repeat steps i) and ii) five times.

13.37 Patient Support Wheel

SMI Part #: SVC-00368
GEHC Part #: 5326159
Time required: 50 minutes
Tools required: Non-magnetic medium slot screwdriver
Personnel required: 1

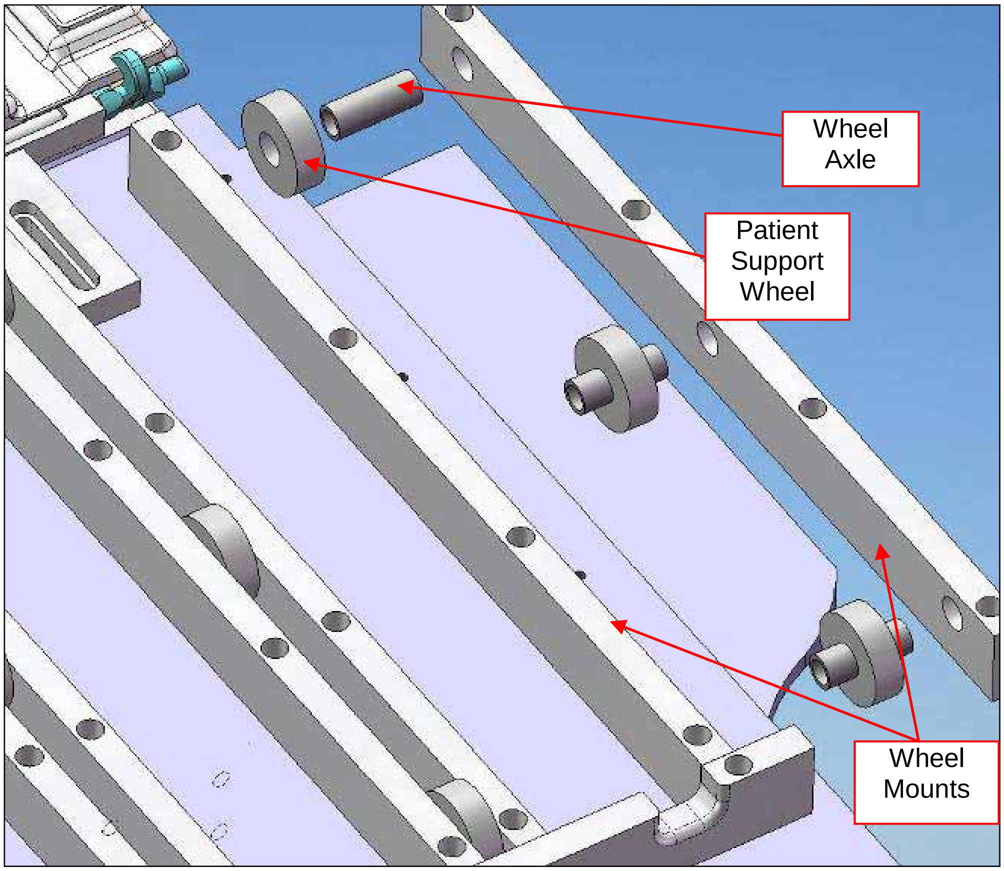


Figure 132

1. Access underside of Patient Support.
2. Remove all fasteners wheel mounts adjacent to the wheel required for replacement.
3. Separate wheel mounts from wheels and axles.
4. Inspect wheels and replace as required.
5. Re-assemble wheels on axles between wheel mounts.
6. Replace Wheel Mounts on Patient Support and screw in fasteners.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan. Ensure that the Patient Support remains stable and does not wobble as it moves into the bore.

iii. Return the patient support into the home position. Ensure that the Patient Support remains stable and does not wobble as it moves out of the bore.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps five times.

13.38 L-R Centering Bumper

SMI Part #: SVC-00339
GEHC Part #: 5326142
Time required: 40 minutes
Tools required: ¼" non-magnetic Allen wrench
Personnel required: 1

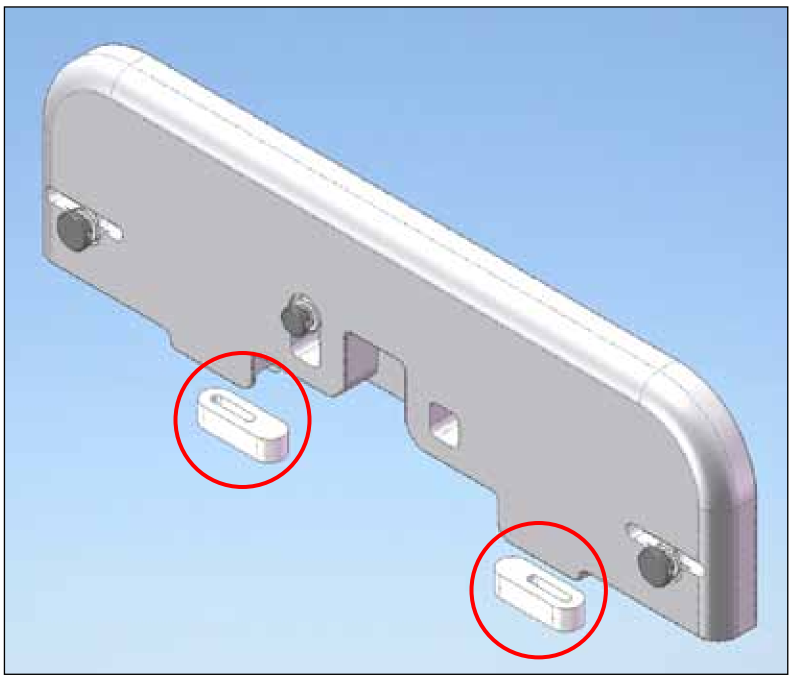


Figure 133

1. Note position of Centering Bumper and remove from bottom of Bumper by unscrewing mounting screw.
2. Insert mounting screw in Centering Bumper and screw into Bumper and measure to ensure correct position.

Finalization Steps

i. Advance the stretcher onto the scanner, ensuring that it is correctly centered.

ii. Dock the stretcher to the scanner.

iii. Advance the patient support to scan.

iv. Return the patient support into the home position.

v. Undock the stretcher from the scanner.

vi. Pull the stretcher away from the scanner.

vii. Repeat finalization steps five times.

13.39 Caster and Caster Boot

SMI Part #: Caster: SVC-00353
Caster Boot: SVC-00355
GEHC Part #: Caster: 5326150
Caster Boot: 5408273
Time required: 40 minutes
Tools required: 1½” non-magnetic wrench (supplied with the table, see *Section 3.1 Primary Tools*), 5/32” Allen wrench, Scissor Jack part 5437045*
Personnel required: 2

* A car jack that meets the follwing requirements may be used (see images of an acceptable jack below):

- lift rated minimum of 1000 lbs
- lift range of 3 ¾” - 14”
- must have a flat surface for lifting and groove

The jack is **ferrous** and all caster and caster boot replacements **must** be performed outside of the magnet room.

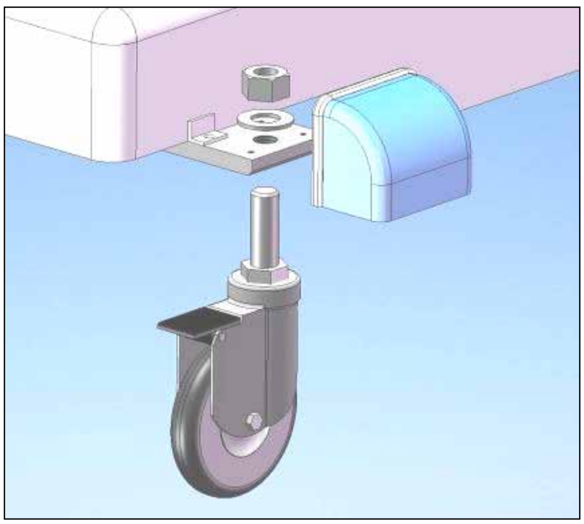


Figure 134

1. **Lock the casters at the opposite end of the stretcher from where the service is being performed. If caster does not successfully lock chock the wheel.**
2. Remove the Patient Support from the foot end by following the steps outlined in *Section 7.1 Initial Alignment, steps 3 to 9.*

HEAVY ITEM

WARNING

Use proper heavy lifting techniques and be careful when lifting the patient support. Ensure that two people are lifting, because the patient support weighs approximately 55lbs.

3. Loosen the mounting screws and remove the Wheel Covers.
4. Note the position of the caster’s fine height adjustment by counting turns or threads.
5. Loosen the 1.5” locknut.

Figure 135

Figure 136

6. Raise the stetcher 30 cm (12") from the floor so the wheel is between 7cm and 10cm off the ground (3" to 4"). You can do this using the FRU scissors jack or car jack. Ensure that the jack is not placed on the shroud, but is actually contacting the aluminum frame (as shown in Figure 135 for the UBS end of the table and Figure 136 for the LBS end). This will leave the wheel that is being changed floating, allowing it to be changed.

⚠ WARNING **When using a jack to raise the stretcher, you must perform this procedure outside of the MR suite.**

⚠ WARNING **It is <u>never</u> safe to put any tools or body parts under the stretcher while it is propped up. Ensure that your feet and hands are clear of the bottom of the stretcher and the casters at all times. This applies to both the person changing the caster assembly and the person assisting.**

PINCH HAZARD

⚠ CAUTION **Exercise caution when operating or replacing part as there may be a potential pinch point. Keep hands and fingers clear of the mechanism while it is moving or in use.**

7. Unscrew the caster assembly completely from the plate.
8. Remove the Caster boot (not shown) and replace with new boot if necessary.
9. Screw in new Caster to correct position and tighten nut very tight.
10. Replace Wheel Cover and tighten mounting screws.
11. Lower the scissor/car jack.
12. Replace the patient support by following the steps outlined in *Section 7.1 Initial Alignment, steps 3 to 9* in reverse.

Finalization Steps

i. Advance the stretcher onto the scanner, ensuring that it is correctly centered.
ii. Dock the stretcher to the scanner.
iii. Advance the patient support to scan.
iv. Return the patient support into the home position.

v. Undock the stretcher from the scanner.

vi. Pull the stretcher away from the scanner.

vii. Repeat finalization steps five times.

13.40 Cable Elbow 105°

SMI Part #: SVC-00359
GEHC Part #: 5408275
Time required: 40 minutes
Tools required: Non-magnetic adjustable wrench
Non-magnetic pliers, needle nose, 6"
GE Only: Non-magnetic pliers, side cutter, titanium
Note: This FRU contains one cable and two cable ends
Personnel required: 2

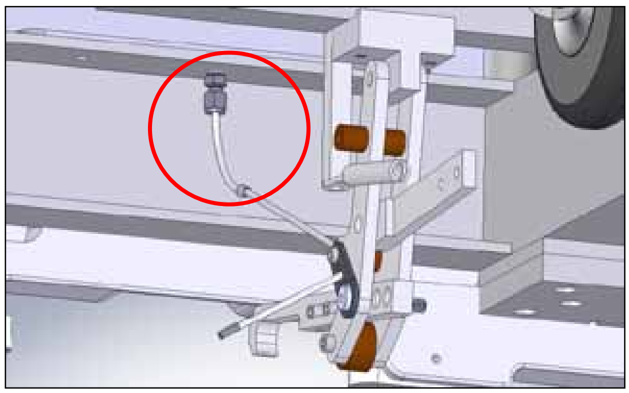


Figure 137

1. Remove the drawers from the stretcher.
2. Remove the Patient Support from the foot end by following the steps outlined in *Section 7.1 Initial Alignment, steps 3 to 9.*

HEAVY ITEM

WARNING

Use proper heavy lifting techniques and be careful when lifting the patient support. Ensure that two people are lifting, because the patient support weighs approximately 55lbs.

3. If the old cable is too frayed to be reused, it must be replaced with the new cable included in this kit. Remove the cable end from the old cable and loosen the screw that is holding the cable to the swing assembly. Unscrew the white screw from Latch tab and push cable up and pull out completely through Latch Tab.

   OR

   If the old cable is in an acceptable condition, remove the cable end from the old cable. Loosen the screw that is holding the cable to the swing assembly and push the cable up above the 105° cable elbow.
4. Unscrew Cable Elbow Assembly from mounting point with adjustable wrench.
5. Insert new Cable through new Cable Elbow Assembly and screw into mounting point with adjustable wrench.
6. Fasten cable to Swing Assembly and refer to Section 7.10 to re-set mounting point. If a new cable was used, cut the cable if it is too long using non-magnetic side cutter pliers.
7. Terminate cable by crimping on a new cable end using non-magnetic needle nose pliers.
8. Place the top drawer back onto the LBS.
9. Ensure that the stretcher docks properly after the installation of the new parts.

10. Replace the patient support by following the steps outlined in *Section 7.1 Initial Alignment, steps 2 to 9* in reverse.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps two times.

13.41 Cable Elbow 75°

SMI Part #:	SVC-00358
GEHC Part #:	5326138
Time required:	40 minutes
Tools required:	Non-magnetic adjustable wrench and 3/16" Allen wrench Non-magnetic pliers, needle nose, 6" GE Only: Non-magnetic pliers, side cutter, titanium
Note:	This FRU contains one cable and two cable ends
Personnel required:	2

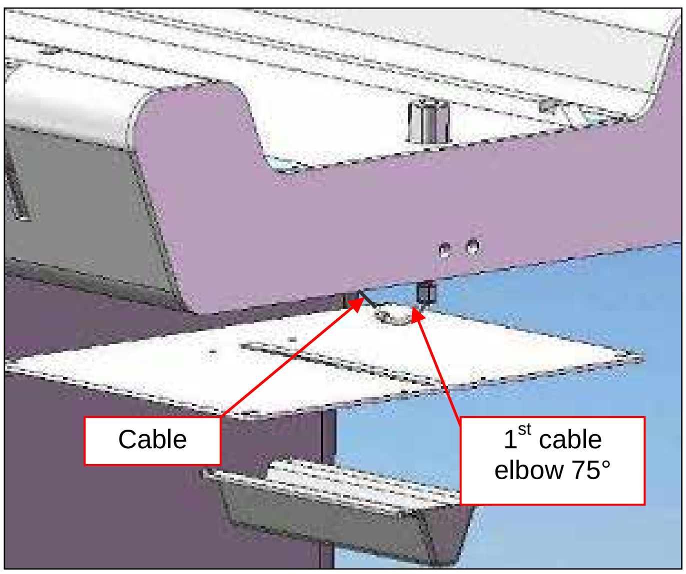


Figure 138

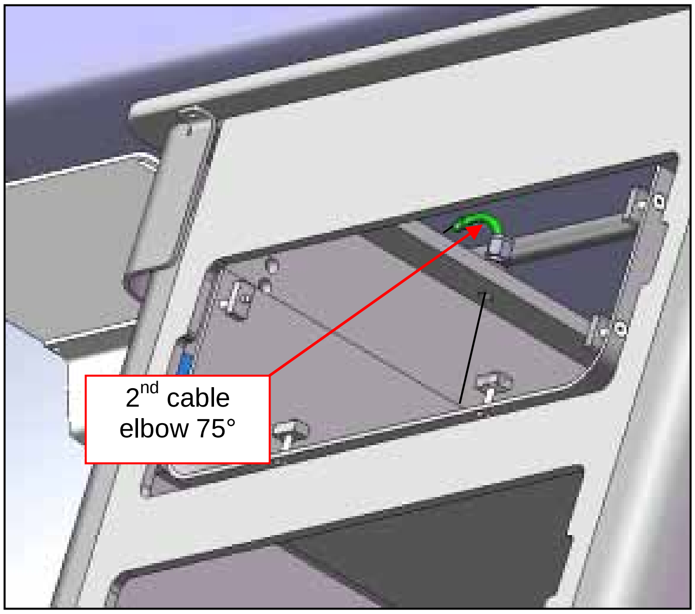


Figure 139

1. Remove the Patient Support from the foot end by following the steps outlined in *Section 7.1 Initial Alignment, steps 3 to 9.*

HEAVY ITEM

WARNING

Use proper heavy lifting techniques and be careful when lifting the patient support. Ensure that two people are lifting, because the patient support weighs approximately 55lbs.

2. Unscrew four mounting screws to remove Cable Elbow Guard to expose the first 75° cable elbow.
3. Remove the top drawer from LBS to expose the second 75° cable elbow.
4. Detach cable from the Swing Assembly.

5. If the old cable is too frayed to be reused, it must be replaced with the new cable included in this kit. Remove the cable end from the old cable and loosen the screw that is holding the cable to the swing assembly. Unscrew the white screw from Latch tab and push cable up and pull out completely through Latch Tab.

   OR

   If the old cable is in an acceptable condition, remove the cable end from the old cable. Loosen the screw that is holding the cable to the swing assembly and pull the old cable out through the Latch Tab.
6. Unscrew Cable Elbow Assembly from mounting point with adjustable wrench.
7. Insert new Cable through new Cable Elbow Assembly and screw into mounting point with adjustable wrench.
8. Fasten cable to the Swing Assembly and refer to Section 7.10 to re-set mounting point. If a new cable was used, cut the cable if it is too long using non-magnetic side cutter pliers.
9. Terminate the cable by crimping on a new cable end using non-magnetic needle nose pliers.
10. Place the top drawer back onto the LBS.
11. Reposition the Cable Elbow Guard and fasten the four mounting screws.
12. Replace the patient support by following the steps outlined in *Section 7.1 Initial Alignment, steps 2 to 9* in reverse.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps two times.

13.42 Cable for Stretcher

SMI Part #: SVC-00360
GEHC Part #: 5408276
Time required: 30 minutes
Tools required: Non-magnetic adjustable wrench and 3/16” Allen wrench
Personnel required: 2

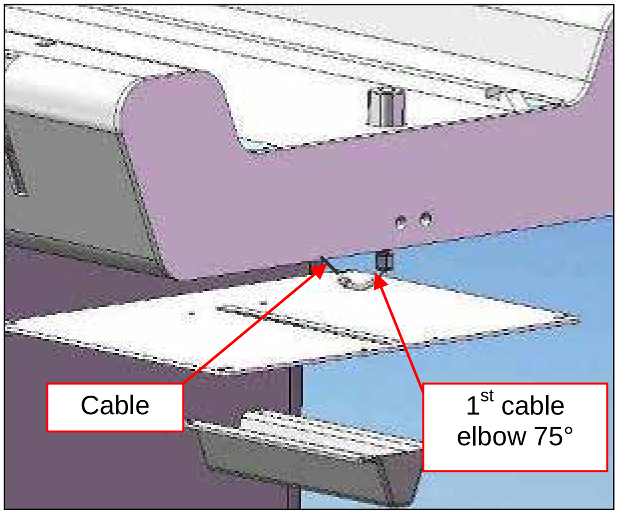


Figure 140

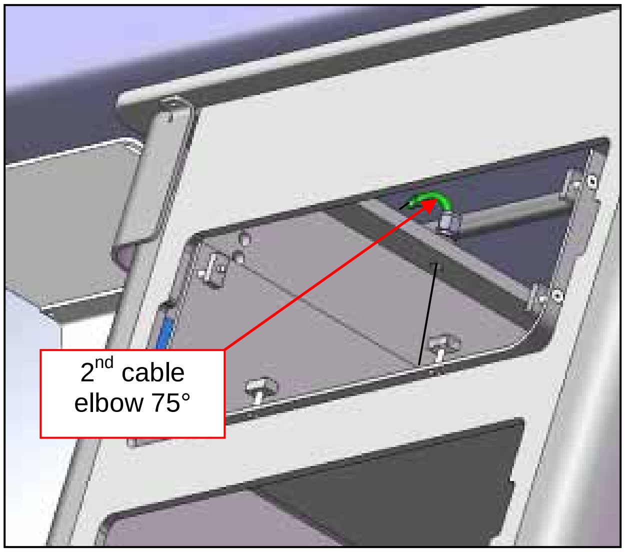


Figure 141

1. Remove the Patient Support from the foot end by following the steps outlined in *Section 7.1 Initial Alignment, steps 3 to 9.*

HEAVY ITEM **WARNING** ***Use proper heavy lifting techniques and be careful when lifting the patient support. Ensure that two people are lifting, because the patient support weighs approximately 55lbs.***

2. Unscrew four mounting screws to remove Cable Elbow Guard to expose the first 75° cable elbow (Figure 140).
3. Remove the top drawer from LBS to expose the second 75° cable elbow (Figure 141).
4. Remove cable end, then detach cable from Swing Assembly.
5. Unscrew white screw from Latch tab and push cable up and pull out completely through Latch Tab.
6. Insert new Cable through Latch Tab, first 75° Cable Elbow, second 75° Cable Elbow, then the 105° Cable Elbow.
7. Fasten cable to Swing Assembly and refer to Section 7.10 to re-set mounting point.
8. Place the top drawer back onto the LBS.

9. Reposition the Cable Elbow Guard and fasten the four mounting screws.
10. Replace the patient support by following the steps outlined in *Section 7.1 Initial Alignment, steps 2 to 9* in reverse.

Finalization Steps

i. Dock the stretcher onto the scanner.

ii. Advance the patient support to scan.

iii. Return the patient support into the home position.

iv. Undock the stretcher from the scanner.

v. Repeat finalization steps two times.

14 Appendix A

GE Vanguard 8ch Breast Table Installation and PM Verification

Install Date:	__ __ __ __ __ 20__ __	Location:	______________________________ , __ __
Customer Name:	____		
Vanguard Serial #:	__ __ __ __	Configuration:	**8 Channel**
Scanner:	**__.__T GE Signa**	Software Rev:	__ __.__ __ __ __ __ __ __

After completion of the installation or PM, the following items must be checked to verify function of the Vanguard on a GE Signa MRI scanner.

Stretcher

- Bridge guide block is centered with square opening in LBS.
- Bridge slides freely without rubbing and latches closed.
- LBS forward/backward and coarse height adjustment screws (4+4) are tight.
- UBS forward/backward and coarse height adjustment screws (4+4) are tight.
- Latch tab is flush with LBS surface when system is docked.
- Latch tab moves freely in patient support latch fixture when undocked.
- No-undock hook is not contacting pedal arm in down position.
- No-undock hook cable is screwed tightly to no undock hook.
- No-undock hook is clear of pedal – contacting hard stop when actuated.

Docking Mechanism

- Stretcher docks/undocks with light foot pressure (20-30lbf).
- No harsh mechanical noises heard or felt through pedal when docking the stretcher.
- Docking force fully compresses both bumper springs and resists undocking attempts.
- Drive link is not loose or rubbing other parts, shoulder bolt is tight.
- Pedals have no excessive left-right play.
- Docking hook height is >¼" above docking fixture hook, not contacting dock ceiling.
- Eccentric dock hook height adjuster mounting screw is tightened firmly.
- Docking hook length adjustment screws (2) are tight.
- Cables and elbows are not damaged.
- Latch tab cable clamp screw is tight.

Bumper and Alignment

- Bumper is vertically and horizontally aligned to docking fixture.
- Bumper mounting screws (4) are tight.
- Spring bumpers and Sensor Actuator plastic Hex nuts are tight.
- Bottom left-right fine adjustment bumper screws (2) are tight.
- Rear left-right fine adjustment bumper screws (2) are tight.
- Stretcher and scanner guide rails are collinear.
- 1"-1.25" gap from edge of scanner bridge to edge of LBS.

Casters

- Casters are firmly mounted, rolling/swiveling smoothly and brakes effectively locking.
- Caster mount fine adjustment nuts are very tight.
- Caster covers are tight and flush with frame.
- No bumping or rattling noises when transporting stretcher.

Patient Support

- Patient support rolls into bore with no noise/vibration with significant load applied
- Emergency release works with <110 degree turn and light pull force.
- No chips on patient support surface >1cm^2.

Doghouse

- Doghouse mechanism retains patient support on stretcher when latch tab is retracted.
- Doghouse mechanism plunger nut is tight (check using a wrench).
- Business card slips undamaged between brass screw and GE doghouse plunger.
- GE doghouse hook has play with patient support latch fixture in latched home position.
- Patient Support remains latched to doghouse if motion is halted by resistance.

General

- All interlocks function as detailed in PM section of Service Manual.
- Side handles slide without interference and lock in extended position.
- System cleaned with alcohol wipes and free of fingerprints or debris.
- Lighting system fuse inserted, lights are not dim/discoloured, all LEDs functioning.
- UBS/LBS surface on which patient support rolls is clean, smooth and undamaged.
- Mirrors clean, aligned and fastened to stretcher with Velcro.
- All coils snap in and hold firmly in the compression frames.
- Compression sliders slide smoothly and locks (2) work without excessive force.
- Cable tray/coil connector cable housings are not separated from connectors.
- All padding is present and free of damage.

SNR/Channel Verification (1.5T)

- 8ch BL SNR: __ __ __ (>403 for 1.5T), 8 channels functioning.
- 4ch BL SNR: __ __ __ (>333 for 1.5T), 4 channels functioning.
- 2chL UL SNR: __ __ __ (>280 for 1.5T), 2 channels functioning.
- 2chR UL SNR: __ __ __ (>280 for 1.5T), 2 channels functioning.

SNR/Channel Verification (3T)

- 8ch BL MCQA Result: Pass / Fail (Circle Result)
- 4ch BL MCQA Result: Pass / Fail (Circle Result)

SMI ONLY	• Any results not in specification reported immediately to SMI Service – 1-866-735-3744. • For any failing results, DICOM data saved to USB/DVD storage.

Pre-Departure

- Customer requested (if available) to report any user issues.
- All parts and accessories shortages reported immediately.

SMI ONLY	• Fill out calling card and leave in a visible location • Any items not checked must be reported immediately by phone to SMI Service Department. • After completing this checklist, please provide a signed copy to the SMI Service Department.

Name:	Position:	Company:
Signed:	Date: D D M M M 20 Y Y	
Notes:		

GEHC ONLY	·· After completing this checklist, please store a signed copy on site.